K-Bound: Knowing When Not to Adapt

A Label-Free Adapt/Freeze/Abstain Certificate for Safe Test-Time Adaptation

Pratik Niroula
Minnesota State University, Mankato
pratik.niroula@mnsu.edu

Abstract

Test-time adaptation can silently degrade a deployed model on unlabeled target data, and with no target labels the system cannot tell a helpful shift from a harmful one. Deciding *whether* to adapt is possible only relative to a declared bound on calibration drift—a drift budget β—so the question that decides the method is where that budget comes from. This paper answers it, and the answer is that it cannot come from the deployment.

We prove that for every declared class of target laws the exact minimax label-free budget is the fibre radius Γ_z, attained by a constant audit that reads none of its data; for the deployment class $\mathcal{C}_\beta$ the fibre radius equals β exactly, so the best possible label-free audit returns its own input, and the number it returns is simultaneously the radius of the band on which a strict adapt-or-freeze commitment is unsound. Two consequences follow. Any δ-valid label-free audited rule commits with probability at most δ over the unrestricted class—decision yield is bounded by the error budget, at every batch size and for every evidence map. And a *fully labeled* calibration sample from any other domain leaves the minimax budget unchanged, absent a separately declared coupling. One lemma—that the label kernel on the disagreement region is unconstrained by every label-free observable—carries the matched-evidence impossibility, the abstention band, and the budget impossibility as readings of the same one-parameter family at three radii.

We then measure the prediction. Declaring β as a high quantile of realized drift on source-like development cells, exactly as this paper's own method section prescribes, fails on 6,885 real evaluation cells: on CIFAR-10-C the declared budget is 1.4× to 50× smaller than soundness requires, 24–73% of deployment cells fall outside the declared class, and committed actions are wrong at 0.4–16.6% where the frontier promises 0%; on ImageNet-C a large enough budget is derivable and returns zero commitments on all 405 cells in five of ten configurations. This is not a quantile-estimation failure: across 470 splits the conformal order statistic that replaces the plug-in quantile differs from it by a median 0.34%, with a median coverage difference of exactly zero. We withdraw the operational reading of the frontier; the proofs are untouched.

What escapes the impossibility is a population of deployments rather than a deployment. Under retrospective outcome logging and exchangeability with K logged episodes, β is identified as an episode quantile and estimable by a conformal order statistic, with the episode requirement bracketed to a factor of e: $1/(e\alpha) - 1 \leq K^\star \leq 1/\alpha - 1$, between 3 and 9 logged deployments at $\alpha = 0.10$, and first feasible on real data at exactly $K = 9$. The escape is average-over-episodes instead of supremum-over-the-fibre, paid for with historical labels, and neither ingredient suffices alone; the guarantee it buys is marginal across deployments, not conditional on the one in hand. The assumption that buys it is the one a shifted deployment violates: on CIFAR-10-C the episode budget attains its nominal 0.900 coverage under an exchangeable control (0.905) and along nuisance axes, and collapses to 0.626 when the corruption family changes and 0.454 when severity changes, with half the severity splits admitting no budget at any declared level; weighted conformal restores coverage to 1.000 by driving decision yield to 0.000.

What survives is a finite-sample certificate that trades decision yield for a false-adapt guarantee, and we measure that frontier exactly. Knowability-Guided Adaptation wraps any adapter (Tent,

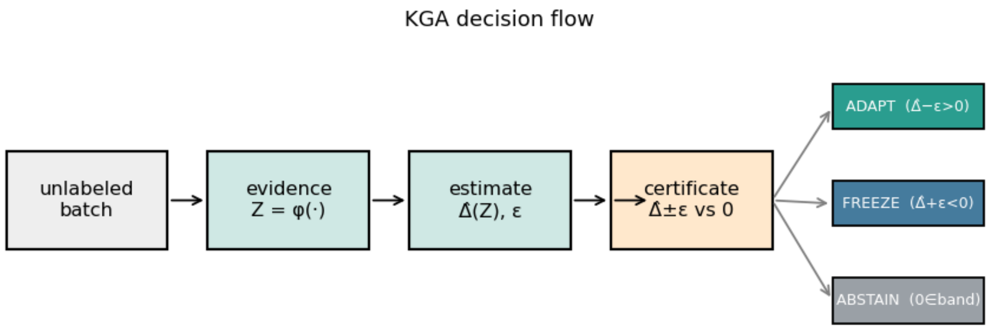

Figure 1: **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (certified helpful), FREEZE (certified harmful), or ABSTAIN (no supported strict commitment). Under interval coverage it controls the marginal events $\mathbb{P}(\text{ADAPT}, \Delta \leq 0)$ and $\mathbb{P}(\text{FREEZE}, \Delta \geq 0)$ at level α; no conditional-error guarantee is claimed.

EATA, SAR), estimates the benefit $\Delta = R_T(f_0) - R_T(f_a)$ from label-free evidence, and commits only when a calibrated interval $\widehat{\Delta} \pm \varepsilon$ excludes zero; ε is a conformal radius and is *not* an estimate of β. On the CIFAR-10-C stress grid it commits on 66.8% of 6,480 cells at zero false adaptations over 1,113 and 1,244 ADAPT decisions (Clopper–Pearson 95% upper bounds 0.0027 and 0.0024 on the conditional rate), cuts regret 5.00× below the better fixed policy and 3.1× below a hindsight-tuned label-free drift heuristic held to the same budget, and places its commitments on cells whose true effect is 24.5× larger than the cells it declines. Under leave-one-corruption-out calibration the radius inflates 4.3×–6.9×, the commitment rate falls from 0.51–0.60 to 0.40–0.42 and regret worsens 3.4×–7.6×, and the false-adapt count stays at zero in all ten runs; under that partition the certificate still beats always-adapt on all five Tent runs and loses to it on all five EATA runs, so what survives leave-one-corruption-out is the safety property and not the routing-utility one. On four one-sided natural tracks (Camelyon17, iWildCam, RxRx1, Office-Home) the certificate ties the better fixed policy at zero observed false adaptation, but three of those zeros are weak—Camelyon17's promoted subset contains no harmful cell, and RxRx1 and iWildCam make 0 and 1 ADAPT decisions—and the held-out record file behind the promoted iWildCam triple is absent from the release, so that row is a sealed summary rather than a reproduced result. On PACS, ImageNet-R and CIFAR-10.1 the certificate is a conservative null or fails the declared transfer bar, and we report those outcomes rather than withdraw the tracks; on ImageNet-R it is 1.76× worse than always-adapt. K-Bound is a safety, validity and abstention layer for test-time adaptation; the drift budget its own theory requires is not obtainable at a deployment, and the certificate is what remains when that is taken seriously.

Keywords: test-time adaptation, safe adaptation, abstention, selective prediction, distribution shift, identifiability, label-free risk estimation

1 Introduction

Test-time adaptation (TTA) updates a model on the unlabeled test stream and, under favorable distribution shift, recovers accuracy. Under unfavorable shift, the same objective silently degrades it, and with no target labels the system usually cannot tell which case it faces. The decision that matters therefore comes *before* the update: given a frozen baseline f_0 and an adapted candidate f_a, choose **adapt** when adaptation is knowably helpful, **freeze** when knowably harmful, or **abstain** when label-free evidence cannot identify the sign of the benefit. The field's default assumption is that adaptation should be made robust enough to always run; we replace it with an explicit, certified decision about whether to run it at all.

Some cases are information-theoretically unknowable. It is established that label-free quantities cannot decide adaptation without an assumption on the shift: identifying OOD accuracy is as hard as identifying the optimal predictor [11], and the optimal predict-versus-abstain rule under a shift budget has a known form [12]. We localize this landscape to the paired benefit sign, and the localization is a single statement in five links. *One.* On the disagreement region the label kernel is a free parameter that no label-free observable constrains (Lemma 1); read at one radius that is the matched-evidence impossibility (Theorem 1), which forces abstention at rate $\geq 1-2\alpha$ on that evidence fibre (Corollary 1) and yields the strict-commitment criterion $|M| > \beta$ (Theorem 2). A label-free adapt/freeze decision therefore exists only relative to a *declared* drift budget β, and the question that decides the method is where that budget comes from. That configuration is not only constructed: §4.7 exhibits it in logged deployments, and measures that it is 39× *rarer* than chance — the evidence is informative and merely insufficient. *Two.* The same lemma, read at its extreme radii, answers it: for every declared class the exact minimax label-free budget is the fibre radius Γ_z, attained by a constant audit that reads none of its data (Theorem 3), and for the deployment class $\Gamma_z(\mathcal{C}_\beta) = \beta$ exactly (Corollary 3) — the best possible label-free audit returns its own input, and the number it returns is the radius of the band on which the frontier must abstain. *Three.* A fully labeled calibration sample from any other domain moves that value by exactly zero absent a separately declared coupling (Theorem 4). That is a prediction about our own method section, and §5 measures it on 6,885 real evaluation cells: declaring β from source-like development data fails, the frontier's operational reading is withdrawn, and the theorem says why. *Four.* What escapes is a population of deployments rather than a deployment: under retrospective outcome logging and episode exchangeability, β is identified as an episode quantile and estimable from K logged deployments bracketed to a factor of e (Theorems 7, 8, 9) — and the assumption that buys it is the one a novel deployment violates, which we also measure (§6.8). *Five.* What remains deployable is a finite-sample certificate that trades decision yield for a false-adapt guarantee (Theorem 11). KGA (Algorithm 1) realizes it: a benefit estimator and a conformal radius wrapped around any adapter (Tent, EATA, SAR), committing only when a calibrated interval excludes zero — a direct benefit estimate with a calibrated radius, not a numerical $|M|>\beta$ test.

The evidence follows the regime map in Table 1, and the primary finding is a safety one. On four one-sided natural tracks—hospital (Camelyon17), wildlife-camera (iWildCam), laboratory-batch (RxRx1), and domain (Office-Home) shifts—KGA is *no-harm*: it ties the better fixed policy and averts the failure of the wrong one, with zero observed false adaptation, and it abstains rather than overclaim where the evidence is weak. Three of those four zeros are weak, and we say so where we report them: Camelyon17's OOD test subset contains no harmful cell, and RxRx1 and iWildCam make 0 and 1 ADAPT decisions. Only Office-Home (22 adapts) carries a non-trivial adapt count among the four, and its Clopper–Pearson bound 0.127 still exceeds α. On the remaining three natural tracks the outcome is different and we report it as such: PACS is a conservative null in which KGA's mean regret (0.0431) is 2.45× always-adapt's (0.0176); ImageNet-R is a weak-evidence null in which KGA (0.0151) is worse than always-adapt (0.0064) on 7 of 10 backbones; and CIFAR-10.1 fails the declared transfer bar at $\mathrm{FA_u} = 0.167$. No-harm is therefore a claim about the one-sided locked tracks, not a universal property of the panel. Where it does hold it is the contribution, not a shortfall: a valid label-free certificate cannot beat a fixed policy that is already optimal, so tying it while avoiding the opposite failure is exactly correct behavior. A three-way router can beat both fixed policies only where helpful and harmful conditions coexist and the evidence separates them, in the operational sense defined in §8.6; we report those beats-both results as secondary mechanism confirmation, on stress and constructed regimes only, never as single-dataset natural-shift dominance. Because an ABSTAIN here realizes the frozen model, always-abstain is numerically always-freeze, so no-harm is *not* free; §9.7 quantifies what the certificate buys instead—on CIFAR-10-C, commitment on 66.8% of 6,480 cells at zero false adaptations, regret 5.0× below the better fixed policy and 3.1× below a hindsight-tuned label-free drift heuristic at the same false-adapt budget, with commitments landing on cells whose true effect is 24.5× larger than those it declines—and where it buys nothing, which on ImageNet-R is a 1.76× loss to always-adapt. The strongest single stress result the paper owns sits behind that number and belongs in the introduction: refitting the whole CIFAR-10-C pipeline leave-one-*corruption*-out, so the estimator never sees the corruption it is scored on, inflates the conformal radius 4.3×–6.9× and worsens regret 3.4×–7.6×, and the false-adapt count is 0 in all ten runs (two adapters × five seeds). The safety property survives that partition; the routing-utility property does not, and we report both: under leave-one-corruption-out the certificate beats always-adapt on all five Tent runs and loses to it on all five EATA runs. K-Bound therefore provides a safety, validity, and abstention layer for TTA rather than an accuracy boost, and does not claim universal improvement.

Plain-language takeaway. The question that matters at deployment is not how hard to adapt but whether to adapt at all. K-Bound answers it with a certificate: adapt only when label-free evidence certifies benefit, freeze when it certifies harm, and ABSTAIN otherwise (the maximal sound three-way rule when $|M| \le \beta$, and the budget β is not obtainable at the deployment, §4; KGA implements the distinct finite-sample interval decision). On the one-sided natural-shift datasets this returns *no-harm*—safety or abstention—which is exactly the point; on the domain-generalization and rendition tracks it returns a conservative null that costs regret relative to always-adapt, and we report that too. Beating both fixed policies additionally requires a mixed regime the evidence can separate (§8.6) and is reported as a secondary check, with any constructed pooled mixture labeled as such, never as single-dataset natural-shift dominance.

Contributions.

1. **One lemma, three radii.** We isolate the construction the paper previously carried out three times: on the disagreement region the label kernel is a free parameter that no label-free observable constrains (Lemma 1). Read at radius δ it is the matched-evidence impossibility (Theorem 1); at radius β it is the abstention band of the strict-commitment frontier (Theorem 2); at the extreme points it is the vacuity of label-free budget audits (Corollary 2). Naming it once is what makes the rest of the paper a single statement rather than three.

2. **The exact minimax label-free drift budget, and the identity that closes the loop.** For every declared class $\mathcal{C}$ the least label-free-certifiable budget is the fibre radius $\Gamma_z(\mathcal{C})$, and it is attained by a constant audit that reads none of its data (Theorem 3(a),(b)). For the paper's own deployment class, $\Gamma_z(\mathcal{C}_\beta) = \beta$ exactly (Corollary 3): *the best possible label-free audit of the drift budget returns the budget you declared, and that number is the radius of the band on which the frontier must abstain.* Two riders make it operational rather than philosophical: no label-free audit ever converts an abstention into a commitment (Theorem 3(c)), and over the unrestricted class the audited rule's decision yield is at most the false-commitment budget δ, at every batch size, every calibration sample size and every evidence map (Theorem 3(d)).

3. **Labels at the wrong domain buy nothing, and the three currencies that a budget costs.** A fully labeled calibration sample of any size leaves the minimax budget at $\frac{1}{2} + |M|$ unless a coupling to the deployment domain is separately declared (Theorem 4), which is

strictly stronger than audit vacuity and is the statement that bears on our own measurements. The bound $|\gamma_T| \le |\gamma_{\mathrm{cal}}| + \Delta_{\mathcal{F}}(\mu_{\mathrm{cal}}, \mu_T) + \rho$ separates the three (Proposition 2); exactly one term is label-free computable, and declaring ρ instead of β is a renaming, because the budget-declaration problem is a fixed point of the same construction (Remark 4).

4. **The theorem explains our own negative experiment.** We ran the declaration procedure this paper prescribes on 6,885 real evaluation cells and it fails (§5): 1.4×–50× too small on CIFAR-10-C, 24–73% of cells outside the declared class, commit errors of 0.4–16.6% against a promised 0%, and zero commitments on all 405 ImageNet-C cells in five of ten configurations. Theorem 4 predicts exactly this: the procedure supplies term 1 of Proposition 2 and sets terms 2 and 3 to zero by an undeclared exchangeability assumption. We check the prediction quantitatively—under the label shuffle of our own ablation the budget must collapse to $q_{0.90}(|\Delta - \mathbb{E}\Delta|) = 0.2376$, and it measures 0.2468–0.2606 across six channels—and we report the one place the prediction fails (§5, the ImageNet-C sign reversal). It is also not a quantile-estimation failure: across 470 splits the conformal order statistic differs from the plug-in by a median 0.34% at a median coverage difference of exactly 0.

5. **Where a budget can come from, what it costs, and why that does not help a shifted deployment.** β is a property of a population of deployments. Under retrospective outcome logging (E1) and episode exchangeability (E2) it is identified as an episode quantile (Theorem 7(a)) and estimable by a conformal order statistic (Theorem 8); it is *not* identified from unlabeled episode observables at any K, so the binding resource is historical *labels*, not historical data (Theorem 7(b)). We locate the escape from Corollary 2 to one logical move—average over the episode law instead of supremum over the fibre—and prove neither ingredient suffices alone (Proposition 3); we bracket the episode requirement to a factor of e, $1/(e\alpha) - 1 \le K^\star \le 1/\alpha - 1$ (Theorem 9, Corollary 4), verified at exactly $K = 9$ on real data; and we price the labels, with the calibrated score acting as a control variate that makes estimating γ strictly cheaper than estimating $\bar{a} - \frac{1}{2}$ (Proposition 4). Then we measure E2 and it fails on the axis a shifted deployment lives on: coverage 0.905 under an exchangeable control and 0.626/0.454 when the corruption family or severity changes, with weighted conformal restoring coverage to 1.000 at decision yield 0.000 (§6.8).

6. **What remains: a finite-sample certificate whose yield/safety frontier we measure.** Under interval coverage the rule controls the marginal false-adapt and false-freeze events at level α (Theorem 11); we do not claim the interval construction as novel machinery, and ε is not an estimate of β. What is new is the measured exchange rate. On CIFAR-10-C the certificate commits on 66.8% of 6,480 cells at $\mathrm{FA_u} = 0/6480$ over 1,113 and 1,244 ADAPT decisions (Clopper–Pearson 95% upper bounds 0.0027/0.0024 on the conditional rate), cuts regret 5.00× below the better fixed policy and 3.1× below a hindsight-tuned label-free heuristic at the same budget, and commits on cells whose true effect is 24.5× larger than those it declines (permutation $p < 0.0002$). Under leave-one-corruption-out recalibration the radius inflates 4.3×–6.9×, the commitment rate falls from 0.51–0.60 to 0.40–0.42, regret worsens 3.4×–7.6×, and the false-adapt count is 0 in all ten runs. Where it buys nothing we say so: ImageNet-R, 1.76× worse than always-adapt, with per-backbone value per committed decision exactly 0 on 9 of 10 backbones.

Table note. The uniform panel below is the sole empirical index; diagnostic variants and historical runs are not additional evidence for the headline claims.

Table 1: **Regime map: expected behavior vs. result.** Each regime, an example dataset, the behavior the frontier predicts for a valid certificate, and what was measured. "No-harm" means matching the better fixed policy while avoiding the worse; comparisons with both fixed policies are secondary routing-utility diagnostics. The evidence tier in Table 23 states whether that point verdict is locked, reconciled, provisional, or diagnostic. Regime assignment follows Definition 13, applied before outcomes are read.

Regime	Example	Expected	Result
Helpful-dominated	Camelyon17 OOD	tie always-adapt	no-harm, and vacuous: recomputable, but 18/18 cells are helpful and KGA adapts 18/18
Harmful-dominated	RxRx1; iWildCam/Office-Home (OOF lock)	tie always-freeze, beat always-adapt	locked no-harm; 0/1/22 adapts
Domain-gen. natural	PACS (4 LODO splits)	no-harm or abstain	**null; loses to always-adapt** 2.45×
Mixed + detectable	CIFAR-10-C Tent/EATA	adapt helpful, freeze harmful, abstain near zero	locked; 0 false adapts in 2357; CI utility vs. both
Mixed, *not* detectable	ImageNet-C Tent (56% harmful)	abstain / tie freeze	ties freeze; 0 adapts
Weak evidence	ImageNet-R, CIFAR-10.1	abstain / conservative	**diagnostics; ImageNet-R loses to always-adapt on** 7/10 **backbones, CIFAR-10.1 fails the bar**
Constructed mixture	three-source OOF stream	route per condition	locked routing result, not transfer
Budget declarable?	any	no label-free audit beats Γ_z (Thm. 3)	measured: $\widehat{\beta}$ 1.4–50× too small (§5)

Scope of the guarantee. K-Bound does not guarantee that a newly encountered deployment is exchangeable with the calibration data, that label-free evidence is risk-aligned, or that a fitted benefit estimator transfers. Its guarantee is conditional: *if* the declared benefit interval attains marginal target coverage, *then* strict adaptation on $\widehat{\Delta} - \varepsilon > 0$ controls the unconditional false-adapt event at level α, and strict freezing on $\widehat{\Delta} + \varepsilon < 0$ the false-freeze event. Deployment diagnostics can identify support shift, instability, leakage and insufficient effective sample size, but passing them does not prove coverage. When required checks fail or cannot be evaluated, the prescribed output is ABSTAIN or FREEZE — not a certified adaptation decision.

2 Related Work

The closest literature improves test-time updates, monitors adaptation failures, estimates target performance without labels, or characterizes abstention under shift. K-Bound differs in the decision object: it certifies the sign of the frozen-versus-adapted risk difference before the candidate state is committed, and it prices the one input that any such certificate needs.

Test-time adaptation. Tent [1], SHOT [2], TTT [3], EATA [4], SAR [5], CoTTA [6], and MEMO [7] improve adaptation through entropy minimization, filtering, regularization, sharpness-aware updates, or temporal stabilization (survey: [33]). These methods primarily answer *how* to adapt. Even when the update rule is more stable, the deployment system generally still commits to adaptation rather than certifying that the candidate has positive target benefit.

Guarded and monitored adaptation. POEM [10], sequential risk monitors [8, 9], PeTTA [40], and sample-level filters such as DeYO [41] guard an adaptation stream during or after updating; reset-timing rules for long-horizon streams [42] decide when to *restart* adaptation once it has already run. These approaches are complementary to K-Bound. They typically monitor a proxy or intervene in one failure direction, whereas K-Bound poses a pre-commitment three-way decision over the benefit sign and explicitly represents an unknowable region.

Label-free performance estimation. ATC [11], agreement-on-the-line [14, 13], AETTA [15], and related confidence-style estimators [32, 31] estimate target accuracy or error without target labels. Estimating $R_T(f)$ is not identical to certifying $\operatorname{sign}(R_T(f_0) - R_T(f_a))$: a deployment controller needs the sign of a paired risk difference and must account for uncertainty around zero. K-Bound uses this difference as the estimand and abstains when the available evidence does not identify its sign.

Impossibility, abstention, and risk control. General label-free target-risk identification requires assumptions on the shift [11, 16, 17, 18]. Kalai and Kanade [12] characterize a related abstain-versus-predict decision under a shift budget, while selective prediction [19], Chow's reject option [50], and learning-to-defer / reject-option classification [51, 52] abstain on individual labels. Our empirical uncertainty layer connects to split conformal prediction, conformal under shift, RCPS, and Learn-then-Test [34, 36, 35, 38, 39]; the episode-level budget of §6 is a domain-level conformal construction in the sense of that line, and we claim no new machinery for it (Remark 8).

Relation to classical DA and label-shift non-identifiability. Theorem 1 constructs two target laws with identical unlabeled observables and opposite label-dependent functionals. That is structurally the same move as the classical impossibility results for domain adaptation [17, 47] and the identification arguments underlying label-shift estimation [48, 49], and we do not claim the move as new. Three things are specific to our setting. (i) The estimand is a *paired sign* $\operatorname{sign}(R_T(f_0) - R_T(f_a))$, not a risk level: it survives any monotone recalibration that preserves the disagreement-region ordering, so estimator-quality arguments do not dispose of it. (ii) The ambiguity is localized to the disagreement region D, which shrinks the construction from the full risk range to a band of width 2β on a single conditional expectation. (iii) The class is parameterized by a *declared* budget, which turns an all-or-nothing impossibility into a frontier with an operational knob — and §4 then computes the exact minimax value of that knob under every label-free procedure, which is where the content of this paper's theory sits. Readers from the label-shift literature should read Theorem 1 as a localization of a known phenomenon, and Theorem 3 with Theorem 4 as the part that is not already implied by it.

Surviving gap, and what is not ours. Existing work provides stronger adapters, reactive monitors, label-free accuracy estimates, and general abstention theory. The remaining gap is not that nobody certifies the benefit sign; it is that *nobody has priced the budget that any such certificate needs*. Every label-free adapt/freeze rule with a validity claim is implicitly a statement about a declared class, and the literature does not say what a procedure can learn about that declaration, what labeled data at the wrong domain buys, or what structure makes the declaration estimable at all. §4 and §6 answer those three questions, and §5 and §6.8 measure the answers.

We state the other half explicitly, because it is easy to overstate. The interval machinery is *not* our contribution: confidence-sequence and anytime-valid risk monitoring for adaptation streams already exists [8, 9, 10], and the unsupervised-accuracy-estimation literature already states the no-assumption impossibility informally [11]. Our certificate shares machinery with that line; what we add is the pre-commitment decision object, the explicit abstention region, the exact minimax value of the budget it requires, and the accounting of when the resulting guarantee is actually tested — not the interval construction.

3 Problem Formulation and Validity

3.1 Deployment Setting and Adaptation Benefit

Source data follow $P_S(X, Y)$ and may be labeled during training and development. Deployment data follow $P_T(X, Y)$, but only target inputs and model outputs are available when the decision is made. A frozen predictor f_0 serves as the safe fallback, and an adapter $\mathcal{A}$ proposes a temporary candidate $f_a = \mathcal{A}(f_0, X_{1:m})$ from an unlabeled target batch. For loss ℓ, the target risk and adaptation benefit are

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)], \qquad \Delta = R_T(f_0) - R_T(f_a). \tag{1}$$

Thus $\Delta > 0$ favors committing the candidate, $\Delta < 0$ favors retaining the frozen state, and $\Delta = 0$ makes the two actions risk-equivalent. Target labels reveal Δ only during offline auditing; they are not available to the deployment rule.

3.2 Target-Label-Free Evidence and Decision Semantics

The controller observes a label-free evidence vector $Z = \phi(X_{1:m}, f_0, f_a)$ containing quantities such as prediction entropy, confidence, class-balance statistics, frozen-versus-adapted disagreement, output-distribution drift, and update magnitudes. The deployment rule returns

$$g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}.$$

ADAPT commits f_a; FREEZE rejects the proposal and retains f_0 (or the latest certified state in continual operation); and ABSTAIN declines to change the model, serves the frozen prediction, and may request more evidence or human review. Abstention therefore applies to the update decision, not to prediction itself.

3.3 What "Label-Free" Means

K-Bound is target-label-free rather than supervision-free. Source labels and a held-out development set may be used to fit the benefit estimator and calibrate its uncertainty radius. No target-test label may influence the adapter configuration, evidence map, benefit estimator, radius, action threshold, or model selection. Target-test labels are opened only after all decisions are frozen, for reporting accuracy, regret, and false-adapt events.

3.4 Assumptions and Validity Scope

The theory and the finite-sample certificate rely on distinct assumptions. *Risk alignment* determines whether the evidence can identify or bound the benefit sign over the declared target class. *Coverage* determines whether the empirical interval contains the true benefit at the stated rate. *Exchangeability* (or an explicit shift correction) connects calibration residuals to deployment residuals. The adapter, evidence schema, estimator, and calibration protocol must be fixed before held-out scoring. If these conditions cannot be justified, the theorem does not license either strict action; deployment must use a separately declared operational fallback, such as abstention or serving the frozen model. This

Table 2: **Protocol discipline.** Top: which labels each stage may see; target-test labels are forbidden until offline audit. Bottom: the conditions the certificate rests on, what each one buys, and the safe response when it cannot be justified.

Stage	Source labels	Dev labels	Target-test labels
Source training	yes	optional	no
Benefit/radius calibration	optional	yes	no
Deployment decision	no	frozen artifacts only	no
Offline evaluation	no	no	yes, after decisions

Condition	Role	If unsupported
Risk-aligned Z	Benefit sign is recoverable over the declared class	abstain; do not interpret a point estimate as a certificate
Coverage of $\widehat{\Delta} \pm \varepsilon$	Controls marginal false-adapt/freeze	recalibrate, widen the radius, or use the declared fallback
Exchangeable/corrected residuals	Transfers calibration to deployment	use shift-aware calibration or abstain
Bounded loss/benefit	Supports concentration and finite-sample analysis	clip/redefine the loss or state no bound
Frozen protocol	Prevents target-test leakage and selection bias	rerun with a locked protocol

Table 3: Core quantities. The population drift budget β and empirical radius ε are related but not interchangeable.

Symbol	Meaning	Available at deployment?
f_0, f_a	frozen and candidate models	yes
Δ	true target adaptation benefit	no
D	disagreement region	yes
M	observable population margin	through label-free evidence
γ	realized calibration drift	no
β	declared bound on $\lvert\gamma\rvert$	supplied pre-deployment
Z	label-free evidence vector	yes
$\widehat{\Delta}$	estimated adaptation benefit	yes
ε	calibrated residual radius	fixed pre-deployment
α	marginal error level	chosen pre-deployment

is a safety policy, not a theorem that freezing is optimal. Risk alignment is an assumption on the deployment class, not a property certified by fitting a benefit regressor: support and residual diagnostics may reject implausible evidence maps, but they do not verify it universally. These conditions are named (A1)–(A6) and stated as a single contract in §7.2; §7.9 makes them an operational gate, and §7.11 records the resulting assumption state per track.

3.5 Notation and Observability

Table 3 lists the core quantities and marks which are observable at deployment.

3.6 Formal Setup

Source law $P_S(X,Y)$ supplies labels at train/validation; target law $P_T(X,Y)$ exposes only $P_T(X)$ at deployment. Fix loss ℓ, frozen predictor f_0, candidate f_a, and target measure μ_T on inputs. Write target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)]$ and the *adaptation benefit*

$$\Delta \;:=\; R_T(f_0) - R_T(f_a) \qquad (\Delta > 0 \;\Leftrightarrow\; \text{adapting helps}).$$

The *disagreement region* $D := \{x : f_0(x) \neq f_a(x)\}$ is observable from (f_0, f_a).

Assumption 1 (Deployment setup)**.** Unless stated otherwise: binary label $Y \in \{0,1\}$, 0/1 loss ℓ, and $\mu_T(D) > 0$. Fix a source-calibrated score $s : \mathcal{X} \to [0,1]$ estimating the candidate's *pointwise correctness* (e.g. a Platt-scaled confidence that f_a is right at x), and write the target pointwise correctness $\eta_a(x) := \mathbb{P}_T(f_a(X){=}Y \mid X{=}x)$ on D, so that $\mathbb{E}_{\mu_T}[\eta_a \mid D] = \bar{a}$. Define the *observable margin*

$$M \ := \ \mathbb{E}_{\mu_T}[s(X) \mid D] - \tfrac{1}{2},$$

the *calibration drift* (unobserved at deploy)

$$\gamma \ := \ \mathbb{E}_{\mu_T}[\eta_a(X) - s(X) \mid D],$$

and label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ for any measurable ϕ using only unlabeled batches and model outputs (entropy, confidence, drift, disagreement, update norms). A *label-free rule* is any measurable map $g : \mathcal{Z} \to \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ that does not use target labels on the deployment batch.

Remark 1 (γ is a residual, so $M + \gamma = \bar{a} - \frac{1}{2}$ is an identity)*.* The drift γ is *defined* as $\bar{a} - \frac{1}{2} - M$: it is whatever is left of the disagreement-region accuracy after the observable part M is removed. Consequently $M + \gamma = \bar{a} - \frac{1}{2}$ by construction, and the reduction of Lemma 2 is a bookkeeping identity rather than a derived fact about the shift. We state this explicitly because it determines what the frontier of Theorem 2 does and does not assert. Sufficiency ($|M| > \beta \Rightarrow \operatorname{sign}\Delta = \operatorname{sign} M$) is then interval arithmetic: the interval $[M-\beta, M+\beta]$ has a determined sign exactly when $|M| > \beta$. The mathematical content sits in the *necessity* clause, which requires exhibiting two admissible target laws that induce the same evidence law and have opposite nonzero benefits (Theorem 1), and in the audit-vacuity theorem (Aud-A), which shows the budget β that makes the decomposition operational cannot itself be audited from label-free data. What the decomposition buys is organizational, and that is not nothing: it names, isolates, and prices the one unobservable quantity a deployment must bound.

Definition 1 (Risk alignment)**.** Evidence Z is *risk-aligned on a class* $\mathcal{P}$ if no two laws $P_T^1, P_T^2 \in \mathcal{P}$ with $\operatorname{sign}\Delta_1 \neq \operatorname{sign}\Delta_2$ satisfy $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$.

All frontier claims stated in terms of M additionally require M to be determined on each evidence-law fibre; equivalently, take the population evidence to be $\widetilde{Z} = (Z, M)$. Without this requirement, $|M| > \beta$ proves sign stability *given* M, not identifiability from $\mathrm{Law}(Z)$ alone. Fitting a benefit regressor does not certify risk alignment: support, dependence, and residual diagnostics may falsify a proposed evidence map but cannot verify the assumption uniformly over the declared deployment class.

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if the population frontier certifies $\Delta > 0$ ($\textsc{adapt}$); *knowably harmful* if it certifies $\Delta < 0$ ($\textsc{freeze}$); *unsupported for strict commitment* if the same augmented-evidence law is compatible with distinct signs, including zero versus a strict sign ($\textsc{abstain}$).

For a supplied drift budget $\beta \geq 0$, the *declared deployment class* is

$$\mathcal{C}_\beta \ := \ \{P_T : \ |\gamma(P_T)| \leq \beta\}.$$

Identifiability statements below are *over* $\mathcal{C}_\beta$, not over arbitrary shifts.

Definition 3 (Strict directional soundness and maximality)**.** Fix an augmented-evidence law z and let $\mathcal{C}_\beta(z)$ be the laws in $\mathcal{C}_\beta$ that induce z. A strict action $a \in \{\textsc{adapt}, \textsc{freeze}\}$ is *uniformly directionally sound at z* when

$$a = \textsc{adapt} \ \Rightarrow\ \Delta(P) > 0 \quad \text{for every } P \in \mathcal{C}_\beta(z),$$
$$a = \textsc{freeze} \ \Rightarrow\ \Delta(P) < 0 \quad \text{for every } P \in \mathcal{C}_\beta(z).$$

A sound three-way rule is *pointwise maximal* if it commits whenever some strict action is uniformly directionally sound and abstains otherwise. Zero benefit blocks a strict directional claim even though adapt and freeze have equal task risk when $\Delta = 0$; this is an epistemic validity convention, stronger than zero regret at the boundary.

Remark 2 (Four quantities). Do not confuse M (observable margin, deploy-time), γ (realized drift, latent), β (declared bound $|\gamma| \le \beta$ on the deployment class), and ε (finite-sample radius from held-out residuals). The radius ε operationalizes coverage; it is *not* an estimate of β.

Symbol	Meaning	At deploy?	Role
M	observable evidence margin	yes	evidence toward adapt/freeze
γ	realized calibration drift	no	can reverse the benefit sign
β	declared bound on $\lvert\gamma\rvert$	externally supplied	population frontier
ε	residual quantile radius	estimated pre-deploy	finite-sample certificate

What "label-free" means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\widehat{\Delta}$ and ε. Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

K-Bound in one sentence. A strict adapt/freeze commitment is uniformly supportable only when the population evidence margin exceeds the declared drift budget; otherwise abstention is the maximal sound three-way action.

4 The drift budget cannot be supplied label-free

The population frontier below commits exactly when $|M| > \beta$, and β is a *declared* input. This section settles what a procedure can do about that. The answer is sharper than "label-free audits are vacuous": for *every* declared class, the best label-free budget is a constant that ignores the data, that constant is exactly the radius of the class's own abstention band, and any attempt to buy it with labels at a domain other than the deployment domain buys nothing unless a transfer constraint is separately declared. The escape is a matched pair — labels at an anchor, plus a declared coupling — and it carries an $n^{-1/2}$ floor that no structural assumption removes.

Throughout, Assumption 1 is in force: $Y \in \{0,1\}$, 0/1 loss, $\mu_T(D) > 0$, $s : \mathcal{X} \to [0,1]$ a fixed *source*-calibrated score, $u := \eta_a - s \in [-1,1]$ on D, $\gamma = \mathbb{E}_{\mu_T}[u \mid D] = \bar{a} - \frac{1}{2} - M$, and $M = \mathbb{E}_{\mu_T}[s \mid D] - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$. The residual γ is *defined*, not derived (Remark 1); everything below is a statement about what a procedure can learn about it.

4.1 The engine: one lemma, used three times

Definition 4 (Audit data)**.** The *audit data W* is the totality of what a deployment-time procedure may read: the unlabeled deployment batch $X_{1:m} \sim \mu_T$, the fixed maps (f_0, f_a, s) and every label-free statistic $Z = \phi(X_{1:m}, f_0, f_a)$ derived from them, together with any sample S — labeled or not — drawn from a source or calibration law that is specified independently of the target label kernel. An *audit* is a measurable $\widehat{\beta} = g(W, U) \geq 0$ where U is a randomisation seed independent of everything else. W contains *no target labels.* This is exactly the "target-label-free" convention boxed in Assumption 1.

Lemma 1 (Kernel freedom on the disagreement region)**.** *Fix the observables* $(\mu_T, P_S, f_0, f_a, s)$*, hence* D *and* M*, and fix any off-*D *conditional label law* K°*. For every measurable* $\eta : D \to [0,1]$ *there is a target law* P_T^η *on* $\mathcal{X} \times \{0,1\}$ *with input marginal* μ_T*, off-*D *kernel* K°*, and*

$$\mathbb{P}_{P_T^\eta}(f_a(X) = Y \mid X = x) = \eta(x) \quad \textit{for } x \in D.$$

Moreover $\mathrm{Law}(W \mid P_T^\eta)$ *does not depend on* η*, and*

$$\bar{a}(P_T^\eta) = \mathbb{E}_{\mu_T}[\eta \mid D], \qquad \gamma(P_T^\eta) = \mathbb{E}_{\mu_T}[\eta \mid D] - \tfrac{1}{2} - M,$$

so that as η *ranges over all of* $[0,1]^D$*,* γ *ranges over exactly* $\left[-\frac{1}{2} - M,\ \frac{1}{2} - M\right]$*, an interval of length* 1 *containing* 0 *in its interior whenever* $|M| < \frac{1}{2}$*.*

Proof. Existence. On D we have $f_0(x) \neq f_a(x)$ and Y is binary, so $\{f_0(x), f_a(x)\} = \{0,1\}$; setting $K^\eta(x, f_a(x)) = \eta(x)$ and $K^\eta(x, f_0(x)) = 1 - \eta(x)$ therefore defines a valid Bernoulli conditional law on $\{0,1\}$ for each $x \in D$, measurable in x because η, f_0, f_a are. Off D take K°. Then $P_T^\eta(dx, dy) := \mu_T(dx)K(x, dy)$ is a probability law with the stated marginal and pointwise correctness.

Invariance of the audit data. By Definition 4, W is a function of $X_{1:m}$, of the fixed maps (f_0, f_a, s), of S, and of U. The construction changes neither μ_T (so $\mathrm{Law}(X_{1:m})$ is fixed), nor the maps, nor the law generating S, nor U. Hence $\mathrm{Law}(W \mid P_T^\eta) = \mathrm{Law}(W)$ for every η.

Range. $\bar{a} = \mathbb{E}_{\mu_T}[\eta \mid D]$ by definition of $\bar{a}$, and $\gamma = \bar{a} - \frac{1}{2} - M$ by Remark 1. Taking $\eta \equiv c$ for $c \in [0,1]$ gives $\bar{a} = c$, so $\bar{a}$ attains every value in $[0,1]$ and γ every value in $[-\frac{1}{2} - M, \frac{1}{2} - M]$; no larger range is possible since $\bar{a} \in [0,1]$ always. Since $\mu_T(D) > 0$ the conditional expectation is well defined. $\square$

Remark 3 (The three uses are one construction)*.* Lemma 1 restricted to the two-point subfamily $\eta \equiv \frac{1}{2} \pm \delta$ with $\delta < \min\{\beta - |M|, \frac{1}{2}\}$ *is* the proof of the matched-evidence impossibility (Theorem 1). The same lemma at the two *extreme* points $\eta \equiv 1$ and $\eta \equiv 0$ *is* the proof of audit vacuity (Corollary 2). Everything in this section is the same one-parameter family $\{\eta \equiv c\}_{c \in [0,1]}$ read at a different radius. In particular the construction that forces abstention and the construction that forbids auditing the budget which decides abstention are literally the same object; Corollary 3 makes this an equality of numbers, not an analogy. No total-variation trade-off is available to sharpen or weaken it: within a fibre the observable laws coincide *exactly*, which is the degenerate — and strongest — case of a two-point argument, so Le Cam's method has nothing to add here. It reappears, genuinely, only in Theorem 5, where labels make the two laws distinguishable.

4.2 What the engine already proved: the frontier

The three results this paper published previously are readings of Lemma 1 at three radii, and we now present them that way. The first is bookkeeping and we do not present it as a result; the second is the construction; the third combines them. The proof of Theorem 1 is now three lines because the work is done in §4 above.

Lemma 2 (Disagreement-region reduction)**.** *Under Assumption 1, with* $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$*,*

$$\Delta = 2\,\mu_T(D)(\bar{a} - \tfrac{1}{2}),$$
$$\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \tfrac{1}{2}) = \operatorname{sign}(M + \gamma).$$

The identity $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *holds for* every *target law; it does not assume* $|\gamma| \le \beta$*.*

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ,

$$\begin{aligned} M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}, \end{aligned}$$

so $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \frac{1}{2}) = \operatorname{sign}(M+\gamma)$. □

Theorem 1 (Interior matched-evidence impossibility)**.** *Fix* $\beta > 0$*. For any margin with* $|M| < \beta$ *there exist* $P_T^1, P_T^2 \in \mathcal{C}_\beta$ *such that* $\mathrm{Law}(\widetilde{Z} \mid P_T^1) = \mathrm{Law}(\widetilde{Z} \mid P_T^2)$ *for* $\widetilde{Z} = (Z, M)$*,* $M(P_T^1) = M(P_T^2) = M$*, and* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$*. (For* $\beta = 0$ *the class* $\mathcal{C}_0$ *forces* $\gamma = 0$*, so* $\operatorname{sign}\Delta = \operatorname{sign} M$ *is determined and no such pair exists: matched-evidence ambiguity is a strictly-positive-drift phenomenon.)*

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$, holding the off-D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ, and: *class membership,* $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \le \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence,* $\mathrm{Law}(W \mid P_T^\eta)$ does not depend on η by the lemma and $\widetilde{Z} = (Z, M)$ is a function of W, so $\mathrm{Law}(\widetilde{Z} \mid P_T^+) = \mathrm{Law}(\widetilde{Z} \mid P_T^-)$; *opposite signs,* Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\,\theta\delta \neq 0$ with $\operatorname{sign}\Delta_\theta = \theta$. □

The two evidence laws here coincide *exactly*, not merely approximately: the construction changes only the label kernel on D, and Z is a function of unlabeled inputs and model outputs. We therefore do not present this as a Le Cam two-point argument. A Le Cam argument trades off a positive total-variation distance against the achievable error; here the distance is zero, which is the degenerate and strongest case, and the corresponding statement is exact non-identifiability rather than a rate.

Corollary 1 (Matched-evidence abstention lower bound)**.** *Fix* $\alpha \in (0, \frac{1}{2})$*. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule* g *satisfying* $\mathbb{P}[g = \textsc{adapt}, \Delta \le 0] \le \alpha$ *in every admissible world and* $\mathbb{P}[g = \textsc{freeze}, \Delta \ge 0] \le \alpha$ *in every admissible world must satisfy* $\mathbb{P}[g = \textsc{abstain}] \ge 1 - 2\alpha$*.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. □

Proposition 1 (Boundary case and closed-band abstention)**.** *Fix* $\alpha \in (0, \frac{1}{2})$*. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band* $|M| \le \beta$*, and any label-free rule controlling both marginal errors at level* α *abstains there with probability at least* $1 - 2\alpha$*. Opposite nonzero signs are asserted only in the interior* $|M| < \beta$*.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so *both* directional error events $\{\textsc{adapt}, \Delta \le 0\}$ and $\{\textsc{freeze}, \Delta \ge 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\textsc{adapt}] \le \alpha$ and $\mathbb{P}[\textsc{freeze}] \le \alpha$, hence $\mathbb{P}[\textsc{abstain}] \ge 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. □

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

Theorem 2 (Exact strict-commitment frontier)**.** *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \ge 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

(i) ***Reduction.*** *For every P_T,* $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)$ *(Lemma 2).*

(ii) ***Sufficiency.*** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then* $\operatorname{sign}\Delta = \operatorname{sign} M$*.*

(iii) ***Necessity.*** *If $|M| < \beta$ (which requires $\beta > 0$), there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ with the same augmented-evidence law and opposite* nonzero *benefit signs; at $|M| = \beta > 0$ the admissible drift $\gamma = -\operatorname{sign}(M)\beta$ yields $\Delta = 0$, and in the degenerate case $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\Delta \equiv 0$. Hence no* strict *benefit direction is uniformly supported by* $\operatorname{Law}(\widetilde{Z})$ *over $\mathcal{C}_\beta$ whenever $|M| \le \beta$.*

(iv) ***Strict-commitment iff.*** *At the fixed augmented-evidence law, a strict action in $\{\textsc{adapt}, \textsc{freeze}\}$ is uniformly directionally sound in the sense of Definition 3 if and only if $|M| > \beta$; on the closed band $|M| \le \beta$ (including $M = \beta = 0$, where $\Delta \equiv 0$) abstention is the pointwise maximal sound action.*

The population rule $M > \beta \Rightarrow \textsc{adapt}$, $M < -\beta \Rightarrow \textsc{freeze}$, $|M| \le \beta \Rightarrow \textsc{abstain}$ is the pointwise maximal sound certificate on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \le \beta$, then $M + \gamma > 0$, so $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\operatorname{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. □

Region	Meaning	Action
$M > \beta$	allowed drift cannot flip the positive sign	ADAPT
$M < -\beta$	allowed drift cannot flip the negative sign	FREEZE
$\lvert M\rvert < \beta$	drift can reverse the strict sign	ABSTAIN
$\lvert M\rvert = \beta > 0$	drift can reduce benefit to zero	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.3 The exact minimax label-free budget

Definition 5 (Fibre and fibre radius). Let $\mathcal{C}$ be a declared class of target laws and let z denote a value of Law(W). The *fibre* of $\mathcal{C}$ at z is $\mathcal{C}(z) := \{P \in \mathcal{C} : \mathrm{Law}(W \mid P) = z\}$, assumed nonempty, and the *fibre radius* is

$$\Gamma_z(\mathcal{C}) \;:=\; \sup_{P \in \mathcal{C}(z)} |\gamma(P)| \;\in\; [0, \tfrac{1}{2} + |M|].$$

An audit $\widehat{\beta}$ is *δ-valid on $\mathcal{C}(z)$* if $\mathbb{P}_P(|\gamma(P)| \le \widehat{\beta}) \ge 1 - \delta$ for *every $P \in \mathcal{C}(z)$*.

Theorem 3 (Label-free budget audits are declarations). *Fix z with $\mathcal{C}(z) \neq \emptyset$ and $\delta \in [0, 1)$.*

(a) ***Floor.*** *Every δ-valid audit satisfies $\mathbb{P}(\widehat{\beta} \ge \Gamma_z(\mathcal{C})) \ge 1 - \delta$, and this probability is the same under every $P \in \mathcal{C}(z)$.*

(b) ***Attainment.*** *The constant audit $\widehat{\beta} \equiv \Gamma_z(\mathcal{C})$ is 0-valid. Hence* $\inf\{\operatorname{ess\,sup} \widehat{\beta} : \widehat{\beta}$ *0-valid*$\} = \Gamma_z(\mathcal{C})$*, and the infimum is attained by an audit that reads none of its data.*

(c) ***Decision inertness.*** *On the event of (a), the audited rule (ADAPT iff $M > \widehat{\beta}$, FREEZE iff $M < -\widehat{\beta}$, else ABSTAIN) commits only where the* a priori *rule with declared budget $\Gamma_z(\mathcal{C})$ commits. No label-free audit ever converts an abstention into a commitment relative to that declaration.*

(d) ***Yield is bounded by the error budget.*** *If $|M| \le \Gamma_z(\mathcal{C})$ then $\mathbb{P}[$the audited rule commits$] \le \delta$.*

Proof. By Lemma 1 and Definition 5, Law$(W \mid P) = z$ for every $P \in \mathcal{C}(z)$; since $\widehat{\beta} = g(W, U)$ with U independent, Law$(\widehat{\beta})$ is one and the same law ν under every $P \in \mathcal{C}(z)$. Write $\mathbb{P}_\nu$ for it.

(a) Fix $P \in \mathcal{C}(z)$. Its drift $\gamma(P)$ is a deterministic number, so δ-validity reads $\mathbb{P}_\nu(\widehat{\beta} \ge |\gamma(P)|) \ge 1 - \delta$. Taking the supremum over the fibre: for every $c < \Gamma_z(\mathcal{C})$ there is $P \in \mathcal{C}(z)$ with $|\gamma(P)| > c$, whence $\mathbb{P}_\nu(\widehat{\beta} \ge c) \ge \mathbb{P}_\nu(\widehat{\beta} \ge |\gamma(P)|) \ge 1 - \delta$. Now $\{\widehat{\beta} \ge \Gamma_z\} = \bigcap_{n \ge 1}\{\widehat{\beta} \ge \Gamma_z - 1/n\}$ is a decreasing intersection of events each of probability $\ge 1 - \delta$, so $\mathbb{P}_\nu(\widehat{\beta} \ge \Gamma_z) \ge 1 - \delta$ by continuity from above. (No attainment of the supremum is needed.)

(b) Immediate from the definition of Γ_z: $|\gamma(P)| \le \Gamma_z$ for every $P \in \mathcal{C}(z)$, deterministically. Combined with (a), no 0-valid audit has essential supremum below Γ_z, and the constant achieves it.

(c) On $\{\widehat{\beta} \ge \Gamma_z\}$, $|M| > \widehat{\beta}$ implies $|M| > \Gamma_z$.

(d) A commitment requires $|M| > \widehat{\beta}$; if $|M| \le \Gamma_z$ this forces $\widehat{\beta} < \Gamma_z$, an event of probability at most δ by (a). ☐

Corollary 2 (Vacuity of label-free audits: Aud-A, recovered and sharpened). *Let $\mathcal{C}^{\mathrm{all}}$ be the class of all target laws matching the observables. Then $\Gamma_z(\mathcal{C}^{\mathrm{all}}) = \frac{1}{2} + |M|$, so every δ-valid label-free audit has $\widehat{\beta} \ge \frac{1}{2} + |M| > |M|$ with probability $\ge 1 - \delta$; the audited frontier's decision yield is at most δ, for every deployment batch size m, every calibration sample size, and every choice of evidence map ϕ.*

Proof. Lemma 1 gives the achievable drift set $[-\frac{1}{2} - M, \frac{1}{2} - M]$, whose largest absolute value is $\max\{|\frac{1}{2} - M|, |\frac{1}{2} + M|\} = \frac{1}{2} + |M|$ because $|M| \le \frac{1}{2}$. Apply Theorem 3(a) and (d), using $\frac{1}{2} + |M| > |M|$. ☐

Corollary 3 (The least auditable budget is the declared budget). *For the declared deployment class $\mathcal{C}_\beta = \{P : |\gamma(P)| \le \beta\}$ with $0 \le \beta \le \frac{1}{2}$, the fibre radius is $\Gamma_z(\mathcal{C}_\beta) = \beta$ exactly. Consequently:*

(i) *the smallest budget any label-free audit can certify over $\mathcal{C}_\beta$ is β itself, so the best possible audit returns its own input; and*

(ii) *the fibre radius is simultaneously the abstention radius: strict commitment is uniformly directionally sound at z if and only if $|M| > \Gamma_z(\mathcal{C}_\beta)$, which is Theorem 2 (iv).*

Proof. By Lemma 1 the achievable drifts inside $\mathcal{C}_\beta$ are $G = [-\beta, \beta] \cap [-\frac{1}{2} - M, \frac{1}{2} - M] = [-\min(\beta, \frac{1}{2} + M),\ \min(\beta, \frac{1}{2} - M)]$. Since $|M| \le \frac{1}{2}$ and $\beta \le \frac{1}{2}$ we have $\beta \le \frac{1}{2} + |M|$, so $\Gamma_z = \max\{\min(\beta, \frac{1}{2} - M), \min(\beta, \frac{1}{2} + M)\} = \min(\beta, \frac{1}{2} + |M|) = \beta$. Part (i) is Theorem 3(a)–(b). For (ii), $\operatorname{sign}(M + \gamma)$ is constant and nonzero over $\gamma \in G$ iff $0 \notin [M + \inf G,\ M + \sup G]$; for $M > 0$ this is $\min(\beta, \frac{1}{2} + M) < M$, i.e. $\beta < M$ (as $M \le \frac{1}{2} < \frac{1}{2} + M$), and symmetrically for $M < 0$; for $M = 0$ it fails for every $\beta \ge 0$. So the condition is $|M| > \beta$, which by the first part is $|M| > \Gamma_z(\mathcal{C}_\beta)$; the equivalence with uniform directional soundness is Theorem 2 (iv). □

Corollary 3 is the precise sense in which the two halves of the paper are one statement. *The number that bounds the region on which the frontier must abstain, and the number that is the best a label-free audit of the budget can ever return, are the same number.* Auditing β from unlabeled deployment data is not merely hard; it is the identity map on the declaration.

The one hypothesis that can be attacked, and where we attack it. Theorem 3 requires validity *uniformly over the fibre $\mathcal{C}(z)$* — a supremum. Section 6 constructs a budget that is valid *on average over a law from which the deployment is drawn*, and that budget is computed from a history independent of the current episode's label kernel, so it is fibre-blind in the sense of Definition 8 and Theorem 3 still applies to it verbatim. The two statements are consistent only through that one substitution, and Proposition 3 isolates it: the episode budget does *not* satisfy fibre-uniform validity, and what it delivers instead is an episode-marginal guarantee. A reader who suspects that §6 contradicts this section should read §6 — the subsection titled "Exactly where the impossibility is escaped" — first.

4.4 Labels at the wrong domain buy nothing

Corollary 2 is often read as "labels fix it". They do not, unless they are labels of the deployment domain or a coupling to it is declared.

Theorem 4 (Anchor destruction)**.** *Suppose the audit additionally observes a* fully labeled *sample $S_{\mathrm{cal}} = (X_i, Y_i)_{i \le n_{\mathrm{lab}}}$ from a calibration law P_{cal}, and that the declared class $\mathcal{C}$ imposes no constraint relating the deployment drift function u_T to u_{cal} (formally: $\mathcal{C}$ is a product, $\mathcal{C} = \mathcal{C}_{\mathrm{cal}} \times \mathcal{C}_T$ with $\mathcal{C}_T$ containing, for each of its members, the whole family $\{P_T^\eta\}_{\eta \in [0,1]^D}$ of Lemma 1). Then $\Gamma_z(\mathcal{C}) = \frac{1}{2} + |M|$ and Theorem 3 applies verbatim: the labeled calibration sample changes the minimax budget by exactly zero, for every n_{lab}.*

Proof. S_{cal} is part of W (Definition 4) and its law is specified by P_{cal}, which the fibre construction of Lemma 1 leaves untouched: the construction alters only the target label kernel on D. Hence $\mathrm{Law}(W)$ is constant on the fibre, and the fibre still contains $\{P_T^\eta\}_{\eta \in [0,1]^D}$, so $\Gamma_z = \frac{1}{2} + |M|$ as in Corollary 2. □

Proposition 2 (Three-term decomposition: what each ingredient buys)**.** *Let $\mathcal{F}$ be a symmetric class of functions $D \to \mathbb{R}$ with $u_{\mathrm{cal}} \in \mathcal{F}$, let $\Delta_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} |Pf - Qf|$, and put $\rho := \mathbb{E}_{\mu_T}[\,|u_T - u_{\mathrm{cal}}| \mid D\,]$. Then, exactly,*

$$|\gamma_T| \;\le\; \underbrace{|\gamma_{\mathrm{cal}}|}_{\textit{needs labels at the anchor}} + \underbrace{\Delta_{\mathcal{F}}(\mu_{\mathrm{cal}}, \mu_T)}_{\textit{label-free computable}} + \underbrace{\rho}_{\textit{declared invariance defect}} .$$

Every audit in Appendix P is an instance of this decomposition with ρ set to 0 by declaration and the middle term estimated in a class-specific way; none of them omits the first term.

Proof. $\gamma_T = \mathbb{E}_{\mu_T}[u_T \mid D] = \mathbb{E}_{\mu_T}[u_{\text{cal}} \mid D] + \mathbb{E}_{\mu_T}[u_T - u_{\text{cal}} \mid D]$. The second summand is bounded in absolute value by ρ by Jensen. For the first, $\mathbb{E}_{\mu_T}[u_{\text{cal}} \mid D] = \gamma_{\text{cal}} + (\mathbb{E}_{\mu_T} - \mathbb{E}_{\mu_{\text{cal}}})[u_{\text{cal}} \mid D]$ and, since $u_{\text{cal}} \in \mathcal{F} = -\mathcal{F}$, the bracket is at most $\Delta_{\mathcal{F}}(\mu_{\text{cal}}, \mu_T)$ in absolute value. Triangle inequality. □

Remark 4 (Reading of Proposition 2). The three terms are bought by three different currencies and no currency substitutes for another. Only the middle term is label-free computable, and Theorem 3 applied to the first and third says why: both are fibre quantities. The third term ρ is a drift budget of exactly the same type as β — apply Lemma 1 with u replaced by $u_T - u_{\text{cal}}$, which is equally unconstrained by W, to get $\Gamma_z = \frac{1}{2} + |M|$ for ρ as well. Declaring ρ instead of β is therefore a renaming, not a reduction: *the budget-declaration problem is a fixed point of the audit construction.* This is not a defect of the paper's audits; it is the reason they must declare (Lip), (Idx), the witness class, or exchangeable calibration domains, and the reason those declarations are falsifiable but never verifiable.

4.5 The escape, and its floor

Theorem 5 (Root-n floor with deployment labels)**.** *Consider the most favourable labeled setting: the audit observes $n \geq 2$ i.i.d. correctness indicators $\mathbf{1}\{f_a(X_i) = Y_i\}$ with $X_i \sim \mu_T(\cdot \mid D)$, i.e. labels at the deployment domain itself, so that $\rho = 0$ and $\Delta_{\mathcal{F}} = 0$ in Proposition 2. Let $\mathcal{U}$ be the declared class of drift functions on D and let $\tau(\mathcal{U}) := \sup\{|c| :$ the constant function $c \in \mathcal{U}\}$. Suppose $\widehat{\beta}$ is δ-valid on $\mathcal{C}_{\mathcal{U}}$ with $\delta \leq \frac{1}{6}$ and is κ-*useful*, meaning $\mathbb{P}_{P_0}(\widehat{\beta} \leq \kappa) \geq \frac{2}{3}$ at a law P_0 with $\mu_T = \mu_{\text{cal}}$ and $\gamma = 0$. Then*

$$\kappa \;\geq\; \min\Big\{\tau(\mathcal{U}),\; \tfrac{1}{4},\; \tfrac{1}{2\sqrt{2n}}\Big\}.$$

In particular, whenever nonzero constant drifts are admissible at scale $\tau(\mathcal{U}) \geq (2\sqrt{2n})^{-1}$, no valid audit is useful at any rate faster than $n^{-1/2}$; matched from above by Theorem 13 (i), whose width is $\sqrt{2\ln(2/\delta)/n}$. (Proof in Appendix C.)

Theorem 6 (Dichotomy)**.** *Fix a declared drift class $\mathcal{U}$ and the observables at z. Exactly one of the following regimes obtains for a δ-valid audit.*

- (N) No deployment labels, no declared coupling. *By Theorems 3 and 4 the minimax budget is $\Gamma_z(\mathcal{C})$, a constant that does not depend on the deployment batch size m, on the calibration sample size n_{lab}, or on the evidence map.* ***There is no rate.*** *If moreover $|M| \leq \Gamma_z(\mathcal{C})$ — automatic for the unrestricted class, where $\Gamma_z = \frac{1}{2} + |M|$ — the decision yield is at most δ.*
- (P) Labels at an anchor, plus a declared coupling $(\mathcal{F}, \rho)$. *Proposition 2 composed with Theorem 13 (i) and its witness-class extension gives a valid $\widehat{\beta} = |\widehat{\gamma}_{\text{cal}}| + O\big(\sqrt{\ln(1/\delta)/n_{\text{lab}}}\big) + \widehat{\Delta}_{\mathcal{F}} + O\big(\sqrt{(v + \ln(1/\delta))/n}\big) + \rho$, and by Theorem 5 the $n_{\text{lab}}^{-1/2}$ term is unimprovable whenever $\tau(\mathcal{U}) \geq (2\sqrt{2n_{\text{lab}}})^{-1}$. The irreducible offset is ρ, which is declared, not estimated.*
- (D) Degenerate declaration. *If $\tau(\mathcal{U}) < (2\sqrt{2n_{\text{lab}}})^{-1}$ the class already pins the drift scale more tightly than n_{lab} labels can resolve; the "audit" is then a restatement of the class declaration and the labels are inert.*

Proof. (N) is Theorem 3(a),(d) together with Theorem 4; the sample-size independence is the statement that Γ_z is defined from $\mathcal{C}(z)$ alone. (P) is Proposition 2 composed with the cited upper bounds, and Theorem 5 for the lower bound. (D) is the second branch of the minimum in Theorem 5. The three cases are exhaustive and mutually exclusive by the trichotomy on (i) presence of deployment-domain labels and (ii) the comparison $\tau(\mathcal{U})$ versus $(2\sqrt{2n_{\text{lab}}})^{-1}$. □

Remark 5 (Where this is *not* sharp). Theorem 6 closes the rate in n_{lab} and closes the labels/no-labels axis exactly. It does *not* close the transfer axis: the $O(\sqrt{v/n})$ complexity term of the witness-class audit has no matching lower bound here, and whether a rotation-closed class strictly larger than adaptive MSW_1-ridge admits a useful root-n audit is open (long manuscript). Nor does anything above forbid an audit valid *at the realised P_T only*; all statements require uniform validity over the declared class, which is the standard — and the only checkable — requirement.

4.6 Honest scope of this section

Remark 6 (Honest scope). Lemma 1 is elementary and the fibre argument of Theorem 3 is the standard observation that an unidentified parameter cannot be estimated, specialised to a fibre on which the observable laws coincide exactly. What is not standard, and what the section is for, is (i) the identification of the exact minimax value $\Gamma_z(\mathcal{C})$ for every declared class rather than an impossibility for one class, (ii) Corollary 3, which shows the abstention radius and the least auditable budget are the same number, (iii) the yield $\leq \delta$ exchange rate of Theorem 3(d), and (iv) Theorem 4, which is strictly stronger than Corollary 2 and is the statement that actually bears on the measured record. A reader who finds (i)–(iv) obvious should nonetheless note that the paper's own appendix presents audits that read as though the budget is computed from data; Proposition 2 says precisely which term each of them computes and which two they declare.

4.7 Is the impossibility configuration observed?

Theorem 1 is proved by constructing two target laws. A referee is entitled to ask whether that configuration is a worst case that practice never visits. We measured it. On the 6,480 logged CIFAR-10-C and 405 logged ImageNet-C deployment cells — the same artifacts that carry §5 and §9.7 — we ran two analyses that share no estimator: a per-pair test for deployments that are indistinguishable in label-free evidence but have opposite-signed benefit, and a population-level estimate of the conditional law of $\operatorname{sign}\Delta$ given evidence at its own noise floor. Throughout, B (realized benefit), a_{oracle}, the oracle action and the regime label are *labels*: they are the quantity whose conditional behaviour is being estimated and are never inputs to any distance, neighbourhood, cross-fitting group or matching rule.

The verdict, stated before the numbers. The impossibility configuration is *observed*, but it is *rare*, and the rarity is the more surprising half. Matched-evidence pairs disagree in benefit sign far *less* often than chance — 39× less on CIFAR-10-C and 1.5× less on ImageNet-C, with permutation p_{below} at the floor of 0.0005 in every specification we ran. Label-free evidence is therefore highly informative about the sign of the benefit and still not sufficient to determine it, which is exactly the gap Theorem 1 describes and exactly the reason a declared β cannot be replaced by an audit. We report this as a *bound on the ambiguity*, not as a demonstration that ambiguity is common.

Existence, at single-deployment and at population resolution. A *law* is a deployment condition modulo the replicate index; its cells are i.i.d. draws, so within-law scatter *is* the sampling noise of the evidence. Distances are Mahalanobis in the pooled within-law covariance, so $d{=}1$ means "one noise standard deviation apart", and two deployments are called matched at level q iff d_{ij} is below the q-quantile of the within-law distances of *their own two laws* — a nonparametric statement that the two are no further apart than two repeats of either one of them. A cell's benefit sign is called resolvable only if $|B| \geq 0.01$, its sign agrees with its law's mean, and the law's mean is significantly nonzero across replicates; the conformal interval on $\widehat{\Delta}$ is deliberately *not* used for this, since it is an interval on the prediction, not on the realized benefit.

The strongest witness is a pair of ImageNet-C TENT deployments, `gaussian_noise|s1|small|aggressive|imba` against `impulse_noise|s3|small|aggressive|iid`, five replicates each. Their evidence is indistinguishable in the strong sense: Hotelling $d^2 = 2.34$ on 9 d.f. ($p = 0.985$), no coordinate discrepancy exceeding 1.01 standard errors, and a 95% upper confidence bound on the noncentrality of 3×10^{-61} — a positive equivalence result rather than a failure to reject. The assumption-free version is sharper still: enumerating all $\binom{10}{5} = 252$ re-splits of the two laws' pooled replicates, 99.2% give a *larger* d^2 than the observed one (median re-split $d^2 = 6.89$), so the two laws' evidence is more similar than a random re-split of their own replicates. Their benefits are -5.33 [-8.11, -2.55] and $+3.50$ [$+2.34$, $+4.66$] accuracy points, with perfect rank separation across replicates (exact Mann–Whitney $U = 0$, $p = 0.0079$, the minimum attainable at 5 versus 5). The mechanism is visible in one line: the two deployments differ by 24 points of base accuracy (0.682 versus 0.441) and their entropy- and confidence-based evidence still cannot be told apart, because a_0 is a label quantity and the audit cannot see it. This is Lemma 1's free label kernel, instantiated.

On CIFAR-10-C the cleanest witness is a same-method, same-corruption, same-severity pair (TENT, `jpeg_compression|s5`) differing only in the stream's class balance: $d = 1.66$ against a q=0.25 pair-specific floor of 2.64, with 97.8% of same-law replicate pairs farther apart, base accuracies 0.7290 and 0.7295, and law-mean benefits $+1.58$ [$+0.82$, $+2.34$] and -1.64 [-2.82, -0.45] accuracy points. KGA abstained on both cells (BLIND zone) — the correct behaviour, and precisely the decision yield the certificate pays for it. On ImageNet-C, 365 of the opposite-sign matched pairs differ *only* in corruption type and/or severity with every deployment knob held fixed, so they are pure distribution-shift witnesses rather than artifacts of varying the protocol.

Rarity, and the clean negative. Table 4 gives the rates against a permutation null that holds the matched-pair set and the entire evidence geometry fixed and permutes only the cell-to-benefit assignment, so it destroys any evidence-to-sign relation while preserving both marginals. The observed rate is *below* the null everywhere, by a factor of 39 on CIFAR-10-C and 1.5 on ImageNet-C, and a random-pair control with no matching at all returns 0.2723 and 0.2232, consistent with the nulls. We ran the excess-over-chance test the lemma's existence claim does not require, it came out the other way, and the number belongs here as the honest calibration of how rare the failure is: matching on label-free evidence *suppresses* benefit-sign disagreement strongly, which is a statement that Z is informative, not that it is sufficient.

The ambiguity is real, certified, and narrow — and its width is β. The second analysis asks not whether a pair exists but how wide the blind spot is. It first bounds what any frontier rule could extract: the best cross-fitted label-free predictor of $\operatorname{sign}\Delta$ from the evidence reaches AUC 0.983 and accuracy 0.931 against a 0.740 majority baseline on CIFAR-10-C (and 0.970 / 0.910 held out across corruption families; permuted-label null AUC 0.504). It then fixes a radius at the median within-cell replicate distance — pure evidence noise, since replicates are independent draws of the same condition — and estimates $\mathbb{P}(\operatorname{sign}\Delta = +1 \mid Z \in N)$ in each matched neighbourhood with a cluster bootstrap over distinct deployment conditions, widened to a Clopper–Pearson interval at the cluster count so that neither ambiguity nor determinacy can be certified by a degenerate interval. On CIFAR-10-C, 29.4% of the 4,949 evaluable cells sit in neighbourhoods whose interval lies entirely inside $[0.05, 0.95]$, and 20.2% entirely inside $[0.25, 0.75]$: both signs occur at $\geq 25\%$ among deployments the evidence cannot tell apart, and this survives 135 matched observations drawn from 28 distinct conditions, so it is ambiguity in the conditional law and not in the estimate. Under the

Table 4: **Matched-evidence benefit-sign disagreement, observed against a geometry-fixed permutation null.** A pair is matched at level q iff its evidence distance is below the q-quantile of the within-law replicate distances of its own two laws; q=0.95 is the formal non-rejection level and q=0.25 is strictly stronger. "Law" rows average replicates and apply a Hotelling test at $\chi^2_9(0.95) = 16.92$. Rates are over pairs whose *both* benefit signs are individually resolvable. Null is 2,000 permutations of the cell-to-benefit assignment within benchmark × method with the matched set held fixed; $p_{\text{below}} = 0.0005$ (the permutation floor) in every row. A random-pair control with no matching gives 0.2723 (CIFAR-10-C) and 0.2232 (ImageNet-C). *Every row is below its null.*

Benchmark	Level	Pairs	Opp. sign	Observed	Null [95% CI]
CIFAR-10-C	cell, q=0.25	30,555	13	0.00043	0.187 [0.176, 0.199]
CIFAR-10-C	cell, q=0.95	252,277	1,194	0.00473	0.185 [0.178, 0.193]
CIFAR-10-C	law	1,172	**0**	0.000	0.164 [0.138, 0.190]
ImageNet-C	cell, q=0.25	867	62	0.0715	0.250 [0.208, 0.288]
ImageNet-C	cell, q=0.95	9,901	1,588	0.1604	0.240 [0.225, 0.256]
ImageNet-C	law	169	**3**	0.0178	0.218 [0.156, 0.274]

same pipeline with the labels permuted and the geometry held fixed the figure is $85.4 \pm 0.9\%$. The fraction moves only between 0.29 and 0.34 across the 25th, 50th and 75th percentile radii and under a strict variant that forbids matching a deployment to the same corruption and severity run by a different adapter. Expressed over *all* 6,480 cells the figure is 22.5%, but that restatement counts every non-evaluable cell as unambiguous; the evaluable subset is itself mildly selected (mean $|\Delta|$ 0.116 evaluable against 0.197 non-evaluable), so 22.5% is a conservative convention rather than a measurement and the claim we make is the 29.4% over evaluable cells.

The width of that ambiguous band is the number the spine predicts. Table 5 places three quantities side by side that were computed without reference to one another. The local dispersion of the benefit over evidence-matched deployments, the global residual budget the same artifact needs, and the episode-conformal $\widehat{\beta}$ that Theorem 8 returns from labelled history all land between 1 and 2 accuracy points, agreeing to within a factor of 1.75. Half of the residual benefit dispersion that a declared budget must cover (0.519 of $\operatorname{Var}\gamma$) survives conditioning on the label-free evidence at its own noise floor. That is Corollary 3 as a measurement rather than an identity: the radius of the ambiguity is the budget, and the budget is the radius of the band on which the rule must abstain. It also brackets decision yield — a label-free rule must abstain where the fibre is certified ambiguous and may commit where it is certified determined, giving yield ≤ 0.706 with ≥ 0.543 certifiably committable. The deployed certificate's measured yield with the same benefit model, 0.594–0.657 across the eight split protocols, sits inside that bracket near its top — but this is indicative and not a matched comparison: the bracket is derived from evidence neighbourhoods and the yield is measured under a different splitting protocol, and it is a range of per-protocol means, not of individual splits (0.246–0.987).

Where the two analyses agree, where they do not, and which we credit. They agree on the conclusion — evidence is highly informative about the benefit sign and is not sufficient to determine it — on the rarity, and, independently, on a structural defect in the logged evidence vector recorded below. They disagree on *which benchmark* carries the evidence, and the two disagreements do not have the same status. The pair analysis finds its population-level witnesses only on ImageNet-C and none on CIFAR-10-C; the neighbourhood analysis certifies ambiguity only on CIFAR-10-C and none on ImageNet-C. We credit the CIFAR-10-C zero and discount the

Table 5: **Three unrelated routes to the same drift budget on CIFAR-10-C.** The first two are local and global dispersions of the benefit residual $\gamma = B - M(Z)$ over evidence-matched deployments with no episode structure; the third is a conformal order statistic over *episodes* with outcomes revealed retrospectively (Theorem 8), which shares neither estimator nor sample construction with the first two. They agree to a factor of 1.75. *The agreement is conditional on the benefit model.* All four rows use the gradient-boosted M; the same episode-conformal estimator run with the alternative confidence-based M on the same 135 splits returns 0.0536, a factor of 4.9 rather than 1.75. The residual $\gamma = B - M(Z)$ is defined relative to M, so a weaker M leaves a larger residual and needs a larger budget; the three routes agree for the M this paper deploys, and we do not claim more.

Quantity	Value
Local fibre radius on B, median over matched neighbourhoods	0.0130
Local fibre radius on γ, median	0.0108
Global $q_{0.9}\lvert\gamma\rvert$ over all cells	0.0155
Episode-conformal $\widehat{\beta}$ (gradient-boosted M), mean over 135 splits	0.0190
Within-fibre share of $\operatorname{Var}\gamma$	0.519
Ratio $\widehat{\beta}$ / local fibre radius	1.75

ImageNet-C one. The ImageNet-C zero is a coverage limitation: with 81 deployment conditions, only 52.6% of cells have enough distinct matched neighbours to be evaluated at all, and the evaluable subset is selected in exactly the adverse direction (97.7% of evaluable cells have $\Delta > 0$ against 51.0% of the non-evaluable ones), so the method measures the sign-pure half of that artifact and correctly reports it as sign-pure. Its β comparison inherits the same defect (ratio 16.8 against 1.75) and we do not quote it. The CIFAR-10-C zero is a measurement: all 229 of its witness law pairs become separable once ten replicates are averaged.

What that structural finding says about the spine. On CIFAR-10-C the non-identifiability is entirely a finite-sample property of *single* deployments and it dissolves under a population of them; on ImageNet-C — the harder artifact — three pairs survive averaging. That is the spine's own architecture appearing in the data: what escapes the single-deployment impossibility is a population of deployments, which is what Theorem 7 formalises and prices. It also sharpens the caveat on that escape. Averaging replicates of a *fixed* condition is not the same operation as averaging over a population of *distinct* deployments, and on the harder benchmark it does not suffice; the escape route remains the one Theorem 7(b) describes, whose binding resource is historical *labels*, not more unlabeled repetitions.

Threats we could not remove, and one disclosure. "Indistinguishable" means "not distinguishable at ten (CIFAR-10-C) or five (ImageNet-C) replicates". Only the headline pair carries a tight equivalence bound; the other two surviving ImageNet-C pairs ($d^2 = 12.8$ and 15.3, exact re-split 13.5% and 14.3%) are consistent with a real but undetected evidence gap and we do not quote them as population-level witnesses. Screening covered 209,628 CIFAR-10-C and 3,240 ImageNet-C law pairs, so three survivors is unremarkable under multiplicity alone — which is why the headline pair rests on an exact 252-fold enumeration and an exact Mann–Whitney rather than on a screening p-value. The witnesses are also adapter-specific and disjoint across benchmarks (CIFAR-10-C: EATA 884, TENT 310, SAR 0; ImageNet-C: TENT 1,586, SAR 2, EATA 0), so whatever generates a witness is not a property of the evidence vector alone. Most importantly, this measurement speaks only to *the evidence vector this system logs.* A richer label-free statistic could shrink the band; bounding what *any* label-free statistic can do is the job of Theorem 3 and Corollary 3, and no experiment can do it. Finally, the disclosure both analyses returned independently: the logged evidence vector is *nine*-dimensional, not eleven. Two of its coordinates are *exact* linear functions of the others — `entropy_drop` = `pre_entropy` − `post_entropy` and `pbal_drop` = `pre_pbal` − `post_pbal`, maximum residual exactly 0.0 — so every covariance of the logged Z is exactly singular and two of the audit's eleven coordinates carry no information beyond the other nine. That does not affect any result in this paper, whose estimators are invariant to it, but it is a defect in the evidence map as documented and we record it here.

4.8 Extensions and deferred results

Additional minimax, anytime-valid, multicandidate, and one-bit results are retained in the long manuscript (`kbound.tex`) and are not used by the short-paper claims; the *positive* side of the budget question — which structural purchases yield a computed budget, and at what rate — is stated in Appendix P and analysed as an instance of Proposition 2. The saved iWildCam streaming script computes its betting increments from frozen and adapted window accuracies, which require target labels. We therefore retain that run only as a label-informed offline stress diagnostic (on the native 35,360-image iWildCam test stream, continual TENT collapses to a cumulative macro-F1 of 0.02 versus 0.26 frozen, with predictions degenerating to a handful of classes); it is not a label-free KGA deployment result and no streaming empirical claim is made in this paper.

4.9 Claim scope

The population theorem is conditional on the declared class $\mathcal{C}_\beta$, and the finite-sample guarantee is conditional on valid interval coverage. The theorem controls the marginal probabilities $\mathbb{P}(\textsc{adapt}, \Delta \le 0)$ and $\mathbb{P}(\textsc{freeze}, \Delta \ge 0)$; it does not automatically control the conditional error among committed actions. Theorem 3 proves that β is not identifiable from unlabeled deployment data, and Theorem 7(b) proves that it is not identifiable from unlabeled *historical* data either; the framework does not claim that the empirical radius ε estimates β, or that calibration coverage transfers to arbitrary unseen domains without an assumption or correction.

5 The measured consequence: a β sweep (negative result)

Theorem 4 predicts that a budget bought with labels at a domain other than the deployment domain is worth nothing absent a declared coupling. **This section is that prediction, measured.** We derived β the way §7 of this paper recommends — as a high quantile of the realized drift on source-like development cells — and ran the population rule against realized benefit signs on 6,885 real evaluation cells. The result is negative: the frontier does not operationalize, because β cannot be declared from development data in a way that is simultaneously sound and useful. We report it in full, and narrow the corresponding claim rather than hedge it (§11).

What is empirically testable, given that the decomposition is an identity. γ is *defined* as the residual $\bar{a} - \frac{1}{2} - M$ (Remark 1), so the sufficiency half of Theorem 2 is interval arithmetic and no experiment can refute it. The empirical content is exactly two claims: **(C1)** the label-free margin M carries usable sign information about Δ; and **(C2)** a budget β declared from data the deployer actually has bounds $|\gamma|$ on the deployment cells. If (C2) fails, clause (ii) is true but vacuous—the world the deployer is in is not the class they declared. We measure both. Proposition 2 says exactly what (C2) asserts: the declaration procedure supplies the first term $|\gamma_{\text{cal}}|$ and sets the transfer term $\Delta_{\mathcal{F}}(\mu_{\text{cal}}, \mu_T)$ and the invariance defect ρ to zero, so (C2) is the claim that those two terms are zero and the measurement below is a test of that claim.

Design. 6,885 real evaluation cells, no retraining: the CIFAR-10-C stress grid (432 conditions $\times$ 3 adapters $\times$ 5 seeds = 6,480; 6 corruption families, 2 severities; $\mathbb{P}(\Delta > 0) = 0.740$, mean $|\Delta| = 0.1355$) and the ImageNet-C multi-seed grid ($27 \times 3 \times 5 = 405$; 3 families, 3 severities; $\mathbb{P}(\Delta > 0) = 0.756$, mean $|\Delta| = 0.0509$). Recomputing the oracle action from sign Δ reproduced the recorded `oracle_action` on 6,885/6,885 cells. M is built from the 11-dimensional label-free evidence vector Z only, in three variants of increasing strength—M_{DoC} (difference of confidences, ridge), M_{ATC4} (confidence gap, entropy drop, high-confidence fraction, marginal KL; ridge), M_{GBM} (all 11 coordinates, gradient boosting)—plus $M := \widehat{\Delta}$, the shipped estimator, used only as a *rule-form* control that holds the evidence channel and the calibration protocol fixed. Four splits: leave-one-corruption-family-out (LOCO, primary), leave-one-seed-out (LOSO), a source-like holdout fit on the mildest severity and mildest update strength (SRCLIKE, the deployment-realistic calibration), and SHIPPED. β is declared as $\widehat{\beta} = q_{0.90}(|\Delta - M|)$ on the source-like dev cells with M computed out of fold; no target cell enters the declaration.

Label firewall. B, a_{oracle}, `oracle_action` and `regime` are loaded into a separate structure that no feature builder and no estimator design matrix touches; labels enter only as the regression target on disjoint calibration folds (the labeled dev data a deployer is assumed to have—the same assumption $\widehat{\Delta}$ already makes) and inside the evaluation functions. An adversarial checker verifies this on all six (dataset $\times$ estimator) configurations: permuting the labels of the *scored* fold leaves M bit-identical; permuting the labels of the *training* folds changes it; and under full label permutation the out-of-fold AUC collapses to chance (0.495–0.501 on CIFAR-10-C, 0.474–0.496 on ImageNet-C, over 10 permutations each).

Scale convention (stated, not glossed). These artifacts do not record $\mu_T(D)$, so M, γ and β here are on the *benefit* scale, $\Delta = M + \gamma$ with $\gamma := \Delta - M$. Since $2\mu_T(D) > 0$ the sign identity and the frontier's form are preserved exactly; what is not preserved is that a single benefit-scale β corresponds to a *cell-varying* disagreement-scale budget $\beta/(2\mu_T(D_i))$.

(C1) The evidence channel is real, and weaker than its AUC suggests. Out-of-fold (LOCO) AUC against sign Δ is 0.754/0.875/0.974 (DoC/ATC4/GBM) on CIFAR-10-C and 0.351/0.739/0.938 on ImageNet-C; the shipped $\widehat{\Delta}$ reaches 0.982 and 0.893. Two qualifications belong next to those numbers. The base rate is 0.740, so M_{ATC4}'s raw sign agreement of 0.808 is only +6.8 points over the constant "always adapt." And difference-of-confidences—the single most standard label-free accuracy-shift statistic—is *anti-predictive* on ImageNet-C: AUC 0.351 (LOCO), 0.264 (LOSO), 0.373 (SRCLIKE), consistently below 0.5. The frontier's evidence channel is not benchmark-agnostic.

(C2) The declared budget does not bound the realized drift. Table 6 is the result of the exercise. Against $\widehat{\beta}$ we report $\beta^{\min}_{\text{sound}}$, the smallest budget for which the population rule makes *zero* committed sign errors—an oracle diagnostic, i.e. the number the deployer would have had to declare. The two benchmarks fail in opposite directions and neither leaves a usable operating point. On CIFAR-10-C the declared budget is 1.4× to 50× too small; between 24% and 73% of deployment cells fall outside the declared class $\mathcal{C}_\beta$, and committed actions are correspondingly wrong at rates of 0.4% to 16.6% where the theorem promises 0%. On ImageNet-C a large enough budget *is* derivable

Table 6: **The β-sweep frontier test: what the deployer gets at the budget the declaration procedure returns.** $\widehat{\beta} = q_{0.90}(|\Delta - M|)$ on source-like dev cells with out-of-fold M; no target cell is used. $\beta^{\min}_{\text{sound}}$ (oracle diagnostic) is the smallest budget giving zero committed sign errors. "$\mathbb{P}(|\gamma| \leq \widehat{\beta})$" is the fraction of deployment cells that satisfy the *declared* class. Theorem 2 promises a commit-error rate of exactly 0 whenever that class holds. Rows with 0 commitments are sound vacuously. *Read the last column against its null, not against* 1*:* $\widehat{\beta}$ is a 0.90 quantile, so $\mathbb{P}(|\gamma| \leq \widehat{\beta})$ is exactly 0.900 on the dev cells by `np.quantile`'s definition, and an adversarial re-run with the labels shuffled returns 0.890–0.902 on the target cells. The informative CIFAR-10-C entries are the ones far *below* 0.90; the ImageNet-C entries (0.896–0.985) sit at or above the null and carry almost no information, which is why the ImageNet-C conclusion is drawn from the yield column instead. Source: `experiments/kbound/frontier_sweep_v1/beta_sweep_results.json` (seed 20260726); null values from `adversarial_ablations_results.json`.

split	M	AUC	$\widehat{\beta}$	$\beta^{\min}_{\text{sound}}$	ratio	yield	#commit	**commit-error**	$\mathbb{P}(\lvert\gamma\rvert \leq \widehat{\beta})$
CIFAR-10-C stress grid, n = 6,480									
LOCO	DoC	0.754	0.0460	0.2375	5.2×	0.691	4478	**0.166**	0.452
LOCO	ATC4	0.875	0.0360	0.5575	**15.5**×	0.472	3057	**0.031**	0.756
LOCO	GBM	0.974	0.0213	0.0433	2.0×	0.634	4108	0.007	0.607
LOSO	DoC	0.772	0.0421	0.2363	5.6×	0.716	4637	**0.165**	0.478
LOSO	ATC4	0.860	0.0326	0.6319	**19.4**×	0.487	3154	**0.039**	0.727
LOSO	GBM	0.954	0.0092	0.4600	**50.1**×	0.742	4807	0.022	0.661
SRCLIKE	DoC	0.822	0.0482	0.1518	3.1×	0.694	3371	**0.108**	0.376
SRCLIKE	ATC4	0.901	0.0345	1.4099	**40.9**×	0.745	3620	**0.075**	0.274
SRCLIKE	GBM	0.969	0.0408	0.0563	1.4×	0.707	3438	0.032	0.386
SHIPPED	$\widehat{\Delta}$	0.982	0.0097	0.0257	2.7×	0.749	4852	0.004	0.707
ImageNet-C multi-seed grid, n = 405									
LOCO	DoC	0.351	0.2395	0.0166	0.1×	**0.000**	0	—	0.956
LOCO	ATC4	0.739	0.2036	0.1167	0.6×	**0.000**	0	—	0.956
LOCO	GBM	0.938	0.0360	0.0343	1.0×	0.314	127	0.000	0.896
LOSO	DoC	0.264	0.2373	0.0154	0.1×	**0.000**	0	—	0.956
LOSO	ATC4	0.733	0.2305	0.1918	0.8×	**0.000**	0	—	0.956
LOSO	GBM	0.946	0.0420	0.0403	1.0×	0.331	134	0.000	0.923
SRCLIKE	DoC	0.373	0.1994	0.0752	0.4×	**0.000**	0	—	0.985
SRCLIKE	ATC4	0.826	0.2156	0.2394	1.1×	0.007	2	**1.000**	0.956
SRCLIKE	GBM	0.882	0.0302	0.0847	2.8×	0.230	62	**0.532**	0.511
SHIPPED	$\widehat{\Delta}$	0.893	0.0487	0.0468	1.0×	0.252	102	0.000	0.921

and it is useless: in 5 of 10 configurations the declared $\widehat{\beta}$ produces zero commitments on all 405 cells. Of the 20 configurations, 3 **deliver the promised zero commit-error rate at nonzero yield** (all ImageNet-C, all using the full-Z GBM or the shipped estimator; *none* using an ATC-style margin, *none* on CIFAR-10-C), 5 **deliver it vacuously** at zero yield, and 12 **commit at error rates between** 0.4% **and** 100%. The most adverse single row is IMAGENETC | SRCLIKE | M_{GBM}—exactly the honest source-only calibration our own text recommends—which declares $\widehat{\beta} = 0.0302$, commits on 62 cells, and gets 53.2% of them wrong while formally "certified." Finding a β that is both sound and high-yield on CIFAR-10-C requires already knowing the answer: $\beta^{\min}_{\text{sound}}$ is computed from labels. The parameter is not declarable.

The confusion matrix for the soundness prediction. Table 7 splits every cell at $\widehat{\beta}$ into committed/abstained × sign M correct/wrong. The theorem's prediction is that the top-right entry is empty and the bottom row is a coin flip. On the primary CIFAR-10-C configuration the top-right entry contains 94 cells, and the bottom row is not a coin flip: M still ranks the sign at AUC 0.728 inside the band it declares unidentifiable, and 36.5% of in-band cells have $|\Delta| > 0.02$. Sign balance inside the band does approach 50/50 on CIFAR-10-C (0.506–0.570 across configurations), which is

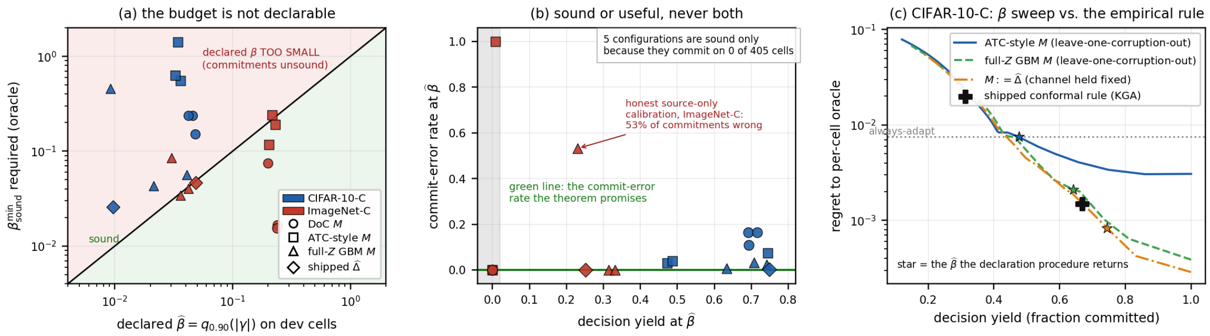

Figure 2: **The population frontier does not operationalize.** *(a)* The budget a deployer can declare from dev data, $\widehat{\beta}$, against the budget soundness actually required, $\beta^{\min}_{\text{sound}}$; points above the diagonal are configurations whose committed actions are unsound. All ten CIFAR-10-C configurations sit above it, by 1.4× to 50×. *(b)* At $\widehat{\beta}$: decision yield against realized commit-error rate. The theorem promises the green line. The five configurations on it at yield 0 achieve soundness by never committing on any of the 405 ImageNet-C cells. *(c)* CIFAR-10-C regret–yield curves for the population rule swept over β, against the shipped conformal rule. With a genuinely held-out evidence margin (blue, green) the population rule is dominated at every $\beta > 0$—the green curve reaches the conformal rule only at its $\beta \to 0$ endpoint, the degenerate never-abstain plug-in; the one curve that beats the conformal rule at a nonzero budget (orange) is the control in which M is set to the shipped $\widehat{\Delta}$, i.e. the evidence channel is held fixed and only the rule form changes. Regenerated by `figures/source/make_frontier_sweep_figs.py` from `beta_sweep_results.json`.

Table 7: **Confusion matrix for the soundness prediction**, at the declared $\widehat{\beta}$. Theorem 2 predicts "committed, sign M wrong" $= 0$ and no usable signal in the abstained row. Neither holds. "in-band AUC" is the AUC of M against sign Δ *inside* the abstention band.

		committed ($\lvert M\rvert > \widehat{\beta}$)		abstained		
config	$\widehat{\beta}$	correct	**wrong**	M right	M wrong	in-band AUC
c10c LOCO ATC4	0.0360	2963	**94**	2270	1153	0.728
c10c LOCO GBM	0.0213	4079	**29**	1866	506	0.877
c10c SHIPPED	0.0097	4834	**18**	1157	471	0.805
i-C SRCLIKE GBM	0.0302	29	**33**	188	20	0.927
i-C SHIPPED	0.0487	102	0	249	54	0.820

a point *for* the theorem's picture; the ranking result is the point against it. Because abstention resolves to FREEZE, the forfeited benefit is expensive: at $\widehat{\beta} = 0.0360$ the population rule's regret is 0.0076 with a condition-clustered 95% interval $[0.0065, 0.0087]$—an interval that *contains* the regret of simply always adapting (0.0075) and excludes the shipped conformal rule's 0.0015 $[0.0013, 0.0017]$.

Sensitivity: β is a free knob, not a specification. Sweeping $\beta \in [0, 0.30]$ on CIFAR-10-C moves decision yield from 1.000 to 0.120 and spans regret by 26× (M_{ATC4}, LOCO) and 192× ($M := \widehat{\Delta}$). Worse, at a *single fixed* $\widehat{\beta}$ the operating point is unstable across adapters and seeds: over the 15 (method, seed) cells, CIFAR-10-C LOCO/ATC4 at $\widehat{\beta} = 0.0360$ gives yield 0.424–0.637 and commit accuracy 0.851–1.000, and ImageNet-C SHIPPED at $\widehat{\beta} = 0.0487$ gives yield 0.000–0.519. A deployment parameter that cannot be estimated to within an order of magnitude, moves regret by two orders across its plausible range, and leaves yield varying by 21 to 52 points at a fixed value, is not a specification.

Head-to-head against the empirical certificate. With a genuinely held-out M (LOCO, LOSO, SRCLIKE), **no $\beta > 0$ matches the shipped conformal rule on both yield and regret in** 15 **of the** 18 **configurations swept.** The three exceptions are all the full-Z GBM channel on the 405-cell, 3-family ImageNet-C grid at $\beta = 0.02$—not a primary result, and in none of them is 0.02 the value the declaration procedure returns. Two further configurations "match" only at $\beta = 0$, the degenerate never-abstain plug-in the paper itself declines to call a guarantee. Compared *at matched decision yield*, the population rule's regret ratio against the empirical rule is 1.1× to 8.1×, with conditional false-adapt rates of 0.7% to 51% against the empirical rule's 0.0%–1.0%.

The one positive result, stated precisely. When the evidence channel and the calibration protocol are held fixed—$M := \widehat{\Delta}$, the same leave-one-cell-out estimator the empirical rule uses—β-thresholding *does* beat the conformal interval on CIFAR-10-C at the honestly-derived $\widehat{\beta} = 0.0097$: yield 0.749 vs. 0.668 at regret 0.00081 vs. 0.00150, with non-overlapping condition-clustered intervals (k=432: $[0.0007, 0.0009]$ vs. $[0.0013, 0.0017]$). We report the two facts that scope it. First, under corruption-family clustering (k=6) the intervals overlap, $[0.0004, 0.0012]$ vs. $[0.0006, 0.0026]$, and a six-cluster interval is not a primary unit. Second, the mechanism is a yield/soundness trade, not a vindication: the population rule buys its extra 8 points of yield by discarding the conformal safety margin, committing on 4,852 cells against 4,330 and making 18 sign errors (10 false ADAPT, 8 false FREEZE) against the empirical rule's 1 (0 false ADAPT). Soundness is precisely what the theorem sells and precisely what is given up. The defensible claim is therefore narrow: *with the evidence channel held fixed, thresholding the plug-in margin at a small budget raises decision yield from* 0.668 *to* 0.749 *at lower oracle regret than the conformal interval, at the cost of a* 0.3% *conditional false-adapt rate*—supported at k=432 condition clusters and not supported at k=6 family clusters.

What this does and does not touch. Nothing here bears on either proof. Sufficiency is arithmetic; necessity rests on the matched-evidence construction of Theorem 1, which is untouched and remains the mathematically substantive content. What the experiment establishes is that the theorem's *operational* reading does not survive contact with data, and we withdraw that reading: the frontier is a statement about a declared class whose antecedent we have now measured and found violated on 24–73% of CIFAR-10-C cells, and the operational certificate is the conformal interval, not $|M| > \beta$. Consequences are recorded in §11 and §O.3. Limitations of this test itself: two benchmarks with 6 and 3 corruption families (family-level intervals are reported but are too few to be primary); the benefit scale rather than the disagreement scale; $\beta^{\min}_{\text{sound}}$ is an oracle quantity; the SHIPPED split is leaky by family *by design* (it is a rule-form control, not a deployment estimate); regret resolves ABSTAIN to FREEZE, which penalizes abstention heavily—the soundness conclusions do not depend on that choice, the regret comparisons do; and M is fit against realized benefit on disjoint calibration folds, so a reader who rejects that assumption should read the AUCs above as upper bounds. Four further caveats, all found by an adversarial re-run of this experiment and recorded in `experiments/kbound/frontier_sweep_v1/adversarial_ablations_results.json`. *(i)* For the two cross-fit splits the source-like dev cells used to declare $\widehat{\beta}$ are also inside the scored target set (1,620/6,480 and 135/405); excluding them moves CIFAR-10-C LOCO coverage from 0.452/0.756/0.607 down to 0.303/0.708/0.509 (DoC/ATC4/GBM) and leaves commit-error unchanged to within 0.5 points, so the published table understates the failure. The SRCLIKE rows are already disjoint. *(ii)* The ImageNet-C zero-yield rows have the same signature the null produces: with labels shuffled, M collapses toward a constant, $\widehat{\beta} \to q_{0.90}(|\Delta|)$, and yield goes to 0 in 8 of 12 configurations. Those rows are therefore evidence that a *weak* M makes $\widehat{\beta}$ useless, not an independent second failure mode of the frontier. *(iii)* The chance level for LOCO AUC is not 0.5 on CIFAR-10-C: replacing Z with noise gives 0.426, because a per-fold constant predictor is anti-correlated with the held-out fold's mean. On ImageNet-C the same control gives 0.486–0.540, so the DoC anti-predictivity reported above is real and not a fold artifact. *(iv)* The GBM rows move with the estimator's random seed: over five seeds CIFAR-10-C LOCO/GBM gives $\widehat{\beta} \in [0.0172, 0.0217]$ and coverage $\in [0.579, 0.629]$, and the ImageNet-C LOCO/GBM commit-error is 0 on four of five seeds and 0.008 on the fifth, so the "3 configurations at zero commit-error" count is 2–3, not exactly 3.

It is not a quantile-estimation failure. The obvious rebuttal is that $\widehat{\beta} = q_{0.90}(|\gamma|)$ is a plug-in quantile and that a conformal order statistic would have fixed it. It would not. Replacing the plug-in by the conformal budget of Theorem 8 across all 470 splits of §6.8 moves the budget by a *median relative gap of* 0.34% and the coverage by a median of *exactly* 0.000 (`beta_estimability/compare_to_sweep_results.json`; the maximum relative gap is 52%, on the smallest splits). At K in the thousands the conformal correction moves one order statistic and nothing else. What the sweep measured was the failure of exchangeability between development and deployment cells, not the failure of a quantile estimator, and no conformal machinery repairs it.

Remark 7 (What the theorems predict about this experiment, and what they do not). The β-sweep is *not* a label-free audit: its budget $\widehat{\beta} = q_{0.90}(|\gamma|)$ is computed on labeled source-like development cells. Corollary 2 therefore does not apply to it, and we do not claim it as a confirmation of Corollary 2. What applies is Theorem 4 with Proposition 2: the procedure supplies the first term and *sets the other two to zero by an undeclared exchangeability assumption between development and deployment cells.* The theory then predicts under-coverage of unbounded size, and that is what is measured.

- *Coverage.* $\widehat{\beta}$ is a 0.90 quantile on development cells, so $\mathbb{P}(|\gamma| \leq \widehat{\beta})$ has a by-construction null of 0.900 on the deployment cells whenever the two pools are exchangeable (0.900 for CIFAR-10-C and 0.8963 for ImageNet-C, recorded as `coverage_on_dev_cells_BY_CONSTRUCTION`). CIFAR-10-C measures 0.274–0.756, far below that null: the exchangeability declaration is falsified, so $\rho + \Delta_{\mathcal{F}} > 0$ was set to 0, and the committed error rate is correspondingly 0.4%–16.6% where Theorem 2 promises 0%. ImageNet-C measures 0.904–0.985 in five of the six configurations recorded in `adversarial_ablations_results.json`, at or above the null, which *carries no information*; and its three `loco` rows additionally have `dev_and_target_disjoint=false`. The single ImageNet-C configuration whose coverage is a genuine measurement is `srclike|M_gbm`, at 0.470 in that re-run and 0.511 in Table 6 (which uses the primary $\widehat{\beta} = 0.0302$); it is also the configuration with commit-error 0.532 on 62 committed cells (sign match 0.468, 95% CI $[0.340, 0.599]$).

- *Sup versus quantile.* Theorem 3 bounds any valid budget by a *supremum* over the fibre, not a development quantile. Measured on CIFAR-10-C: the maximal realised $|\gamma|$ on deployment cells exceeds the maximal $|\gamma|$ on development cells by factors of 2.4–28.4 across the ten configurations (`gamma_absmax_target` versus `beta_hat_max` in `beta_sweep_results.json`).

- *Anchor destruction, quantitatively.* Theorem 4 says an uninformative anchor leaves only a class-scale constant. Under the global label permutation of `adversarial_ablations.py`, any estimator fit on the permuted benefit column converges to the constant $\mathbb{E}[\Delta]$, so the declaration procedure must return approximately $q_{0.90}(|\Delta - \mathbb{E}\Delta|)$ — a number computable from the raw artifacts with no model fitting. On CIFAR-10-C that prediction is 0.2376; the six measured shuffled budgets are 0.2468–0.2606, ratios 1.04–1.10, against real budgets of 0.0178–0.0482 (inflation 5.2×–14.1×), with zero commitments in 5/5 replicates for the `M_doc` and `M_atc4` channels. On ImageNet-C the prediction is 0.0689 against measured 0.0742–0.1171 (ratios 1.08–1.70; $n = 405$). Script and output: `frontier_sweep_v1/beta_impossible/{check_anchor_collapse.py,anchor_collapse_check.json}`.

- *A prediction that fails, stated as such.* On ImageNet-C the anchored budget for the `M_doc` and `M_atc4` channels is *larger* than the anchor-free constant (0.199–0.239 real versus 0.074–0.090 shuffled), which is the reverse of the pattern the mechanism above describes. Nothing in §4 forbids it: Theorem 3 lower-bounds *valid* audits and says nothing about how far an invalid heuristic may overshoot. The mechanism is visible in the same artifact — difference-of-confidences is anti-predictive on ImageNet-C (AUC 0.351), so its residuals exceed the marginal spread of the benefit. The operational reading is that on that benchmark the anchored declaration is beaten by the trivial constant, and those are exactly the rows with decision yield 0.000. We report this as a failed prediction rather than omit it.

6 Where a budget can come from, what it costs, and why it does not help here

Corollary 2 says a label-free audit of the drift budget is vacuous, and Theorem 4 says labels at the wrong domain do not repair it. This section asks the complementary question — under what structure is β *estimable*, and at what price — and answers it in one direction: β is a property of a *population of deployments*, not of a deployment. Under exchangeability across deployments plus retrospectively revealed outcomes it is identified and estimable at the conformal rate; without both ingredients it is not identified at all. We then measure, on the K-Bound artifacts, how far real deployment populations are from that structure. The answer is: far enough that the estimator is valid along nuisance axes (fitting seed, adaptation method) and fails along the axis a deployment actually cares about (a corruption family or severity never seen before).

6.1 The episode model

Definition 6 (Deployment episode)**.** An *episode* e is one deployment: a target law P_e, a frozen model f_0^e, a candidate f_a^e, a source-calibrated score s^e, and the induced disagreement region D_e with $\mu_{P_e}(D_e) > 0$. Write the *observable part* $W_e = (Z_e, M_e)$ — the label-free evidence and margin of Assumption 1 — and the *latent part* $\gamma_e = \bar{a}_e - \frac{1}{2} - M_e$. Put $V_e = (W_e, \gamma_e)$.

Assumption 2 (E1: retrospective outcome logging)**.** For each past episode $e \le K$, a sample of n units drawn i.i.d. from $\mu_{P_e}(\cdot \mid D_e)$ has its labels revealed *after* that deployment concluded, yielding

$$\widehat{\gamma}_e = \frac{1}{n} \sum_{i=1}^{n} \big(\mathbf{1}\{f_a^e(X_i) = Y_i\} - s^e(X_i)\big) \in [-1, 1].$$

No labels are available for the current episode $K + 1$.

Assumption 3 (E2: episode exchangeability)**.** $(V_1, \ldots, V_{K+1})$ is exchangeable. Write Q for the common marginal law of V.

E1 is an accounting assumption: it says outcomes are eventually observed and logged, which is true of any deployment with a downstream ground truth (delivered claims, resolved tickets, confirmed diagnoses) and false of deployments whose outcomes are never revealed. E2 is a much stronger statement about the deployment pipeline; §6.7 argues it is the binding constraint and §6.8 measures its failure.

Definition 7 (Episode budget)**.** For $\alpha \in (0,1)$, $\beta^\star(\alpha) := \inf\{b \ge 0 :\ Q(|\gamma| \le b) \ge 1-\alpha\}$, the $(1-\alpha)$-quantile of $|\gamma|$ under the episode law.

6.2 Identification, and what it costs

The binding resource is historical *labels*, not historical data. Part (b) below is the statement that matters operationally: no quantity of logged unlabeled deployments moves the budget at all, so an organisation that keeps telemetry but never resolves outcomes has bought nothing.

Theorem 7 (Episode identification and its price)**.** *Under Assumption 3:*

(a) ***(Positive.)*** *$\beta^\star(\alpha)$ is a functional of the episode law of (W,γ), and it is the smallest budget b for which the declared class holds with episode probability at least $1-\alpha$, i.e. $Q(P \in \mathcal{C}_b) \ge 1-\alpha$. Consequently the frontier rule of Theorem 2 run at $b = \beta^\star(\alpha)$ has episode-marginal false-strict-commitment probability at most α.*

(b) ***(Negative: the price is historical labels, not historical data.)*** *$\beta^\star(\alpha)$ is* not *a functional of the episode law of W alone. For every $b \in [0,\frac{1}{2}]$ there is an episode law Q_b with the same W-marginal and $\beta^\star_{Q_b}(\alpha) = b$. No quantity of unlabeled historical deployments identifies β.*

Proof in Appendix D.

6.3 The estimator

Theorem 8 (Episode-conformal budget)**.** *Assume E1 and E2, with a common labeled size n and the same estimator in every episode. Let $S_e = |\widehat{\gamma}_e|$ for $e \le K$, let $\delta \in (0,\alpha)$, $t(n,\delta) = \sqrt{2\ln(2/\delta)/n}$, and*

$$j = \lceil (1-\alpha+\delta)(K+1) \rceil, \qquad \widehat{\beta}_{\text{epi}} = S_{(j)} + t(n,\delta),$$

feasible exactly when $\alpha \ge \frac{1}{K+1} + \delta$. Then $\mathbb{P}(|\gamma_{K+1}| \le \widehat{\beta}_{\text{epi}}) \ge 1-\alpha$, with the probability taken over the joint draw of all $K+1$ episodes and their label samples. Composed with Theorem 14, the audited frontier makes a false strict commitment with probability at most α (population M) or $\alpha+\delta'$ (empirical $\widehat{M}$ from m unlabeled deployment points). Proof in Appendix D.

Remark 8 (Relation to Theorem 15). Theorem 8 is Theorem 15 with the label-free drift predictor set to $\gamma_{\text{pred}} \equiv 0$ and "domain" read as "episode". **We claim no new machinery here.** We restate it because the specialisation is the object the rest of this section analyses and validates: it needs no auxiliary predictor, so its coverage failures isolate the exchangeability assumption rather than confounding it with predictor misspecification. The conformal backbone is the standard split-conformal construction, and the hierarchical domain-level form is due to the conformal-under-hierarchy literature; Theorem 15's assumption that any predictor be fixed independently of the $K+1$ episodes is vacuous here.

6.4 Exactly where the impossibility is escaped

This is the load-bearing check. If Theorem 8 did not violate a hypothesis of Corollary 2, one of the two would be false.

Definition 8 (Fibre-blind budget)**.** Fix observables (μ_T, f_0, f_a, s) and let $\mathcal{C}^{\text{all}}(z)$ be the matched-evidence fibre of Definition 5 for the unrestricted class. A random budget B is *fibre-blind* if $\text{Law}(B)$ is the same for every $P \in \mathcal{C}^{\text{all}}(z)$. Both a deployment-evidence audit $B = g(Z)$ and any budget computed from a history drawn independently of the current episode are fibre-blind.

Proposition 3 (Two ingredients, neither sufficient alone). (a) ***Labels alone do not escape Aud-A.*** *If B is fibre-blind and $\mathbb{P}(|\gamma(P)| \le B) \ge 1-\delta$ for* every *$P \in \mathcal{C}^{\mathrm{all}}(z)$, then $B \ge \frac{1}{2} + |M|$ with probability at least $1-\delta$, and the audited frontier abstains with probability at least $1-\delta$. In particular $\widehat{\beta}_{\mathrm{epi}}$, which is fibre-blind, does* not *satisfy fibre-uniform validity.*

(b) ***The escape is the average.*** *Theorem 8's guarantee is Q-marginal: the deployment law is itself drawn from Q, and the extreme worlds $\bar{a} \in \{0,1\}$ that force Aud-A's bound carry Q-probability at most α. The hypothesis Corollary 2 uses and E2 removes is precisely the* supremum over the fibre*, replaced by an* average over *Q.*

(c) ***The average alone does not escape either.*** *By Theorem 7(b), without historical labels the Q-average budget is as uninformative as the fibre supremum: the W-marginal is consistent with $\beta^\star(\alpha) = b$ for every $b \in [0, \frac{1}{2}]$.*

Proof. (a) Take the extreme world $P^\dagger \in \mathcal{C}^{\mathrm{all}}(z)$ with $\bar{a} \in \{0,1\}$ chosen so that $|\gamma| = \frac{1}{2} + |M|$, which exists by Lemma 1 at $\eta \equiv 1$ or $\eta \equiv 0$. Validity at $P^\dagger$ forces $\mathbb{P}(B \ge \frac{1}{2} + |M|) \ge 1-\delta$; fibre-blindness transfers this statement to every other world in the fibre. Since $|M| < \frac{1}{2} + |M|$, neither branch of $|M| > B$ can fire on that event. (b) and (c) are restatements of Theorem 8 and Theorem 7(b). □

Remark 9 (What the deployer actually gets). Proposition 3(a) is not a technicality. It says the episode budget provides no protection *at* a deployment: conditional on the episode being one of the adversarial ones, the frontier commits and is wrong. What is controlled is the long-run rate across deployments. This is the same marginal-versus-conditional distinction as Remark 13, and it should be stated to any deployer who is told β was "estimated".

6.5 How many episodes, how many labels

Theorem 9 (Episode floor). *Fix $\alpha \in (0, \frac{1}{2})$ and $b \in (0, \frac{1}{2}]$. Consider budgets $\widehat{\beta} = h(\gamma_1, \ldots, \gamma_K)$ computed from K historical drifts observed* without label noise *(the most favourable case), and require validity $\mathbb{P}_Q(|\gamma_{K+1}| \le \widehat{\beta}) \ge 1-\alpha$ for every i.i.d. episode law Q. Then either*

$$h(0, \ldots, 0) \;\ge\; b \qquad \text{or} \qquad \alpha \;\ge\; \frac{1}{K+1}\Big(1 - \frac{1}{K+1}\Big)^K \;\ge\; \frac{1}{e(K+1)}.$$

Equivalently: a budget that returns anything strictly below b on a drift-free history requires $K \ge \frac{1}{e\alpha} - 1$ episodes. Proof in Appendix D.

Corollary 4 (Bracket). *Let $K^\star(\alpha)$ be the least number of episodes admitting a non-vacuous level-α episode budget. Then, ignoring the label-noise allowance δ,*

$$\frac{1}{e\alpha} - 1 \;\le\; K^\star(\alpha) \;\le\; \frac{1}{\alpha} - 1.$$

The upper bound is Theorem 8's feasibility condition; the lower bound is Theorem 9. At $\alpha = 0.10$: between 3 and 9 historical deployments.

Proposition 4 (Label price per episode). *Write $C = \mathbf{1}\{f_a(X) = Y\}$ and $u = C - s(X) \in [-1, 1]$ on D, $\sigma^2 = \mathrm{Var}(u)$. Bernstein's inequality gives, with probability $1-\delta$,*

$$|\widehat{\gamma} - \gamma| \;\le\; \sqrt{\tfrac{2\sigma^2 \ln(2/\delta)}{n}} + \tfrac{4\ln(2/\delta)}{3n}.$$

Hence to keep the label-noise inflation in Theorem 8 below $\kappa\beta$ it suffices to take $n \geq \frac{8\sigma^2 \ln(2/\delta)}{(\kappa\beta)^2} + \frac{8\ln(2/\delta)}{3\kappa\beta}$. Moreover, if s is pointwise calibrated on D, i.e. $\mathbb{E}[C \mid X] = s(X)$ μ_T-a.s. on D, then

$$\sigma^2 = \mathbb{E}[s(1-s)] = \mathrm{Var}(C) - \mathrm{Var}(s) \; < \; \mathrm{Var}(C),$$

so estimating the drift γ is strictly cheaper in labels than estimating $\bar{a} - \frac{1}{2}$ directly, by exactly the variance of the calibrated score. The score acts as a control variate. Proof in Appendix D; the identity is checked numerically in Appendix E.

Proposition 5 (Probe lower bound and break-even)**.** (a) *Fix $\alpha, \eta \geq 0$ with $2\alpha + \eta < 1$ and $\varepsilon \in (0, \frac{1}{4}]$. Any rule that observes k i.i.d. labeled probes from $\mu_T(\cdot \mid D)$, commits strictly with probability at least $1-\eta$, and has false-adapt and false-freeze probability at most α whenever $|\bar{a} - \frac{1}{2}| = \varepsilon$, must use $k \; \geq \; 3(1-2\alpha-\eta)^2/(16\,\varepsilon^2)$. Theorem 13 (i) achieves $O(\varepsilon^{-2}\log(1/\delta))$, so the probe price is $\Theta(\varepsilon^{-2})$ up to the confidence log.*

(b) *Over N future deployments the probe route costs Nk labels; the episode route costs Kn labels once and none thereafter. The routes break even at $N^\star = Kn/k$. With $\alpha = 0.10$, $K = 9$ (Corollary 4), n from Proposition 4 at $\kappa = 1$ and k from part (a),*

$$N^\star \; = \; \frac{384\, K\, \sigma^2 \ln(2/\delta)}{3(1-2\alpha-\eta)^2} \cdot \Big(\frac{\varepsilon}{\beta}\Big)^2,$$

which is $\Theta\big(K\sigma^2 \log(1/\delta)\, (\varepsilon/\beta)^2\big)$: the budget route pays for itself after a number of deployments proportional to the squared ratio of the typical margin to the budget. It never pays for itself if $\beta \ll \varepsilon$.

Proof in Appendix D.

Proposition 5 is the honest ranking, and it is uncomfortable. At a *single* deployment, buying k probe labels strictly dominates estimating a budget: the probe certifies $\mathrm{sign}(M+\gamma)$ directly, whereas the budget route certifies the weaker $|M| > |\gamma|$. The budget is worth estimating only because it is *reused*, and reuse is exactly what E2 licenses. If E2 fails, the budget route loses both its guarantee and its economic rationale.

6.6 Relaxing exchangeability

Theorem 10 (Graceful degradation)**.** *Let the history be drawn from Q and the current episode from Q'.*

(a) ***Bounded episode-law ratio.*** *If $dQ'/dQ \leq \rho$ then the budget of Theorem 8 run at level α satisfies $\mathbb{P}_{Q'}(|\gamma_{K+1}| \leq \widehat{\beta}_{\mathrm{epi}}) \geq 1 - \rho\alpha$. To retain level α under Q', run the history at level α/ρ, which requires $K \geq \rho/\alpha - 1$ episodes: a ρ-fold mismatch costs a ρ-fold increase in the number of historical deployments.*

(b) ***Episode covariate shift.*** *If $Q'(dw, d\gamma) = Q(d\gamma \mid w)\, Q'(dw)$ — the episode observables shift but the conditional drift law does not — then weighted split conformal with weights $w(W) = dQ'_W/dQ_W$ is exactly valid at level α, and w is estimable from* unlabeled *episode observables alone by a two-sample density-ratio fit.*

Proof in Appendix D; (b) is weighted conformal [34], cited rather than reproved.

Corollary 5 (The relaxation is not free)**.** *The conditional-stability hypothesis of Theorem 10(b) is not auditable from label-free deployment data: Corollary 2 applies verbatim at each fixed w, so no label-free statistic distinguishes $Q(\cdot \mid w)$ from any other conditional drift law with the same observables. It is, however,* falsifiable *given retrospective labels on two episode populations, which a declared β at a single deployment is not.* ***That is the whole gain: the assumption moves from unfalsifiable to falsifiable, not from unverified to verified.***

6.7 Is E2 plausible? An honest answer

E2 asserts that the deployment a system is about to face is exchangeable with the deployments it has already logged. Three observations, in decreasing order of comfort.

Where it is plausible. When the episode population is generated by a stable operational process — a fixed fleet of sites, a recurring seasonal cycle, a queue of customers drawn from a stable market — and the model, the candidate update rule, and the scoring function are held fixed, E2 is the same assumption already made whenever a deployment team computes a historical error rate and uses it as a forecast. It is checkable in the weak sense that its consequences (coverage of past budgets on later episodes) can be back-tested.

Where it fails, and it is the interesting case. E2 fails exactly when the current deployment differs from the history in kind rather than in draw: a new corruption mechanism, a new site, a new severity regime, a new sensor. This is the case that motivates test-time adaptation in the first place. A deployer who could guarantee E2 could often also have retrained.

What the theory can and cannot say about that failure. Theorem 10(a) prices the failure if ρ is known, and ρ is not knowable label-free (Corollary 5). Theorem 10(b) removes the need for ρ at the cost of a conditional-stability assumption that is itself unverifiable at deployment. So the honest summary is: **episode structure converts an unfalsifiable per-deployment declaration into a falsifiable population-level one, and buys a marginal (not conditional) guarantee. It does not make β knowable at the deployment that needs it.** The next subsection quantifies the gap.

6.8 Measured: E2 holds on nuisance axes and fails on deployment axes

Protocol. Episodes are the 6480 CIFAR-10-C and 405 ImageNet-C cells (condition $\times$ method $\times$ seed) of §9; $\gamma := \Delta - M$ in benefit units as in the β-sweep. Each split partitions the cells into three disjoint pools: a fit pool F on which the benefit model $M(\cdot)$ is fit, a history H whose labels are revealed, and a deployment set D whose labels are used only for evaluation. M is fit once on F and applied identically to H and D, so the scores on H and D are exchangeable exactly when the episode populations are. Five random F/H splits per protocol; $\alpha = 0.10$, so **the null coverage is** 0.90: any protocol whose deployment set is exchangeable with its history must return ≈ 0.90, and the finding is the deviation. This is the same class of trap as the documented `np.quantile` null in Table 6's caption. Artifacts: `frontier_sweep_v1/beta_estimability/{episode_beta.py,episode_beta_results.json}`.

Five things follow. *(i) The estimator is correct.* On the exchangeable control the realised coverage is 0.905 and 0.912 against a null of 0.900 — which says only that the machinery is not broken — and a K-sweep reproduces the feasibility floor exactly: at $\alpha = 0.10$ the conformal budget is infeasible for $K \in \{3, 5\}$ and first attains 0.907 ± 0.088 at $K = 9 = \lceil 1/\alpha \rceil - 1$, matching Corollary 4. *(ii) Nuisance axes are safe, deployment axes are not.* Coverage is within 0.04 of nominal when the seed, class-balance protocol or batch size changes, and falls to 0.626 and 0.454 when the corruption family or severity changes. The implied ρ rises from 1 to 5.5, and by Theorem 10(a) restoring $\alpha = 0.10$ at $\rho = 5.5$ costs $K \geq 54$ episodes. *(iii) On the hard axes the budget frequently does not exist at all.*

Table 8: **Episode-budget coverage by the axis that changes at deployment** ($\alpha = 0.10$; 32,400 deployment cells per axis, 24,300 for the control). **The null is** 0.900 **by construction** — rows at or near 0.900 are evidence the estimator is not broken, not evidence that it works. Only deviations are findings. $\rho_{\text{impl}} = (1 - \text{coverage})/\alpha$ is the episode-law ratio that Theorem 10(a) would need in order to explain the realised miss rate; it equals 1 under E2. Full tables, ImageNet-C, and the false-adapt columns are in Appendix D and `beta_estimability/tables.md`.

Episode axis perturbed	changes at deployment	coverage	95% CI	ρ_{impl}	yield
CIFAR-10-C, M_{gbm} (11 evidence coordinates, GBM)					
`RAND` (control)	nothing	0.905	[0.901, 0.908]	0.95	0.647
`LOSO`	fitting seed	0.875	[0.872, 0.879]	1.25	0.647
`LOPROT`	class-balance protocol	0.877	[0.874, 0.881]	1.23	0.657
`LOBATCH`	batch size	0.862	[0.858, 0.866]	1.38	0.655
`LOMO`	adaptation method	0.804	[0.800, 0.808]	1.96	0.630
`LOSTR`	adaptation strength	0.721	[0.717, 0.726]	2.79	0.594
`LOCO`	corruption family	0.626	[0.621, 0.631]	3.74	0.645
`LOSEV`	corruption severity	0.454	[0.449, 0.460]	5.46	0.615
CIFAR-10-C, M_{atc4} (4 ATC coordinates, ridge)					
`RAND` (control)	nothing	0.912	[0.908, 0.915]	0.88	0.446
`LOCO`	corruption family	0.866	[0.863, 0.870]	1.34	0.438
`LOSEV`	corruption severity	0.595	[0.589, 0.600]	4.05	0.424

Per split we compute the level a deployer would have had to declare on the history to reach 0.90 coverage on the deployment episodes, and the implied $K = 1/\alpha' - 1$. On the safe axes every split is feasible at K between 7 and 13. On `LOSEV`, 5 of 10 splits are **infeasible outright** — the budget required exceeds every score in the history, so no declared level works at any K — and on `LOCO`, 7 of 30 splits are infeasible with a median required K of 36 on the rest, against a nominal 9. *(iv) A better benefit model transfers worse.* The eleven-coordinate GBM has higher decision yield than the four-coordinate ATC ridge (0.647 vs 0.446) and strictly worse budget transfer on every deployment axis (0.626 vs 0.866 on `LOCO`, 0.454 vs 0.595 on `LOSEV`). Fitting the benefit surface harder shrinks the history residuals without shrinking the residuals on a novel episode, so the quantile it produces is too small. This warning applies to the shipped benefit model as well. *(v) Weighted conformal restores validity by destroying yield.* With label-free evidence weights (Theorem 10(b)) the `LOSEV` coverage goes from 0.454 to 1.000 and the decision yield goes from 0.615 to 0.000: the weights concentrate on the few history episodes that resemble the deployment, the weighted quantile runs off the end of the score list, and the frontier abstains everywhere. This is the same trade Corollary 2 forces on any label-free audit, now appearing inside the method that was supposed to escape it.
One negative result on the other side, which is a point *for* the theory: `LOCO` does not fail on ImageNet-C (0.888 / 0.903). The three held-out ImageNet-C "families" are Gaussian, shot and impulse noise, which are of one kind; exchangeability is a claim about the episode design, and holding out a same-kind family does not break it.

6.9 A label-free screen for E2, and what it cannot certify

Corollary 5 says no label-free statistic certifies the conditional part of E2. The *marginal* part is a different matter: Δ_{sep} = AUC of a cross-fitted classifier separating history from deployment episodes on the evidence Z alone is computable without any labels, and equals $\frac{1}{2}$ under E2. Measured across the 470 splits it correlates with realised coverage at Pearson $r = -0.346$ ($p = 1.2 \times 10^{-14}$), Spearman $\rho = -0.351$, and it reads 0.497 ± 0.033 on the exchangeable control, as it must. Used as a one-sided screen at $\Delta_{\text{sep}} < 0.70$ ($n = 142$): mean coverage 0.905 with 7.7% of splits below 0.85; above the threshold ($n = 328$), mean coverage 0.782 with 43.3% below 0.85. So a low separability reading is informative and a high one is only a warning — which is exactly the asymmetry Corollary 5 predicts, since the residual 65% of the coverage variation lives in the conditional component Corollary 2 makes invisible. **We recommend reporting Δ_{sep} alongside any episode budget; we do not recommend treating it as a certificate.**

6.10 What is left open

Conjecture 1 (Estimable budgets without exchangeability)**.** *Is there a structural assumption, weaker than E2 and* falsifiable from retrospective labels on the history alone*, under which $\beta^\star$ for a novel episode is identified? The two natural candidates both fail to be that: bounded episode-law ratio (Theorem 10(a)) needs an unknowable ρ; conditional stability (Theorem 10(b)) is falsifiable only with labels from the novel population, which is what is missing. A positive answer would need a drift functional that extrapolates off the support of the history, and Theorem 7(b) shows no such functional can be built from unlabeled observables alone.*

This is stated as a conjecture, not a theorem: we neither exhibit such an assumption nor prove that none exists. We did not pursue an anytime-valid within-episode budget from the e-process machinery, and the reason is structural rather than technical: within one episode γ is a fixed unknown number and every observable in the adaptation stream is label-free, so Corollary 2 applies to any stopping-time-measurable statistic of that stream. Sequential machinery buys anytime validity, not identification.

7 What remains: the finite-sample certificate and KGA

Sections 4–6 withdraw the population frontier's operational reading and price its replacement. What survives is the object of this section: a finite-sample certificate that trades decision yield for a false-adapt guarantee. Its radius ε is a conformal quantity calibrated on held-out residuals and it is *not* an estimate of β; nothing in §4 applies to it, because it certifies an interval around $\widehat{\Delta}$ rather than a bound on the latent drift.

7.1 Three coverage statements, kept separate

Much of the difficulty in reading an adaptation certificate comes from one word doing three jobs. We fix the terminology here and hold to it for the rest of the paper. In what follows $[L, U] = [\widehat{\Delta}-\varepsilon,\ \widehat{\Delta}+\varepsilon]$ is the benefit interval of the certificate rule, so that the coverage premise $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \le \varepsilon\,] \ge 1 - \alpha$ of Theorem 11 and the interval statement below are the same hypothesis written two ways.

Definition 9 (Theoretical coverage)**.** The interval has *theoretical coverage* at level $1 - \alpha$ if

$$\mathbb{P}\{\Delta \in [L, U]\} \ \ge\ 1 - \alpha, \tag{2}$$

the probability taken over the common space of calibration, benefit-model fitting and deployment-unit randomness. This is a property of a distribution, not of a run. We assert it only where (A1)–(A6) below are argued to hold, and never on the basis of an observed hit rate.

Definition 10 (Observed empirical coverage)**.** For a completed, labelled track with n independent inference units (Definition 12), the *observed empirical coverage* is

$$\widehat{\mathrm{Cov}} \ =\ \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\Delta_i \in [L_i, U_i]\}, \tag{3}$$

reported with n, the resampling unit, and an interval computed at that unit. It estimates a hit rate on the conditions evaluated, and is not evidence for (2) on any other condition.

Remark 10 (When *not* to attach an interval). Definition 10 applies to a hit rate that is genuinely random. It does *not* apply to a realized calibration-set coverage figure produced by in-sample rank calibration, which is a deterministic function of n and carries no sampling variability at all — attaching a binomial or bootstrap interval to such a figure gives it a frequentist meaning it does not have. §11 identifies the figures in this paper that are arithmetic in this sense, and they are reported without intervals. The discipline this section imposes is that every coverage number states which of the three kinds it is; for the arithmetic ones, the honest annotation is "deterministic in n", not a wider interval.

Definition 11 (Deployment diagnostic)**.** A *deployment diagnostic* is a label-free statistic whose large values are evidence *against* (A1)–(A6). Diagnostics are one-sided by construction: they can falsify and cannot verify. Passing every diagnostic of §7.9 does not establish (2).

Definition 12 (Inference unit)**.** The *inference unit* is the level at which calibration draws are treated as exchangeable and at which uncertainty is resampled: a domain, an episode, a corruption×severity cell, a backbone, or a seed, declared per track. It is the individual example only when examples are independently drawn, which on the tracks of §9 they are not.

7.2 The assumption contract

The paper already records which condition buys what, and what to do when one fails. What that table does not do is give the conditions names that the rest of the paper, the released artifacts and a deployment can all refer to. We do that now, because a condition nobody can cite is a condition nobody checks.

(A1) **Calibration exchangeability.** Calibration units and the deployment unit are exchangeable at the declared inference unit, or an explicit shift correction supplies the same conclusion. (This is the episode-exchangeability condition E2, read at the unit level.)

(A2) **Risk alignment.** The evidence Z is risk-aligned (Definition 1): it retains enough information about Δ for calibration residuals to transfer to the deployment condition.

(A3) **Coverage premise.** The interval satisfies (2) on the target class.

(A4) **Protocol independence.** Target-test labels influence none of: feature selection, candidate selection, threshold selection, calibration, or stopping. (The retrospective outcome-logging condition E1, and the data-access discipline of the protocol tables.)

(A5) **Correct sampling unit.** Calibration and uncertainty calculations operate on independent domains, episodes, corruptions, backbones or seeds, not on correlated examples pooled as though independent.

(A6) **Fixed decision rule.** Estimator, interval construction, α and decision rule are fixed before the promoted target conditions are evaluated.

(A1)–(A4) and (A6) rename conditions the paper already carried: the conditions previously listed as risk-aligned Z, coverage of $\widehat{\Delta} \pm \varepsilon$, exchangeable or corrected residuals, and frozen protocol become (A2), (A3), (A1) and (A4)/(A6) respectively, and the long manuscript's assumptions (i)–(iii) map to (A1), the bounded-benefit regularity condition, and (A2). Bounded loss and benefit remains a regularity condition needed for the concentration arguments; it is not part of the deployment gate because it is a property of the loss we choose, not of the deployment we meet. **(A5) is new in this revision**, and it is the one the empirical sections most often failed to make explicit: a coverage figure computed over correlated cells was reported without the denominator that makes it readable.

Remark 11 (What the contract is not). Theorem 11 is conditional on (A1)–(A6), and K-Bound does not infer, test into, or prove any of them from unlabelled deployment data. (A1)–(A3) are not identifiable label-free; this is the obstruction of §4 applied to calibration transfer rather than to the drift budget. What a deployment can do is detect *some* violations and refuse to certify when it does, which is §7.9.

Remark 12 (Where the guarantee lives). The proof of Theorem 11 uses no property of $\widehat{\Delta}$ beyond its appearance in L and U: a false adapt requires $\Delta \leq 0 < L$, which is the complement of the coverage event, and a false freeze requires $\Delta \geq 0 > U$. So the guarantee is carried entirely by (A3); the fitted benefit estimator contributes decision *yield*, not safety. A better estimator narrows the interval and buys more committed decisions at the same α; a worthless one widens it until the system abstains. That is the intended failure direction, and it is why (A3) — not estimator quality — is the assumption a deployment has to defend.

7.3 Permitted behaviour by assumption state

Where the earlier tables say what to do when a single condition fails, Table 9 says what a deployment is permitted to *emit*, which is the operational form of the same idea and the object §7.9 implements.

Table 9: Fallback ladder. Permitted behaviour is a function of the assumption state, not of the estimated benefit. "Certification disabled" means no certificate is emitted; the system may still serve the frozen model, but no K-Bound claim attaches to that choice.

Assumption state	Permitted behaviour
All required gates pass	Full ADAPT/FREEZE/ABSTAIN certificate
Support warning, no hard violation	ABSTAIN or FREEZE only
Insufficient independent calibration units	Diagnostic output only; no decision
Leakage, failed transfer, or unknown provenance	Certification disabled

Read with Remark 12, this is the honest description of what K-Bound offers: *conditional insurance against detectable adaptation harm*, not an unconditional safety guarantee.

Theorem 11 (Finite-sample adapt/freeze/abstain certificate)**.** *Suppose Δ, $\widehat{\Delta}$, and the nonnegative random radius ε are measurable on a common probability space containing all calibration, benefit-model fitting, and deployment-unit randomness, and that the benefit interval satisfies marginal coverage*

$$\mathbb{P}\big[\,|\widehat{\Delta} - \Delta| \leq \varepsilon\,\big] \;\geq\; 1 - \alpha.$$

Define

$$g(Z) = \begin{cases} \textit{ADAPT} & \widehat{\Delta} - \varepsilon > 0, \\ \textit{FREEZE} & \widehat{\Delta} + \varepsilon < 0, \\ \textit{ABSTAIN} & \textit{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\mathbb{P}[g(Z) = \textit{ADAPT}, \Delta \leq 0] \leq \alpha,$$
$$\mathbb{P}[g(Z) = \textit{FREEZE}, \Delta \geq 0] \leq \alpha.$$

Proof. On the event $\{|\widehat{\Delta} - \Delta| \leq \varepsilon\}$, the condition $\widehat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$, so a false adapt requires the complement of this event, which has probability at most α. The freeze case is symmetric. □

Exchangeability, or an explicit shift correction, is one route to this coverage assumption. Risk alignment is not required for the conditional-on-coverage implication.

Remark 13 (Marginal vs. conditional error). Theorem 11 bounds the *marginal, unconditional* false-adapt and false-freeze probabilities ($\mathrm{FA_u}$). It does *not* bound the conditional rate $\mathrm{FA_c} = \mathbb{P}[\Delta \leq 0 \mid \textsc{adapt}]$, which we report empirically. Risk alignment (Definition 1) is required for the *frontier* identifiability claim; coverage alone certifies the *interval* rule above.

Remark 14 (Fallback when assumptions are unsupported). If interval coverage—or the calibration assumptions supporting it (exchangeability, or an explicit shift correction)—cannot be justified on the deployment batch, Theorem 11 supplies no level-α protection for either strict action. A deployment must then use a separately declared operational fallback, such as abstention or serving the frozen model; this is a policy recommendation, not a conclusion that either fallback is optimal. If instead *risk alignment* (Definition 1) cannot be justified, the population-frontier interpretation and the transferability of the benefit estimator are unsupported, even though the interval implication of Theorem 11 remains valid *conditional on coverage* (cf. Theorem 1).

Theorem 11 is an implication from an assumed premise, and the premise — not the theorem — is where the difficulty lives; §9.6 shows that on our stress grids the premise is supplied by a calibration that makes the conclusion arithmetically unavoidable, and reports what an observed $\mathrm{FA_u} = 0$ does and does not establish.

7.4 System Overview

KGA is the paper's practical finite-sample method—a decision wrapper around a fixed adapter, not a new adaptation objective and not a numerical implementation of the population $|M| > \beta$ rule. The wrapper is *adapter-interface agnostic*: it requires only that a candidate can be proposed, evaluated on the batch, and rolled back. Its calibration is *adapter-specific*: the benefit model and radius are refit per dataset and per candidate adapter, and cross-adapter transfer of a fitted calibration is disallowed because it violates the false-adapt bound (Table 29). We fit the benefit estimator and its radius *out of fold*, so the data used to fit $\widehat{\Delta}$ and the data used to calibrate its residual are disjoint for every point. On the per-cell stress grids (CIFAR-10-C and the synthetic grids), for each cell i a gradient-boosted $\widehat{\Delta}_{-i}$ is trained on the other $N-1$ cells and evaluated on cell i. The radius ε is then an upper empirical quantile of the leave-one-out residuals $\{|\widehat{\Delta}_{-i}(Z_i) - \Delta_i|\}_{i=1}^{N}$—leave-one-condition-out cross-fitted empirical residual calibration, so no cell's residual uses an estimator that saw it. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home) the split is across domains: $\widehat{\Delta}$ and ε are fit once on a dev split and the untouched held-out target-domain conditions are scored a single time. At deploy we extract Z from an unlabeled batch and return adapt, freeze, or abstain (Algorithm 1). Labels enter only through the calibration split; the deployment batch and held-out test scorer never use target labels for the adapter, ε, evidence, or rule selection. The coverage guarantee is split-dependent. Every radius in this paper is the exact rank statistic $\varepsilon = \rho_{(k)}$, $k = \lceil (n_{\mathrm{cal}}{+}1)(1-\alpha) \rceil$, unclamped, with $\varepsilon = +\infty$ (hence ABSTAIN) when $k > n_{\mathrm{cal}}$, and with the scored cell excluded from its own pool (declared once in §8); archived benchmark JSONs generated under the earlier interpolated empirical quantile have been regenerated under this rule, and the regenerated value is what is printed. Under exchangeability and a clean split, this rank statistic supports finite-sample marginal coverage. On the per-cell grids the pool is the leave-one-out residual vector rather than a clean split, so what the grids demonstrate is *calibration-set* coverage close to nominal (realized 0.898 at nominal 0.90, α=0.10)—and, as §9.6 makes explicit, that realized figure is a deterministic function of n under in-sample rank calibration, not a measurement. This is the standard jackknife radius, which is distribution-free only asymptotically under estimator stability; its formal finite-sample counterpart is the jackknife+ of Barber et al. (2021) at level $1-2\alpha$, which we note but do not implement. The decision rule's false-adapt guarantee is stated under the coverage hypothesis (§7), which the empirical coverage approximately meets. This construction sits in a known landscape: weighted conformal corrects coverage under known covariate shift [34], adaptive conformal tracks coverage online [36], and beyond exchangeability the coverage gap is bounded by the total-variation drift of the calibration pairs [35], which is conceptually related to but mathematically distinct from our disagreement-region calibration budget β. The certificate's marginal false-adapt control is risk control in the RCPS/Learn-then-Test lineage [38, 39], specialized to the adapt/freeze/abstain loss; PAC prediction sets give set-valued analogues under covariate shift [37].

7.5 Evidence Map and Benefit Estimator

The evidence map uses only unlabeled batches and model outputs, and is instantiated as two label-free panels matched to the two calibration regimes below. On the controlled stress grids (CIFAR-10-C, ImageNet-C, PACS) the base panel is an 11-dimensional vector computed from the frozen softmax p_0 and the adapted softmax p_a on the adaptation batch: the mean predictive entropy and mean confidence of f_0 and of f_a; the normalized marginal (predicted-class) entropy of each; the entropy and marginal-balance drops between them; the fraction of adapted predictions above confidence 0.9 (a collapse indicator); the KL divergence between the adapted and frozen predicted-class marginals; and the ℓ_2 norm of the adapter's parameter update. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home, RxRx1) a 16-dimensional panel adds distribution- and disagreement-oriented statistics computed from the frozen logits: entropy and confidence quantiles; the frozen/adapted prediction-disagreement rate; the entropy gap; a free-energy source$\rightarrow$target shift; a batch-vs-running BatchNorm-statistic KL; an ATC-style threshold-accuracy estimate (its threshold fixed on *source* labels only, never target labels); and a source$\rightarrow$target confidence drop. The source-calibrated correctness score s of §4 is operationalized by the adapted-confidence and ATC-style features. KGA does not explicitly estimate the theoretical margin M; it learns the benefit $\widehat{\Delta} = h_\theta(Z)$ directly from the full evidence vector Z.

The benefit model predicts $\widehat{\Delta} = h_\theta(Z)$ from labeled development conditions; h_θ is a gradient-boosted regression tree (scikit-learn `GradientBoostingRegressor`, squared error, 250 trees, depth 2, learning rate 0.05, subsample 0.8, seed 0), refit *per dataset and per candidate adapter* rather than shared, so the estimator itself implies no cross-dataset transfer. The coordinate-by-coordinate schema a re-implementer needs—exact per-feature formulas and families, for both the 11-dimensional base panel that drives the stress grids and the 16-dimensional panel the natural-shift protocols use—is Appendix B (Tables 31 and 32), with per-adapter update hyperparameters in Table 33. The calibration and held-out evaluation protocol behind every radius in this paper is Appendix A.

Algorithm 1 KGA: Knowability-Guided Adaptation

Require: Frozen f_0; adapter $\mathcal{A}$; batch $X_{1:m}$; $\widehat{\Delta}$, ε, α
Ensure: Decision $g \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$

$f_a \leftarrow \mathcal{A}(f_0, X_{1:m})$
$Z \leftarrow \phi(X_{1:m}, f_0, f_a)$
$\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$; $L \leftarrow \widehat{\Delta} - \varepsilon$; $U \leftarrow \widehat{\Delta} + \varepsilon$
if $L > 0$ **then**
    $g \leftarrow$ ADAPT
else if $U < 0$ **then**
    $g \leftarrow$ FREEZE
else
    $g \leftarrow$ ABSTAIN
end if
return g

7.6 Out-of-Fold Uncertainty Calibration

For calibration condition i, an estimator that did not train on i produces $\widehat{\Delta}_{-i}(Z_i)$ and residual

$$r_i = |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|.$$

The radius is a pre-specified upper residual quantile. On clean development/test splits, ordinary split conformal provides finite-sample marginal coverage under exchangeability. On stress grids, the reported radius uses leave-one-condition-out cross-fitted empirical residual calibration; its observed coverage does not imply the same finite-sample guarantee as split conformal, and jackknife+ is not claimed. We therefore report the calibration design and guarantee level separately for each track.

On the good event $|\widehat{\Delta} - \Delta| \leq \varepsilon$, committing adapt only when $L > 0$ implies $\Delta > 0$ (Theorem 11). The implication requires measurable marginal coverage; exchangeability or shift correction is used to justify coverage, while risk alignment is relevant to the population interpretation and estimator transfer. No target labels are used at deployment.

7.7 ε is not an estimate of β

KGA's commit boundary is not a swept hyperparameter, and it is not a numerical implementation of the population frontier. §4 and §5 make the distinction load-bearing, so we state it once, exactly. The population frontier (Theorem 2) certifies a strict action over the declared class $\mathcal{C}_\beta$ when the population margin $|M|$ exceeds the *declared* budget β; the realized drift γ is unobserved and, by Theorem 3, unknowable label-free. KGA instead learns $\widehat{\Delta}$ from Z and uses a finite-sample residual-coverage radius ε, fixed on the calibration split before any target-test cell is scored. ε **neither estimates nor substitutes for** β: it is a quantile of held-out $|\widehat{\Delta} - \Delta|$ residuals, and conditional on valid coverage it controls marginal decision errors (Theorem 11) without requiring risk alignment and without any statement about γ. Nothing in §4 constrains it, because it is not an audit of the drift budget.

What separates KGA from "estimate accuracy, then threshold" is therefore not the value of its cutoff. Confidence-thresholding (ATC) and agreement-trend selection (AGL) fit a scalar against a labeled or held-out signal and predict *accuracy*; KGA certifies a strict benefit *direction* and abstains where the interval crosses zero. Finite-sample abstentions may reflect the structural region of Theorem 1 or merely sampling uncertainty, estimator inadequacy, or calibration conservatism,

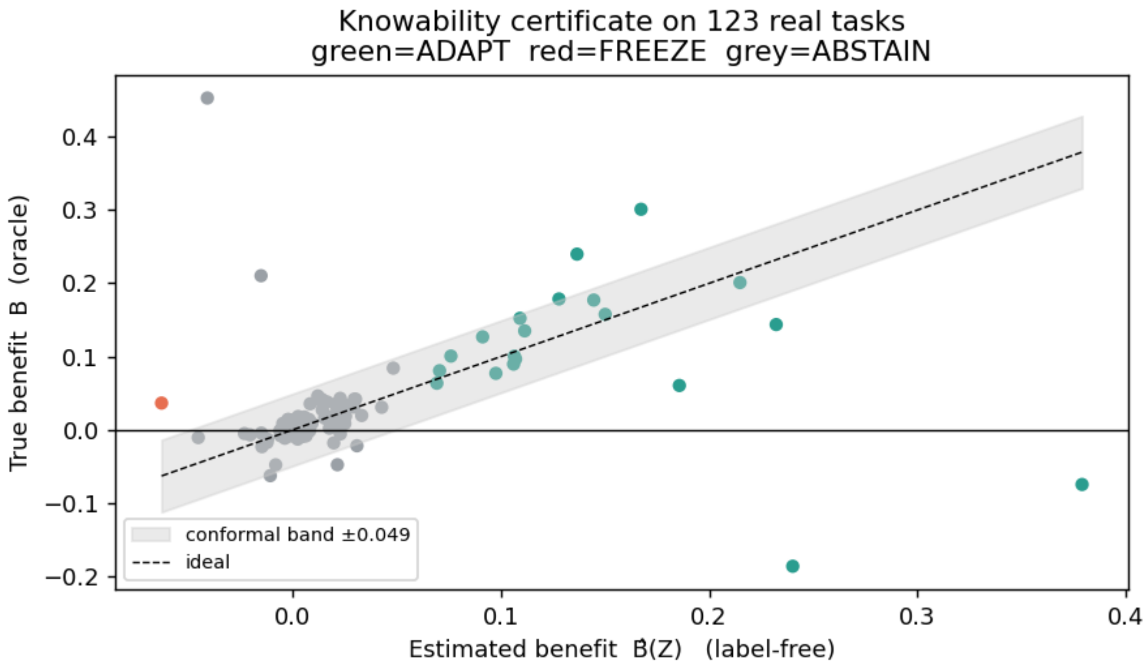

Figure 3: **The certificate rule in one picture.** KGA estimates the label-free benefit $\widehat{\Delta}(Z)$ and forms an interval $\widehat{\Delta} \pm \varepsilon$: if the interval is entirely positive it adapts, if entirely negative it freezes, and otherwise it abstains. The vertical axis here uses true benefit only for offline audit; deployment decisions use only unlabeled evidence and the pre-calibrated radius.

and §9.7 separates those causes empirically. Finally, the two rules are not interchangeable and the population rule is *not* the better one: with a genuinely held-out margin, no positive β matches the conformal rule on both decision yield and regret in 15 of 18 configurations (§5).

7.8 Deployment Semantics and Failure-Safe Behavior

The shadow-state workflow is: (1) checkpoint the active model; (2) propose adaptation on a temporary copy; (3) evaluate frozen and candidate outputs; (4) extract Z; (5) compute $\widehat{\Delta} \pm \varepsilon$; (6) commit or roll back; and (7) log the configuration, evidence, interval, and action. Abstention applies to the adaptation update, not to prediction. Adapt commits the candidate; freeze rolls it back; and abstain retains the frozen state while recording that evidence was insufficient rather than certifiably harmful. The implementation should automatically freeze or abstain when evidence is missing, the batch is too small, a checkpoint or adapter configuration differs from calibration, the evidence lies outside calibration support, numerical values are invalid, or no valid radius is available. In the reference implementation (`kbound_pkg`), the gate is applied per batch (`KGA.decide_from_batch`); on abstain or freeze the adaptation optimizer scales the candidate update by `abstain_scale`=0, i.e. the update is skipped and the frozen state f_0 is served—this is the rollback mechanism. The failure-safe defaults instantiate the diagnostics of Table 30: abstain when the batch is smaller than the calibration batch size, when Z lies outside calibration support, when a checkpoint or adapter-configuration hash differs from the calibrated one, or when the radius or evidence is missing or numerically invalid. In continual operation the last certified state replaces f_0 as the fallback.

7.9 The assumption gate

The failure-safe defaults above abstain when the batch is too small, the evidence lies outside calibration support, a configuration hash differs, or no valid radius exists. Those are the right reflexes, and the gate below is what they look like once they are stated as a precondition on *certification* rather than as guards inside the decision path. Because (A1)–(A6) cannot be verified label-free (Remark 11), a deployment is permitted to issue strict ADAPT/FREEZE decisions only after passing it. Thresholds are fixed before deployment and versioned with the protocol: a gate tuned after seeing the outcome violates (A6) and voids the certificate.

Inputs. Frozen candidate f_a; frozen benefit estimator $\widehat{\Delta}$; calibration evidence and residuals; unlabelled deployment evidence Z; the declared inference unit; thresholds fixed before deployment.

1. **Verify protocol separation.** If target labels affected candidate, feature, threshold or stopping choice: *certification disabled* — (A4).
2. **Verify inference-unit adequacy.** If the effective number of independent calibration units is below the declared minimum: *diagnostic only* — (A5).
3. **Test evidence-support overlap.** If deployment evidence lies outside the calibrated support: ABSTAIN or FREEZE — (A1)/(A2) suspect.
4. **Test calibration stability.** Recompute ε across folds, seeds, domains or time blocks. If the radius or the action proportions are unstable: ABSTAIN or FREEZE.
5. **Check sensitivity.** Vary the declared calibration construction and the clustering unit. If the promoted conclusion changes: *diagnostic only.*
6. **Apply the certified rule.** ADAPT if $L > 0$; FREEZE if $U < 0$; otherwise ABSTAIN.
7. **Monitor sequentially.** If support, evidence distribution or action rates drift beyond predeclared thresholds, suspend strict decisions and revert to FREEZE/ABSTAIN.

Steps 1–2 are protocol checks and are fatal; steps 3–5 are statistical diagnostics and degrade the permitted behaviour along Table 9 rather than aborting; step 7 runs for the lifetime of the deployment. The gate fails closed: an *unevaluated* check counts as a failed check, so a pipeline that does not yet supply deployment evidence or unit labels returns a restricted decision rather than a clean one.

7.10 Diagnostics, and what each one cannot do

Evidence-support overlap. We report the distance to the nearest calibration unit, the fraction of deployment evidence outside the calibration envelope, a cross-fitted domain-classifier AUROC, and the maximum standardised feature deviation. High separability is evidence *against* direct transfer; low separability is not evidence for it, since two conditions can be indistinguishable in Z and still have different Δ — which is exactly the configuration §4 constructs, not a corner case. This is the same one-sidedness already noted for the separability screen in §11.

Radius stability. We recompute ε leaving out one unit at a time and report the mean radius, its range and coefficient of variation, and the decision-disagreement rate against the pooled radius. The number that matters is the fraction of promoted conclusions that change under an alternative *equally valid* split: a conclusion that survives only one admissible clustering is a conclusion about the clustering. The check is not an outlier detector — a contaminated fraction well below α does not move the quantile and should not.

Table 10: The diagnostic suite consumed by the gate. None establishes theoretical coverage (Definition 9); each can only remove a way of being wrong. The right-hand column is as much a part of the method as the left.

Diagnostic	Detects	Cannot show
Evidence-support overlap	deployment evidence outside the calibrated envelope; high calibration-vs-deployment separability	that low separability implies transfer
Radius stability	ε or the action mix depending on an arbitrary choice of fold, seed, domain or block	that a stable radius is a correct radius
Risk-alignment audit	benefit-sign errors, rank decorrelation, coverage varying by evidence region	anything about a condition with no labels; it is retrospective by construction
Leakage audit	target labels or test-set influence entering candidate, threshold or hyperparameter choice	analyst decisions outside the recorded protocol

Risk-alignment audit. Where evaluation labels exist we report benefit-estimation error, rank correlation between predicted and realised benefit, coverage stratified by evidence region, false-adapt by domain or cluster, and whether benefit *sign* prediction transfers across domains. This is a retrospective audit of (A2) on the tracks we evaluated. It is not available at deployment time, which is the entire difficulty.

Leakage audit. For each promoted claim we record when the candidate was fixed, when the calibration design was fixed, whether target labels were accessed, whether the test set influenced hyperparameters, the protocol-lock identifier, and whether failed runs were retained. Chronology is the evidence for (A6); an assertion of pre-registration without a timestamped lock is not, and the audit fails closed on a missing record rather than assuming the benign case.

7.11 The assumption report

Each track emits a fixed-schema record (Table 11) alongside its results, so the assumption state travels with the number instead of living in prose. Every value is computed from raw units by `kga.assumptions`; none is transcribed by hand. The records are released with the artifacts.

Remark 15 (Reading a report)*.* `theoretical_coverage_claimed` is `false` on every track in this paper. That is not a formatting artefact: there is no track on which we are willing to assert (2) for an unseen deployment condition, because (A1)–(A3) are not checkable from the evidence we hold. `observed_coverage` is the statistic of Definition 10, reported with `n_units` and an interval computed at the declared unit, so that a coverage figure computed on correlated cells cannot be read as though it came from that many independent draws.

8 Experimental Setup

8.1 Research Questions

The experiments address eight questions: **RQ1** whether the population frontier $|M| > \beta$ operationalizes on real data—i.e. whether a β declared from development cells bounds the realized drift at

Table 11: Assumption-report schema, version `kbound-assumption-report/1`. A statistic that cannot be computed from the supplied inputs is `null` and adds a string to `limitations`; it is never defaulted to a plausible number. The released records are the authority for every value — nothing here is an example value.

Field	Type / admissible values
`dataset`, `protocol`	`str`
`inference_unit`	`str` (Def. 12)
`calibration_test_separated`	`bool`
`candidate_fixed_before_test`	`bool`
`target_labels_used_for_routing`	`bool`
`coverage_type`	`theoretical` | `observed_empirical` | `diagnostic_only`
`theoretical_coverage_claimed`	`bool`
`observed_coverage`	`float` | `null`
`n_rows`, `n_units`	`int`
`coverage_interval_95`	`[float, float]` | `null`
`coverage_interval_method`	`str`
`support_overlap_status`	`pass` | `warning` | `fail`
`radius_stability_status`	as above
`conclusion_stability_status`	as above
`leakage_status`	`pass` | `fail`
`deployment_gate`	`certify` | `restricted` | `diagnostic_only` | `reject`
`fallback_action`	`adapt_freeze_abstain` | `freeze_or_abstain` | `none`
`alpha`, `thresholds`	`float`, `object`
`diagnostics`	`object` (per-check detail)
`limitations`	`[str]`

deployment, and whether the resulting rule is competitive with the finite-sample certificate (§5; the answer is no, and Theorem 4 says why); **RQ1b** what the certificate buys at its declared operating point, in decision yield, regret, and false-adapt rate, and whether its commitments are better than randomly placed (§9.7); **RQ2** whether the interval controls marginal false adaptation *where that is testable* — wherever a rank radius is computed in-sample it makes $\mathrm{FA_u} \leq \alpha$ an arithmetic identity rather than a measurement (§9.6), so RQ2 is really the question of how many ADAPT decisions a track makes under a leave-one-out radius and whether any of them is wrong; **RQ3** whether KGA assigns ADAPT, FREEZE, and ABSTAIN consistently with helpful, harmful, and low-margin conditions; **RQ4** whether it prevents damage on held-out natural shifts; **RQ5** whether it improves decision validity over simple and neighboring label-free gates; **RQ6** which evidence and calibration choices matter; and **RQ7** what computational overhead the controller adds.

8.2 Datasets and Shift Taxonomy

Table 12 is the track-by-track inventory. The evaluation separates controlled corruption/composition stress, large-scale corruption, natural institutional or geographic shifts, domain generalization, and low-margin diagnostic transfers. This taxonomy matters because mixed conditions test action discrimination, one-sided shifts test safe commitment, and low-margin transfers test abstention. The nine tracks draw on CIFAR-10-C [21], ImageNet [20], the WILDS Camelyon17, iWildCam, and RxRx1 shifts [22, 23, 24], Office-Home [25], PACS via DomainBed [26], and CIFAR-10.1 [28]; models build on RobustBench checkpoints [27] with PyTorch [29] and scikit-learn [30].

Table 12: Dataset roles in the evaluation. The expected regime is a hypothesis fixed by the protocol, not a conclusion inferred from the final score.

Track	Shift type	Candidate role	Evaluation purpose
CIFAR-10-C stress	corruption/composition/update stress	Tent/EATA (SAR withheld)	mixed detectable decision benchmark
ImageNet-C noise	large-scale corruption	Tent/EATA/SAR	scale and candidate dependence
Camelyon17	hospital/institution shift	EATA	helpful-dominated natural safety
iWildCam	geographic camera shift	Tent	harmful-dominated natural safety
Office-Home	visual domain/style shift	SAR	held-out domain safety
RxRx1	experimental/laboratory shift	online adapter	harmful-dominated natural safety
PACS	domain generalization	fixed candidate	breadth and null behavior
ImageNet-R	rendition shift	candidate panel	weak-evidence diagnostic
CIFAR-10.1	natural resampling	TTA candidates	low-margin transfer diagnostic

8.3 Models, Adapters, and Leakage Control

For every track, the source checkpoint, candidate adapter, update mode, learning rate, number of steps, batch construction, adapted parameter subset, and random seeds must be declared before held-out scoring. Adapter selection, evidence selection, benefit-model fitting, and residual calibration use development data only. All choices are frozen before the target test cells are scored once; target labels are opened afterward solely for offline audit. The calibration and scoring configuration is uniform across tracks except where noted in Table 13.

A seed-level heterogeneity we found and did not re-pool. Seed 0 of the `stress_grid_multiseed_v1` artifact is not exchangeable with seeds 1–4: its mean Δ is 0.0895 against 0.128, its helpful fraction 0.614 against 0.767–0.776, and its mean adapter update norm 12.39 against 3.42–3.44, i.e. 3.6× the others. The episode-conformal estimator of §6 detects it independently: the leave-one-seed-out coverage for seed 0 is 0.730/0.731 against 0.885–0.942 for seeds 1–4. This is a pre-existing property of the artifacts, not of any analysis in this paper, and it may affect any number pooled over the five seeds. We disclose it and *do not* silently re-pool or drop the seed; every five-seed figure in this paper includes seed 0. Source: `beta_estimability/episode_beta_results.json`.

The quantile rule, declared once. Every radius reported in this paper is the **exact-rank** order statistic

$$\varepsilon = \rho_{(k)}, \qquad k = \lceil (n_{\mathrm{cal}}{+}1)(1-\alpha) \rceil, \qquad \alpha = 0.10,$$

computed on the residual vector defined by the track's calibration unit (Table 13). There is **no** $\min\{n_{\mathrm{cal}}, \cdot\}$ **clamp**. When $k > n_{\mathrm{cal}}$ the rule is *infeasible* — no finite radius attains $1-\alpha$, because the best attainable coverage is $n_{\mathrm{cal}}/(n_{\mathrm{cal}}{+}1)$ — and the released library (`kga.certificate.split_conformal_rank_radius`) returns $\varepsilon = +\infty$ with a warning, which forces ABSTAIN. At $\alpha = 0.10$ this makes $n_{\mathrm{cal}} \geq \lceil 1/\alpha \rceil - 1 = 9$ the feasibility threshold. Earlier drafts of this paper declared the *clamped* variant $k = \min\{n_{\mathrm{cal}}, \cdot\}$; that variant silently returns the largest residual and attains only $n_{\mathrm{cal}}/(n_{\mathrm{cal}}{+}1) < 1-\alpha$, so it is not the rule we claim under. It survives in the library solely as an opt-in replay flag (`on_infeasible='clamp'`) for reproducing archived numbers, and we do not use it for any number in this paper.

The clamp is never binding here, and we checked. At every calibration size that occurs in this paper, $k \leq n_{\mathrm{cal}}$: $n{=}9 \Rightarrow k{=}9$, $n{=}12 \Rightarrow k{=}12$, $n{=}18 \Rightarrow k{=}18$, $n{=}27 \Rightarrow k{=}26$, $n{=}135 \Rightarrow k{=}123$, $n{=}432 \Rightarrow k{=}390$. So clamped and unclamped agree on every printed value, and the distinction matters for what the paper *claims*, not for what it prints. The one place the threshold does bite is the leave-one-out-of-pool pool of size $n_{\mathrm{cal}} - 1$ (next paragraph): at $n_{\mathrm{cal}} = 9$ the leave-one-out pool has 8

Table 13: Per-track calibration and scoring configuration (α=0.10 throughout; benefit estimator family and calibration recipe are used across tracks, with parameters refit per dataset and candidate adapter, §7). The radius rule is the same unclamped exact-rank order statistic $\rho_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$ ($+\infty \Rightarrow$ ABSTAIN if $k > n$), in every row—only the residual pool differs. The two grid rows exclude the scored cell from its own pool (§8.3); the split-calibrated rows do so by construction. Backbones: ResNet-18 (CIFAR), ResNet-50 (ImageNet-C); the natural-shift backbones follow the WILDS/DomainBed defaults per track.

Track	Residual pool for $\rho_{(k)}$	Test unit
CIFAR-10-C stress	grid cell (leave-one-cell-out)	5×432 cells
ImageNet-C	grid cell (leave-one-cell-out)	27 cells × 5 seeds
PACS (LODO)	source-domain cells	18 cells/target/seed; 3 seeds
Camelyon17	dev domains $\{0, 1\}$, out-of-fold	test $\{2, 3, 4\}$
iWildCam H v2	calibration domain $\{0\}$, out-of-fold	held-out $\{1\}$
Office-Home M v2	calibration $\{0, 1\}$, out-of-fold	target test $\{0, 1\}$
RxRx1 J	development $\{0, \ldots, 4\}$, out-of-fold	test $\{5, \ldots, 9\}$

residuals and $k = 9 > 8$, so leave-one-out-of-pool is *infeasible* on the 9-condition Camelyon17 per-seed pools of Table 22; those are scored in-pool and their $\mathrm{FA_u}$ column is accordingly uninformative (§9.6).

This is the only radius rule used anywhere in the paper; wherever an earlier draft or an archived artifact reported the interpolated empirical quantile `np.quantile`$(\rho, 1-\alpha)$, the number has been regenerated under the rule above and the regenerated value is the one printed. The two rules are not cosmetically different: on ImageNet-C SAR they give pooled regret 0.0289 (exact rank) versus 0.0122 (interpolated), a factor of 2.4, and they differ in the number of ADAPT decisions by 13 versus 61 out of 135. Adopting the exact-rank rule uniformly also moves three other published rows relative to earlier drafts: ImageNet-R 0.0112 → 0.0151, CIFAR-10.1 SAR 0.0045 → 0.0057, and Camelyon17 Table 22 SAR 0.0410 → 0.0425; the CIFAR-10-C rows move in the fourth decimal only (0.0015736 → 0.0015852 Tent, 0.0012676 → 0.0012799 EATA). We report the exact-rank value in every case, including where it is the less favourable of the two.

The pool, declared once: leave-one-out-of-pool. A radius computed from a pool that *contains the residual of the cell being scored* is a function of that cell's label, which is the label the $\mathrm{FA_u}$ guarantee is about. Every per-cell radius in this paper therefore excludes the scored index: cell i's radius is $\rho_{(k)}$ over the other $n_{\mathrm{cal}} - 1$ residuals (`kga.certificate.conformal_radii_loo`). On the split-calibrated natural-shift tracks the question does not arise — the calibration domains and the scored domains are disjoint by construction. The correction is free on CIFAR-10-C and is not free on ImageNet-C: it changes 0 **of** 9,504 committed CIFAR-10-C decisions and leaves $\mathrm{FA_u} = 0$ throughout, while on ImageNet-C SAR it changes 2 of 135 decisions and moves the promoted triple from 0.0264 to 0.0289 with $\mathrm{FA_u}$ from 0 to 1/135 (§9.4). We promote the leave-one-out value on both tracks.

Two caveats attach to the rule rather than to any track. First, on the stress grids the residual pool is leave-one-*cell*-out, i.e. a jackknife, not a clean split; it therefore carries empirical rather than distribution-free coverage (§7), and Algorithm 2 labels each branch accordingly. Second, wherever a radius is nonetheless computed from the same vector it is used to test — which in this paper is only the n=9 Camelyon17 per-seed pools of Table 22, where leave-one-out is infeasible — k equals n exactly, so ε is the maximum residual, $\mathrm{FA_u}$ is forced to 0, and that column carries no information; see §9.6. Exact per-adapter update settings (optimizer, steps, learning rate, and the mild/aggressive operating points) are tabulated per track in Appendix B, Table 33.

8.4 Baselines

We distinguish fixed policies (always-freeze and always-adapt), an offline per-condition oracle, simple label-free gates (confidence, entropy, drift/KL, and ATC-style), KGA without its radius, and neighboring methods or ports inspired by POEM and AETTA. Each comparison must state whether the baseline is an official implementation, a faithful reimplementation, or a protocol-matched port.

8.5 Metrics and Statistical Testing

The primary decision metrics are regret-to-oracle, marginal false-adapt $\mathrm{FA_u} = \mathbb{P}(\textsc{adapt}, \Delta \le 0)$, conditional false-adapt $\mathrm{FA_c} = \mathbb{P}(\Delta \le 0 \mid \textsc{adapt})$, adapt rate, action composition, and decision coverage $\mathbb{P}(\textsc{adapt or freeze})$. Accuracy or AUROC is reported as a task metric, not as the controller's objective. The theorem controls $\mathrm{FA_u}$ under its assumptions; $\mathrm{FA_c}$ is descriptive. Lower regret than both fixed policies is a secondary routing-utility check and does not establish interval coverage or risk alignment. Paired bootstrap confidence intervals, their resampling unit, and seed aggregation are declared per comparison family.

The multiplicity family, declared prospectively. Earlier versions of this work declared the Holm family as "the three beats-both candidates (CIFAR-10-C Tent/EATA versus always-adapt; ImageNet-C SAR versus always-freeze), all of which survive Holm at 0.05" — that is, exactly the three comparisons that came out positive, named after the results were known. That is not a multiplicity correction and we withdraw it. The project's search history is public and large: across the committed campaign artifacts there are 1,387 recorded `beats_both` determinations (326 of them true, a 23.5% base rate) over 70 distinct scored result units and 69 pre-registered protocol files, with directory names (`win_hunt_v2`–`v5`, `win_finder_v*`, `hard_dataset_win_loop_v1`) that describe the search accurately. Most of those rows are exploratory development screens and do not belong in an inferential family, but the paper previously published no arm inventory at all, so a reader had no way to judge.

We therefore define the family *prospectively* as follows. The confirmatory family is the set of **pre-registered beats-both bars** — one per (track, candidate) pair named in a `research_lock/` protocol before any number for that pair was computed — evaluated once on the declared test cells. Rows produced by development screens, superseded protocol versions, or post hoc re-scorings are *not* members of that family and are not eligible for promotion, no matter how they come out. Appendix N publishes the resulting protocol → arms → verdict inventory in full, including the arms that failed, and Holm is applied over that inventory rather than over the winners. Readers who want the raw, uncorrected picture should read the appendix table, not the three intervals in the main text.

8.6 Defining "mixed and detectable"

Earlier versions of this work asserted that KGA beats both fixed policies "whenever helpful and harmful conditions are mixed and detectable" without ever saying what *detectable* means. As written that claim was unfalsifiable: the regimes labelled detectable were exactly the regimes where the win occurred. We replace it with an ex ante criterion, applied to every track before its outcome is discussed.

Definition 13 (Mixed, detectable)**.** A track is **mixed** at threshold p if its harmful base rate—the fraction of evaluation cells with $\Delta \le 0$—lies in $[p, 1-p]$; we fix $p = 0.10$. A mixed track is **detectable** at level α if the cross-fitted certificate strictly commits ($\textsc{adapt}$ or $\textsc{freeze}$, i.e. $|\widehat{\Delta}| - \varepsilon > 0$) on at least 25% of its cells. Both quantities are computable from the logged decisions without target labels beyond the offline audit, and neither refers to the regret comparison.

Applying Definition 13 to the nine tracks: CIFAR-10-C Tent (harmful 0.34, decision coverage 0.68) and EATA (0.27, 0.63) are mixed and detectable; ImageNet-C SAR (0.156, 26/135 = 0.19) is mixed but sits *below* the coverage bar and is therefore **not** detectable by our own criterion despite producing a win — we flag this rather than move the bar; ImageNet-C Tent (0.556, 0/135) is the most mixed track in the paper and is decisively *not* detectable; ImageNet-C EATA (0.02) and Camelyon17 (0.00–0.11 per candidate) are not mixed; PACS is mixed (mean harmful base rate 0.278) and passes the coverage bar (0.815) yet loses to always-adapt — a clean counterexample to sufficiency; ImageNet-R is mixed on only 3 of 10 backbones; RxRx1 and iWildCam are harmful-dominated one-sided tracks. Four tracks meet the mixed bar: CIFAR-10-C Tent and EATA meet detectability and produce a point-estimate beats-both (cluster-robust for Tent only, §9.1); PACS meets detectability and *loses* to always-adapt; ImageNet-C Tent fails detectability outright and produces nothing. ImageNet-C SAR produces a point-estimate beats-both without meeting our coverage bar, and without an interval to support it either. So the criterion is neither necessary nor sufficient on our own data, and the correct statement is the weak one: *a routing win requires mixed conditions, but mixedness plus decisiveness does not deliver one.* We say so rather than tune the thresholds until the sentence becomes true.

8.7 Evidence-Status Policy

Locked results trace to raw cells/seeds and a frozen aggregate; **reconciled** results correct a prior interpretation using raw records; **provisional** summaries require canonical replay; and **diagnostic** runs are complete but deliberately not promoted. Only locked and reconciled rows support the empirical headline. Every evidence-tier cell in the result tables begins with one of these four labels; qualifiers (seed counts, lock type, known caveats) follow in parentheses.

8.8 Hardware and Software

Runs use Python 3.12, PyTorch 2.5.1, and scikit-learn for the benefit estimator. Accelerator and checkpoint details are recorded per run manifest rather than pooled into one hardware claim: CIFAR experiments used the saved ResNet-18 setup; the authoritative ImageNet-C result used ResNet-50 on seeds 0–4. Hardware affects runtime but not the decision definitions or replayed metrics.

9 Results

The results are organized by regime rather than by dataset prestige, and they are read as a safety evaluation. The primary result is that on the four one-sided natural tracks with locked held-out artifacts KGA is no-harm—the deployment guarantee an operator wants—while on the three remaining natural tracks it is a conservative null or a declared failure, which we report in the same table rather than in a footnote. Controlled mixed regimes then test action discrimination and selective coverage and supply the secondary beats-both mechanism confirmation, and low-margin transfers test whether the method declines unsupported claims. Accuracy and regret are reported as descriptive task metrics throughout, not as the controller's objective. This section is the evidence for the last link of the spine: what the surviving certificate buys. §9.1 carries its strongest stress result (leave-one-corruption-out recalibration at zero false adaptations in all ten runs), §9.6 is the paper's own refutation of its most quotable number, and §9.7 measures the yield/safety frontier exactly, including the track on which the certificate buys less than nothing. The evidence for the earlier links is in §5 and §6.8.

9.1 CIFAR-10-C stress grid (5 seeds)

What the grid is. The *stress* grid crosses corruption, severity, batch size, composition, and update aggressiveness (432 conditions/method, ResNet-18; pre-registered as Protocol A; seeds 0–4). Its exact composition is 6 corruptions × 2 severities × 3 batch regimes × 3 compositions × 2 aggressiveness levels × 2 repeats = 432. The six corruptions are `contrast`, `defocus_blur`, `fog`, `gaussian_noise`, `jpeg_compression`, and `pixelate`, at severities $\{1, 5\}$ — that is 6 **of the** 15 **official CIFAR-10-C corruptions and** 2 **of the** 5 **severities**, selected by the driver's `-quick` subset (`"quick": true` appears in every run manifest). Nine corruptions are absent, including the rest of the noise family (`shot_noise`, `impulse_noise`), the weather corruptions (`snow`, `frost`, `brightness`), and `glass_blur`, `motion_blur`, `zoom_blur`, `elastic_transform`. Earlier versions of this work described the grid as covering "the official CIFAR-10-C corruptions"; that was wrong and is corrected here. Note also that the two repeats `r0/r1` are exact design-point replicates and correlate at 0.948 (adapt gap) to 0.999 (freeze gap), so the effective number of independent conditions is at most 216, not 432.

Result. Harmful base rates are 34% (Tent) and 27% (EATA). The primary result is selective routing at 0% false-adapt (Table 15). Among harmful cells KGA adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains. This is the one track in the paper where the false-adapt guarantee is exercised at power: 1,113 (Tent) and 1,244 (EATA) ADAPT decisions, none of them wrong, Clopper–Pearson 95% upper bounds on $\mathrm{FA_c}$ of 0.0027 and 0.0024 (§9.6).

Leave-one-corruption-out recalibration: the strongest stress test in the paper. The radius on this grid is calibrated leave-one-*cell*-out, so the calibration pool for a cell contains the other five severities and repeats of the *same* corruption. That is the most favourable partition available, and it is the obvious place to attack the result. We therefore refit the whole pipeline leave-one-*corruption*-out — six folds, the estimator never sees the corruption it is scored on — for *both* candidates and *all five* seeds, ten runs in total (Table 14). Every intermediate quantity degrades: the conformal radius inflates from 0.0152–0.0219 to 0.0926–0.1122 (4.3×–6.9×), the commitment rate falls from 0.509–0.600 to 0.398–0.417, and regret worsens by 3.4×–7.6×. The safety property does not: $\mathrm{FA_u} = 0.0000$ **in all ten runs**, zero false adaptations in every fold. Source: `experiments/kbound/frontier_sweep_v1/out_cifar_loco_tent_eata.json`. Three things are lost and we state all three. *(a) The routing-utility claim does not survive on EATA.* Against always-adapt, KGA under LOCO wins on all five Tent runs (0.66×–0.82× its regret) and *loses on all five EATA runs* (2.01×–2.94×). It beats always-freeze on all ten. So what survives leave-one-corruption-out is the false-adapt guarantee and the freeze-side utility, not beats-both. *(b) The freeze branch empties.* The inflated radius swallows the harmful cells: 0 freezes in every LOCO run against 22–76 as shipped, so the three-way routing degenerates to adapt-or-abstain. *(c)* Six corruption families is a small number of folds; this is a robustness check on the safety property, not an independent benchmark.

The beats-both claim, separated by candidate. Both candidates beat both fixed policies on the point estimate. They do *not* survive the same interval scrutiny, and we report them separately rather than as one sentence. The 432 conditions are not 432 independent draws: the

Table 14: **Leave-one-corruption-out recalibration, all ten runs** (two candidates × five seeds, 432 cells each). "Shipped" is the leave-one-cell-out calibration used everywhere else in the paper. Regret is against the oracle. **The safety property survives and the beats-both property does not:** against always-adapt (0.00296–0.00833 across runs) KGA wins on all five Tent runs and loses on all five EATA runs; against always-freeze (0.12373–0.13210) it wins on all ten. Source: `out_cifar_loco_tent_eata.json`, exact-rank radius.

	shipped	LOCO	ratio	FA_u (LOCO)
radius ε	0.0152–0.0219	0.0926–0.1122	4.3×–6.9×	—
commitment rate	0.509–0.600	0.398–0.417	0.70×–0.80×	—
KGA regret	0.00101–0.00185	0.00550–0.00997	3.4×–7.6×	—
freeze decisions	22–76	0	—	—
regret vs. always-adapt	wins 10/10	wins 5/10 (Tent only)	—	—
false adaptations	0	0	—	**0.0000** in 10/10

design crosses 6 corruptions with 2 severities and duplicates every design point (`r0`/`r1` correlate at 0.948–0.999), so the honest resampling units are the 216 twin-pairs and, more conservatively, the 12 corruption×severity or 6 corruption-family clusters. Table 16 gives the paired-bootstrap gaps at all four units.

- **Tent survives every unit.** The gap to always-adapt stays negative with the interval excluding zero from 432 cells down to 6 corruption clusters ($[-0.00952, -0.00259]$), and so does the gap to always-freeze. Clustering widens the intervals by 2.3× (adapt side) and 3.9× (freeze side) — a real cost, not a fatal one.

- **EATA does not.** Its adapt-side gap loses the interval as soon as the corruption structure is respected: $[-0.00483, +0.00043]$ at 12 corruption×severity clusters and $[-0.00436, +0.00035]$ at 6 corruption families, both containing zero. **We therefore claim a CI-supported beats-both for Tent only.** For EATA the supportable statement is: a point-estimate beats-both, CI-supported against always-freeze at every unit, and *not* CI-supported against always-adapt once corruption is treated as the cluster.

The per-corruption breakdown says the same thing at the level of the individual family. Tent is worse than always-adapt on one of six corruptions (`gaussian_noise`, adapt gap +0.00189); EATA is worse on *two* (`gaussian_noise` +0.00022, `jpeg_compression` +0.00292). Six families is a small enough number that one or two adverse ones is exactly what a cluster bootstrap should, and does, convert into an interval that touches zero.[1]

Leave-one-corruption-out, intermediate quantities (Tent, seed pool). For the single-run detail behind the paragraph above, the bottom row of Table 16 records residual MAE 0.00959 → 0.03090 (3.2×), R^2 0.993 → 0.895, ε 0.0213 → 0.0988 (4.6×), adapt rate 51.5% → 41.3% and KGA regret 0.00159 → 0.00603 (3.8×), at $FA_u = 0$ under every partition. Table 14 generalises that row to both candidates and all five seeds.

[1] The design-based intervals we reported previously resample the pre-registered mixing-ratio distribution of the stress-grid design (Protocol A; the 11-point Pareto over harmful fraction, N_{boot}=5000): Tent +0.019 $[0.011, 0.026]$ vs. always-adapt, +0.080 $[0.052, 0.107]$ vs. always-freeze; EATA +0.015 $[0.009, 0.021]$ and +0.075 $[0.049, 0.100]$. Those intervals treat the mixing ratio as the only source of variation and are not cluster-robust; where they disagree with Table 16, the cluster-robust row is the one we claim on.

candidate	policy	mean acc	regret↓	FA_u	coverage	lower than both?
Tent	always-adapt	0.786	0.0079	—	1.00	
	always-freeze	0.671	0.1241	—	1.00	
	KGA	**0.793**	**0.0016**	**0**	0.67	**yes**
EATA	always-adapt	0.798	0.0033	—	1.00	
	always-freeze	0.671	0.1314	—	1.00	
	KGA	**0.801**	**0.0013**	**0**	0.63	**yes**

Table 15: CIFAR-10-C stress grid (432 conditions/method, five seeds, α=0.1, Protocol A). "coverage" is decision coverage $\mathbb{P}[\textsc{adapt} \vee \textsc{freeze}]$; both fixed policies commit on every cell. SAR is omitted from this frozen display; its reconciled five-seed current-tree result is reported in the text together with its larger across-seed radius variation.

The withheld SAR arm. SAR on this grid is quarantined and we draw no comparative verdict from it. The five-seed aggregate (0.0015/0.0112/0.1286) is dominated by seed 0, whose harmful base rate is 0.5278 — 5.8× the mean of seeds 1–4 (0.0909). On seeds 1–4 alone KGA's regret (0.0015990) *exceeds* always-adapt's (0.0003099) by 5.2×, and 1 of 5 seeds beats both fixed policies — seed 0, the seed the quarantine exists for. The adapt-gap bootstrap interval is entirely *positive* (i.e. against KGA) at every resampling unit we tried: $[+0.00101, +0.00158]$ over 432 cells, $[+0.00003, +0.00298]$ over 12 corruption×severity clusters, $[+0.00012, +0.00271]$ over 6 corruption families. Two further corrections to earlier text: the aggregate was rebuilt from **four**, not five, saved per-condition seed files for the 432-cell grid (`seed0/` contains no per-condition dump; a fifth complete set exists elsewhere in the tree but is a 270-cell grid), and the earlier claim that its intervals "exclude zero" is withdrawn — they exclude zero on the wrong side. The SAR conformal radius is also far less stable across seeds (c.v. 0.390) than Tent's (0.030) or EATA's (0.068). SAR is therefore reported here as a negative and withheld from Table 15.

9.2 Decision-baseline comparison: empirical safety of the calibrated certificate

KGA's claim goes beyond "better than always-adapt/always-freeze": it is better than *simple label-free decision rules* under the same locked protocol. On the CIFAR-10-C stress grid (Tent candidate, identical cells), we score six rules from the same evidence (Table 17): a confidence gate (adapt iff adaptation raised mean confidence), an entropy gate (adapt iff it lowered entropy—Tent's own objective), a drift/KL threshold, an ATC-style accuracy-prediction gate, KGA *without* its conformal radius (adapt iff $\widehat{\Delta} > 0$), and the full certificate. Only KGA is derived from a coverage-based false-adapt certificate; on this stress grid the leave-one-cell-out radius provides strong *empirical* coverage, not the exact distribution-free validity of clean split conformal or jackknife+. It attains zero observed FA_u on this grid while staying near-oracle on regret (the radius-free variant keeps FA_u=0.049<α empirically, but with no such guarantee). The confidence/entropy gates false-adapt on ∼74% of harmful cells; the drift threshold, tuned to the budget, degenerates to always-freeze. The KGA-without-radius row isolates the radius: removing it lowers regret slightly (0.0017→0.0004) but raises FA_u from 0 to 0.049 (to 0.141 on harmful cells): the radius is what turns a good benefit *estimate* into a false-adapt *guarantee*. Reproduced from the locked per-cell dump (artifact manifest, App. L).

Table 16: **CIFAR-10-C: what the beats-both claim costs when the correlation structure of the grid is respected.** *(a)* Paired percentile bootstrap, 20,000 replicates, seed-averaged 432-vector, declared exact-rank radius; cluster variants resample whole clusters with replacement. * marks an interval excluding zero. **Tent's adapt-side interval survives every unit; EATA's does not survive clustering by corruption**, which is why the two candidates are claimed separately in the text. *(b)* The same pipeline refitted under a coarser calibration partition (Tent, all five seeds; the estimator never sees the corruption it is scored on). $\mathrm{FA_u} = 0$ under all three partitions, and KGA stays below both fixed policies (0.00792 always-adapt, 0.12410 always-freeze) in all three.

(a) gap intervals by resampling unit

resampling unit	units	KGA − always-adapt	KGA − always-freeze
Tent — point estimates 0.00159 / 0.00792 / 0.12410			
432 cells, i.i.d. (as run)	432	$[-0.00787, -0.00489]^*$	$[-0.13738, -0.10810]^*$
216 twin-pairs (`r0`/`r1`)	216	$[-0.00846, -0.00434]^*$	$[-0.14351, -0.10235]^*$
12 corruption × severity clusters	12	$[-0.01157, -0.00158]^*$	$[-0.21426, -0.04230]^*$
6 **corruption-family clusters**	6	$[\mathbf{-0.00952}, \mathbf{-0.00259}]^*$	$[-0.17790, -0.06364]^*$
EATA — point estimates 0.00128 / 0.00327 / 0.13138			
432 cells, i.i.d. (as run)	432	$[-0.00280, -0.00123]^*$	$[-0.14535, -0.11518]^*$
216 twin-pairs (`r0`/`r1`)	216	$[-0.00311, -0.00096]^*$	$[-0.15184, -0.10920]^*$
12 corruption × severity clusters	12	$[-0.00483, +0.00043]$ *(includes 0)*	$[-0.22510, -0.04666]^*$
6 **corruption-family clusters**	6	$[\mathbf{-0.00436}, \mathbf{+0.00035}]$ *(includes 0)*	$[-0.18757, -0.06998]^*$

(b) leave-one-corruption-out calibration (Tent, 5 seeds refitted)

calibration partition	folds	residual MAE	R^2	ε	adapt rate	KGA regret
leave-one-cell-out (as shipped)	432	0.00959	0.9929	0.02132	51.5%	0.00159
leave-one-twin-pair-out	216	0.00981	0.9924	0.02201	51.2%	0.00166
leave-one-corruption-out	6	**0.03090**	**0.8954**	**0.09882**	41.3%	**0.00603**

Per-corruption adapt gaps (negative = KGA better). Tent: contrast −0.00825, defocus_blur −0.01054, fog −0.01029, **gaussian_noise** +0.00189, jpeg_compression −0.00347, pixelate −0.00744. EATA: contrast −0.00264, defocus_blur −0.00542, fog −0.00556, **gaussian_noise** +0.00022, **jpeg_compression** +0.00292, pixelate −0.00153. Tent has one adverse family, EATA two.

Table 17: **Decision-baseline comparison, CIFAR-10-C stress grid** (Tent candidate, n=432 cells, 149 harmful, α=0.1). Lower regret and lower $\mathrm{FA_u}$ are better. Only the conformal certificate keeps $\mathrm{FA_u}$=0 on the full grid and on harmful cells while staying near-oracle.

Decision rule	regret↓	$\mathrm{FA_u}$	$\mathrm{FA_c}$	adapt	cov.	$\mathrm{FA_u}$(harm)
confidence gate	0.0084	0.257	0.301	0.85	1.00	0.745
entropy gate	0.0086	0.255	0.304	0.84	1.00	0.738
drift/KL gate	0.1232	0.000	0.000	0.00	1.00	$0.000^\dagger$
ATC-style gate	0.0045	0.116	0.172	0.67	1.00	0.336
KGA (no radius)	0.0004	0.049	0.071	0.68	1.00	0.141
KGA (certificate)	0.0017	**0.000**	**0.000**	0.51	0.68	**0.000**

†Degenerate: the budget-tuned drift gate never adapts (adapt-rate 0), so its "0 false-adapt" is just always-freeze (regret 0.123).

9.3 Mixed head-to-head vs. POEM-/AETTA-style baselines (pre-registered)

The stress-grid wins above beat the *trivial* policies always-adapt and always-freeze. A harder question is whether KGA improves over strong protocol-matched label-free decision baselines inspired by

Table 18: Pre-registered mixed head-to-head on the locked CIFAR-10-C stress stream. POEM/AETTA rows are protocol-matched style ports, not official reproductions. All policies use the same cross-fitted evaluation cells, candidate adapter, logged label-free evidence, regret-to-oracle metric, and unconditional false-adapt metric.

policy	mean regret↓	FA_u	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM-style	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA-style	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.0000**	0.6847	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

Table 19: Baseline faithfulness audit for POEM/AETTA comparisons.

Baseline	Original method needs	Our locked protocol uses	What is matched	What is not claimed
POEM-style	online entropy/betting monitor over adaptation stream	batch-summary entropy and logged per-condition evidence	same cross-fitted evaluation cells, same candidate, same false-adapt/regret metrics	not official per-sample POEM reproduction
AETTA-style	dropout/disagreement-based label-free accuracy estimation	protocol-matched accuracy/benefit gate from logged evidence	same evidence budget, same decision task, same cross-fitted evaluation stream	not official dropout-AETTA reproduction
KGA	benefit estimator + conformal radius	out-of-fold $\widehat{\Delta}$ and ε	same stream, same oracle/regret/FA_u metric	official method in this paper

POEM [10] and AETTA [15], on a *mixed* harmful+helpful stream. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion *before* computing any number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals Z and benefit B from the locked CIFAR-10-C stress grid (Tent, 432 conditions/seed, five seeds; harmful fraction ≈33%, i.e. the always-adapt FA_u of Table 18); only the decision rule differs. The competitors are *protocol-matched POEM-style and AETTA-style baselines*: ports of the published protectors onto this protocol with documented simplifications (batch-summary entropy rather than per-sample streams). Official per-sample POEM and dropout-based AETTA reproduction remains a camera-ready faithfulness check.

Verdict: WIN. KGA lowers mean regret-to-oracle vs. both ported baselines with paired-bootstrap 95% CIs entirely below zero and Holm correction (α=0.05), at $FA_u=0 \leq \alpha$ (Table 18). The ported POEM-/AETTA-style baselines false-adapt at ∼24–25% on the same stream, consistent with the absence of an explicit marginal false-adapt certificate under this ported protocol. Secondary pooled sets (Tent+EATA, EATA alone) yield the same WIN verdict, and a separate pre-registered recomputation pipeline (paired bootstrap 10^4) reproduces all three configurations internally, with both regret-gap CIs below zero in each and $FA_{KGA}=0$ versus 11–33% for the competitors; both artifacts are indexed in the manifest (App. L).

Baseline faithfulness. Table 19 records exactly what each port matches (evidence, candidate, cross-fitted evaluation cells, metrics) and what it does not claim (POEM's per-sample betting process; AETTA's dropout estimator); the comparison claim is limited to this locked protocol-matched setting.

9.4 ImageNet-C SAR (five seeds, at a declared stress operating point)

On ImageNet-C noise (ResNet-50, 27 cells: three noise corruptions $\times$ severities $\{1, 3, 5\}$ $\times$ three batch-composition regimes (iid, imbalanced, single-class), seeds 0–4, full-scale), SAR is harmful on 15.6% of cell–seed pairs.

The operating point, stated inline. This is not SAR at its published settings. Our re-implementation reproduces SAR's mechanisms (SAM, recovery reset, EMA), but the promoted run uses **learning rate** 4×10^{-3}**, batch size** 16**,** 50 **adaptation steps, and the final ResNet stage (`layer4`) left adaptable**. Niu et al. [5] use 2.5×10^{-4} and freeze that stage; our learning rate is therefore $16\times$ theirs, and our own driver's docstring records the official value. We adopt this point because it is the learning rate shared across all three candidates in this track, which is what makes the Tent/EATA/SAR rows comparable—but it is an aggressive operating point at which SAR is expected to be fragile, and the claim below is a claim about *this* point, not about SAR. A control at official settings (lr $= 2.5 \times 10^{-4}$, `layer4` frozen) is not yet run; until it is, "KGA beats both because SAR collapses" must be read with the knowledge that we chose the regime in which SAR collapses. *Do not* read "mechanism-faithful" as "configuration-faithful"; earlier drafts of this paper used the former phrase in a way that invited the latter reading.

Result. Pooled over five seeds (seed-averaged conditions, the same paired design as the CIFAR-10-C rows), always-adapt drops to 0.385 accuracy while KGA reaches 0.424: regret **0.0289** versus 0.0529 (always-adapt) and 0.0319 (always-freeze), at $\mathrm{FA_u} = 1/135 = 0.0074$ over 13 ADAPT decisions, under the declared exact-rank radius calibrated *leave-one-out-of-pool*: a cell's radius is computed from the other 134 residuals and never from its own (§8). We promote that number rather than the in-pool one. Scoring a cell with a radius fitted on a pool containing that cell's own residual makes ε a function of the very label the $\mathrm{FA_u}$ guarantee attaches to; the in-pool triple for this track is 0.0264/0.0529/0.0319 at $\mathrm{FA_u} = 0$ over 12 adapts, and we report it below as a sensitivity, not as the headline. Two of 135 decisions differ between the two pools.

What this track supports, exactly. The unit of analysis matters here and we previously got it wrong twice — once on the pool, once on the resampling unit. Under the promoted leave-one-out-of-pool radius, the 95% paired-bootstrap gap over the 27 seed-averaged conditions the text describes is $[-0.0085, +0.0038]$ against always-freeze and $[-0.0755, +0.0181]$ against always-adapt: *both include zero.* Even the more generous design that resamples the 135 cell–seed rows i.i.d. gives $[-0.0079, +0.0036]$ on the freeze side. The only interval that excludes zero on the freeze side has 3 clusters (the corruption families of this grid are gaussian, shot and impulse noise) and must not be read as a primary interval. **ImageNet-C SAR therefore supports a point-estimate no-harm statement against always-freeze (**0.0289 **versus** 0.0319**) and *not* a CI-supported beats-both against either fixed policy.** Table 21 gives every resampling unit and both pools, including the ones that flatter us. For the record, the sensitivity in the other direction: under the in-pool radius the freeze-side condition-level interval is $[-0.0092, -0.0023]$ and does exclude zero, and the previously reported adapt-side interval $[-0.052, -0.003]$ came from resampling the 135 cell–seed rows as if independent, which they are not. We do not claim on either.

Per seed. Point estimates improve both fixed-policy regrets on 2 of 5 seeds (seeds 2 and 4), under both pools. On seeds 0, 1, and 3 KGA adapts on no cell at all and its regret is bit-identical to always-freeze; the pooled point estimate rests on 13 ADAPT decisions, 10 of them from seed 2, and the single false adapt in the whole track is also on seed 2 (App. M). An earlier draft of this paper reported "5/5 seeds", which was an artifact of the interpolated quantile rule and is withdrawn.

candidate	policy	mean acc	regret↓	harmful cells	lower than both?
Tent	always-adapt	0.402	0.0191	56%	
	always-freeze	0.406	0.0145		
	KGA	0.406	0.0145		no (ties freeze; 0 adapts)
EATA	always-adapt	**0.440**	**0.0001**	2%	no (trails adapt)
	always-freeze	0.406	0.0342		
	KGA	0.440	0.0007		
SAR	always-adapt	0.385	0.0529	16%	
	always-freeze	0.406	0.0319		
	KGA	**0.409**	**0.0289**		point est. only, both sides

Table 20: ImageNet-C (27 cells: 3 noise corruptions, ResNet-50, α=0.1, exact-rank radius calibrated *leave-one-out-of-pool*); pooled over seeds 0–4 (seed-averaged conditions), superseding the earlier single-seed and 36-cell provisional configurations. All three candidates run at lr $= 4 \times 10^{-3}$, batch 16, 50 steps, `layer4` adaptable (§9.4); this is 16× SAR's published learning rate. Accuracy and regret in this table are both computed under that one rule; an earlier version of this table mixed an interpolated-quantile accuracy column with an exact-rank regret column (which is where the superseded values 0.407/0.0139, 0.440/0.0003 and 0.427/0.0264 came from). The SAR row's advantage over *both* fixed policies is a point estimate: under the promoted pool neither condition-level interval excludes zero (Table 21). Per-seed detail in App. M.

Other candidates. EATA stays helpful-dominated: KGA adapts on 120 of 135 cells and lands at regret 0.0007 against always-adapt's 0.0001 — essentially a tie, and *not* a win; on Tent the harmful regime dominates (55.6% of cell–seed pairs)—the highest harmful fraction anywhere in this paper—and KGA adapts on no cell, tying always-freeze and *not* beating both (Table 20). We flag this explicitly because ImageNet-C Tent is the most mixed track in the paper and is nonetheless a non-detectable regime by the criterion of §8.6: mixedness alone does not produce a routing win. Under the same logged conditions, an ATC-style confidence gate false-adapts on the harmful SAR cells—it adapts wherever confidence is high, which is exactly where SAR collapses—while KGA abstains on them.

9.5 Natural Shifts and Consolidated Summary

All benchmark tracks use the same decision target: the adaptation benefit $\Delta = R(f_0) - R(f_a)$, evaluated against the same per-condition oracle, always-adapt, and always-freeze policies. The candidate adapter is locked using development data. Evidence Z, the benefit estimator $\widehat{\Delta}$, and radius ε are then frozen before scoring the declared test cells; test labels are used only after decisions for offline regret and false-adapt audit. Stress grids use leave-one-cell-out residuals; semantic domain-shift tracks use a dev/test split. We report $\mathrm{FA_u} = \mathbb{P}[\textsc{adapt}, \Delta \le 0]$ and do not interpret $\mathrm{FA_c}$ as a certificate.

Evidence tiers. Tier labels follow the four-level policy of §8.7; only locked and reconciled rows support the empirical headline.

Multi-seed stability (Camelyon17). The one natural-shift track with per-condition logs at multiple seeds is Camelyon17 (4 seeds, 9 composition-stress conditions/seed). Aggregating the deployed per-seed decisions (locked per-seed artifacts, App. L) shows the no-harm outcome is

Table 21: **ImageNet-C SAR: no legitimate design gives this track a CI-supported beats-both.** Paired percentile bootstrap, 20,000 replicates, exact-rank radius. *Upper block* is the promoted leave-one-out-of-pool calibration (point estimates 0.0289 / 0.0529 / 0.0319); *lower block* is the superseded in-pool calibration (0.0264 / 0.0529 / 0.0319), shown so the reader can see exactly what the leakage was worth. The manuscript describes a seed-averaged condition-level design; the shipped script resampled the 135 cell–seed rows i.i.d. * marks an interval excluding zero. Only 3 corruption families exist on this grid (gaussian, shot, impulse noise), so that row is shown for completeness and is not a primary interval. Under the promoted pool every legitimate unit includes zero on *both* sides.

resampling unit	units	KGA − adapt	KGA − freeze
Promoted: leave-one-out-of-pool radius			
27 **conditions, seed-averaged** (as described, and as used for CIFAR-10-C)	27	[−0.0755, +0.0181]	[−0.0085, +0.0038]
135 cell–seed rows, i.i.d. (the generous design)	135	[−0.0491, −0.0013]*	[−0.0079, +0.0036]
cluster = corruption family	3	[−0.0321, −0.0197]*	[−0.0036, −0.0020]*
Superseded: in-pool radius (the scored cell is in its own pool)			
135 cell–seed rows, i.i.d. (as previously coded)	135	[−0.0519, −0.0034]*	[−0.0087, −0.0027]*
27 conditions, seed-averaged	27	[−0.0806, +0.0175]	[−0.0092, −0.0023]*
cluster = seed	5	[−0.0308, −0.0204]*	[−0.0127, +0.0000]
cluster = corruption family	3	[−0.0321, −0.0197]*	[−0.0094, −0.0034]*

candidate-dependent and stable only for the milder adapter (Table 22): with Tent, KGA ties always-freeze and beats always-adapt on all four seeds; with EATA the margin is inconclusive; and with the aggressive helpful-dominated SAR arm KGA over-freezes (regret 0.0425 versus always-adapt's 0.0002). The dominant fact about this table is structural, not empirical: at n=9 per seed the declared exact-rank radius takes $k = \lceil 10 \cdot 0.9 \rceil = 9 = n$, so ε is the *maximum* residual and $\mathrm{FA_u}$ is forced to exactly 0 for any data (§9.6). This is also the one table in the paper that cannot use the declared leave-one-out-of-pool calibration: a pool of 8 residuals needs $k = 9$, which no finite radius supplies, so the released library would return $\varepsilon = +\infty$ and abstain on all 36 cells (§8.3). We report the in-pool scoring and treat the $\mathrm{FA_u}$ column as void rather than as evidence. The $\mathrm{FA_u}$ column of Table 22 therefore carries no information, and the same is true of every $n \leq 12$ track in the panel. Under the archived interpolated quantile the SAR arm's seed 0 showed $\mathrm{FA_u} = 1/9 = 0.111 > \alpha$; we record that exceedance rather than let the rule change hide it, and note that 1/9 has a Clopper–Pearson 95% interval of $[0.003, 0.429]$, so it is well inside binomial noise at this sample size and is not evidence that the bound fails. The over-freezing itself has a simple cause: on this track the realized radius is 0.07–0.09 (SAR) and 0.15–0.37 (Tent) against benefits of order 10^{-3}, so the interval essentially never excludes zero. Per-seed multi-seed replays for iWildCam, Office-Home, and RxRx1 are single-run in the current artifacts and remain future work.

9.6 What an observed $\mathrm{FA_u} = 0$ does and does not establish

Six of the nine panel rows report $\mathrm{FA_u} = 0$. Read naively that is a nine-track confirmation of Theorem 11. It is not, for two separate reasons, and the honest accounting makes the one genuinely strong track visible rather than burying it in a row of zeros.

Table 22: **Multi-seed Camelyon17 (seeds 0–3, 9 conditions/seed).** Regret mean±std across seeds, re-scored under the declared exact-rank radius. AD is the total number of ADAPT decisions over 36 cells. $\mathrm{FA_u}$ is 0 *by construction* at n=9 (§9.6) and is shown only to make that point; it is not evidence. Reproduced from the committed per-seed logs.

candidate	KGA regret	adapt	freeze	AD	verdict
Tent	0.0201±0.0230	0.1380	0.0201	0	ties freeze; never adapts
EATA	0.0393±0.0252	0.0417	0.0424	1	inconclusive
SAR	0.0425±0.0185	0.0002	0.0654	5	over-freezes (helpful-dom.)

(a) On in-*sample* rank calibration, $\mathrm{FA_u} \leq \alpha$ **cannot be violated.** When ε is the k-th order statistic of the same residual vector it is then used to test, the number of residuals exceeding it is identically $n - k$. With $k = \lceil (n+1)(1-\alpha) \rceil$ the miscoverage ceiling is therefore $(n-k)/n$, a function of n alone: 0.0972 at n=432, 0.0370 at n=27, and exactly 0 at $n \in \{9, 12, 18\}$, where $k = n$. All of these are below α=0.10, so under in-sample calibration "$\mathrm{FA_u} \leq \alpha$" holds for *any* data whatsoever and is not a measurement. We verified this ceiling is attained exactly on all 69 shipped per-condition files. The informative statistic is therefore $\mathrm{FA_u} = 0$ *against that ceiling*, not $\mathrm{FA_u} \leq \alpha$; and at $n \leq 18$ with $k = n$ (Camelyon17's Table 22 cells, ImageNet-R conditions, PACS cells) the ceiling is 0 and the statistic carries no information at all.

This identity is exactly what the leave-one-out-of-pool calibration of §8.3 breaks, and breaking it is the point. Once cell i's radius is fitted without cell i's residual, nothing forces the miscoverage count, and a nonzero $\mathrm{FA_u}$ becomes *possible*. It is therefore informative that on CIFAR-10-C the count stays at 0 over all 9,504 committed cells under the leave-one-out pool — and equally informative that on ImageNet-C SAR it does not, moving to 1/135. A reader should read the CIFAR-10-C zero as a measurement and every $n \leq 18$ zero in the panel as arithmetic. For the same reason the realized calibration-set coverage figures (0.898 at n=432, 0.889 at n=27) are deterministic functions of n; earlier versions of our audit artifact attached Wilson binomial intervals to them, which has no frequentist meaning, and those intervals have been removed.

(b) Where the guarantee is actually exercised. A false-adapt bound is only tested by ADAPT decisions. Table 24 reports the action composition of every promoted row together with a one-sided Clopper–Pearson 95% upper bound on the *conditional* rate $\mathrm{FA_c} = \mathbb{P}[\Delta \leq 0 \mid \text{ADAPT}]$ — i.e. how much the observed zero actually constrains anything. On CIFAR-10-C the bound is 0.0027 from 1,113 adapts. On ImageNet-C SAR it is 0.316 — and there the observed conditional rate is not even 0, it is 1/13; on Office-Home the bound is 0.127; on iWildCam 0.950. Three tracks (RxRx1 with 0 adapts, iWildCam with 1, the D33 mechanism check with 9) make fewer than 10 ADAPT decisions and we mark them *guarantee untested*.

A low adapt count is not the only way a zero can be empty, and Camelyon17 is the other way. Its recomputed row makes 18 ADAPT decisions — above the bar — for a CP_{95} of 0.153, but every one of those 18 cells has $\Delta > 0$. There is no harmful cell in the subset, so no ADAPT on it could have been false, whatever the rule had decided. The count is adequate and the test is still empty. We report the row because it is now recomputable from the release, not because it is evidence.

The conclusion we draw is deliberately narrow: **the certificate's safety guarantee is tested with real power on exactly one track, and there it holds**.

Table 23: **Uniform nine-track empirical panel.** Regrets are ordered as KGA / always-adapt / always-freeze; lower is better; all rows use the declared exact-rank radius (§8). Fixed-policy comparisons are secondary. The "AD" column is the number of ADAPT decisions, i.e. the number of opportunities the false-adapt guarantee had to fail; rows marked † have fewer than 10 and their observed $\mathrm{FA_u} = 0$ is uninformative (Table 24). The final column prevents a point estimate, a reconstructed summary, and a locked multi-seed result from being presented as equivalent evidence.

Track	Evaluation unit	AD	Result	Decision interpretation	Evidence tier
CIFAR-10-C stress	5×432 per candidate	1113 / 1244	Tent: 0.0016/0.0079/0.1241; EATA: 0.0013/0.0033/0.1314; SAR withheld; $\mathrm{FA_u} = 0$, $\mathrm{CP_{95}}$ upper on $\mathrm{FA_c}$ 0.0027 / 0.0024	selective routing; **the one powered track**. Cluster-robust beats-both for *Tent only*: EATA's adapt-side CI includes zero once corruption is the cluster (Table 16)	locked (Tent/EATA; five raw seeds); SAR withheld (seed-0 aggregate non-reproducing)
ImageNet-C SAR	27 cells × 5 seeds	13	0.0289/0.0529/0.0319 pooled; $\mathrm{FA_u} = 1/135 = 0.0074$, $\mathrm{CP_{95}}$ 0.316	**point-estimate** no-harm vs. always-freeze; *no* CI-supported beats-both at any legitimate unit	locked (5 seeds; 2/5 seeds beat both, 3/5 tie freeze exactly)
Camelyon17 OOD (EATA candidate)	genuine OOD test $n = 18$ (support $n = 36$)	18	0.0000/0.0000/0.1381; $\mathrm{FA_u} = 0$	adapts on every cell, so it *is* always-adapt here; all 18 cells are helpful, so $\mathrm{FA_u}$=0 is vacuous, not evidence	**recomputed 2026-07-26** from `camelyon17_richZ_F_v1/_partial.json` (App. L)
iWildCam H v2	held-out test $n = 72$	1†	0.0037/0.1028/0.0041; $\mathrm{FA_u} = 0$, $\mathrm{CP_{95}}$ 0.950	no-harm, ties freeze; guarantee untested	**not reproducible from the release**: the held-out record file is absent (App. L)
Office-Home M v2	target test $n = 35$	22	0.0157/0.0468/0.0158; $\mathrm{FA_u} = 0$, $\mathrm{CP_{95}}$ 0.127	no-harm, tiny point edge; CP bound is $1.3 \times \alpha$	locked (OOF no-harm only; LOO BB not promoted)
RxRx1 J	OOD test $n = 60$	0†	0.0000/0.2531/0.0000; $\mathrm{FA_u} = 0$	freezes on every cell; guarantee untested	locked (real ckpt; single-run)
PACS	3 seeds ×4 LODO targets ×18 cells	12	mean regret 0.0431/0.0176/0.0446; pooled $\mathrm{FA_u} = 2/216$, $\mathrm{CP_{95}}$ on $\mathrm{FA_c}$ 0.438	**worse than always-adapt** (2.45×); all 12 adapts come from one domain–seed cell, 2 of them false	locked diagnostic null; no beats-both claim
ImageNet-R D	4 seeds ×10 backbones ×12 conditions	0–46	mean-across-backbone regret 0.0151/0.0064/0.0325; per-backbone breakdown in Table 41	**worse than always-adapt on 7 of 10 backbones**; 4 backbones have a 0% harmful base rate	locked diagnostic null
CIFAR-10.1 K	cross-seed test $n = 48$	18	0.0021/0.0190/0.0017; $\mathrm{FA_u} = 0.167$ (8/48), $\mathrm{FA_c} = 0.444$	fails transfer bar; no claim	locked diagnostic fail

Table 24: **Action composition and the strength of the false-adapt evidence, per track.** $\mathrm{FA_u}$ is the marginal false-adapt rate the certificate bounds; CP_{95} is the one-sided Clopper–Pearson upper bound on the *conditional* rate $\mathrm{FA_c} = \mathbb{P}[\Delta \le 0 \mid \textsc{adapt}]$. † fewer than 10 ADAPT decisions: guarantee untested. Only CIFAR-10-C exercises the guarantee with power.

track	N	AD	FR	AB	false	$\mathrm{FA_u}$	CP_{95}
CIFAR-10-C Tent	2160	1113	358	689	0	0.0000	**0.0027**
CIFAR-10-C EATA	2160	1244	130	786	0	0.0000	**0.0024**
ImageNet-C SAR	135	13	15	107	1	0.0074	0.3163
Office-Home M v2	35	22	12	1	0	0.0000	0.1273
PACS (pooled)	216	12	164	40	2	0.0093	0.4381
iWildCam H v2†	72	1	60	11	0	0.0000	0.9500
Camelyon17 OOD	18	18	0	0	0	0.0000	0.1533
RxRx1 J†	60	0	60	0	0	0.0000	undef.
CIFAR-10.1 K	48	18	24	6	8	0.1667	0.6594
multimodal D33†	130	9	119	2	0	0.0000	0.2831

Camelyon17's counts were dashed until 2026-07-26 because the promoted n=18 triple had no reproducible source; they are now recomputed cell by cell (App. L) and they are 18/18 ADAPT. Its CP_{95} of 0.1533 is the least informative kind of zero in this table: not one of those 18 cells is harmful, so no ADAPT could have been false. Office-Home, iWildCam, RxRx1 and CIFAR-10.1 counts are $n_{\text{test}}\times$ the promoted adapt rate, the CIFAR-10-C, ImageNet-C, PACS and Camelyon17 counts are recomputed cell by cell.

9.7 What the certificate buys: decision yield, safety, and where abstention is correct

A certificate that abstains almost everywhere invites the objection that its no-harm result is vacuous, because no-harm is trivially achievable by abstaining—and on the promoted natural tracks the certificate does commit rarely (Office-Home `tent_online_aggressive`: 0/0/35; RxRx1: 0 ADAPT; iWildCam: 1). This subsection answers the objection with numbers, and concedes the two places where the answer goes against us.

Two things have to hold for the objection to fail. **(i) Abstention must not be free.** In this system an ABSTAIN realizes the *frozen* model, so always-abstain is numerically identical to always-freeze and inherits its regret exactly; whether that is cheap or ruinous is a per-track empirical fact. **(ii) The commitments must be worth something.** A certificate that fires rarely *and at random* is decoration; one that fires rarely but only on high-stakes cells is an instrument. These are distinguishable, and we distinguish them.

Integrity of the replay. The κ-family rule "ADAPT iff $\widehat{\Delta} - \kappa\varepsilon > 0$, FREEZE iff $\widehat{\Delta} + \kappa\varepsilon < 0$" reproduces the shipped `kga_decision` on 7,365/7,365 cells (100.0%) at $\kappa = 1$ across CIFAR-10-C (6,480), ImageNet-C (405) and ImageNet-R (480). $\kappa = 1$ *is* the shipped α=0.10 certificate, so the sweep below is a sweep of the real system. Label hygiene is enforced at run time rather than by comment: every label array is registered on load and every array a policy consumes is checked against the registry by identity, buffer and value; the guard is demonstrated firing on B, on a copy of B, on a view of B, and on a_{oracle}.

The yield/regret/false-adapt frontier. Table 25 and Fig. 4 sweep κ from 0 (commit on the sign of $\widehat{\Delta}$ everywhere) to ∞ (abstain everywhere). On CIFAR-10-C the certificate cuts regret 5.0× below the better fixed policy and 85× below the worse, at $\mathrm{FA_u} = 0/6480$ (Wilson 95% upper bound 0.0006) — and always-abstain ties always-freeze as the *worst* row in the table, so that reduction is

Table 25: **Decision-yield / safety frontier** (pooled per track, radius scaled by κ; κ=1 is the shipped certificate and reproduces its decisions on 7,365/7,365 cells). Regret is to the per-cell oracle; ABSTAIN and FREEZE both realize f_0, so always-abstain is numerically always-freeze. "best label-free heuristic" is the minimum over four single-signal thresholded scores (`marginal_KL` as a batch-statistics drift proxy, `entropy_drop`, confidence gain, `pbal_drop`) whose thresholds are chosen *with* the test labels subject to $\mathrm{FA_u} \le \alpha$; the comparison is therefore biased in the heuristic's favour. Source: `decision_value_results.json`.

	CIFAR-10-C (6480)		ImageNet-C (405)		ImageNet-R (480)	
policy	yield	regret	yield	regret	yield	regret
$\kappa = 0$	1.000	0.00029	1.000	0.00595	1.000	0.00126
$\kappa = 0.5$	0.763	0.00074	0.751	0.00575	0.812	0.00465
$\kappa = 0.9$	0.684	**0.00134**	0.551	**0.00763**	0.588	0.00939
$\kappa = 1$ **(shipped)**	**0.668**	**0.00150**	**0.521**	**0.00829**	**0.535**	**0.01120**
$\kappa = 2$	0.546	0.00312	0.400	0.01260	0.098	0.02686
always-adapt	1.000	0.00748	1.000	0.02403	1.000	**0.00636**
always-freeze	1.000	0.12797	1.000	0.02685	1.000	0.03253
always-abstain	0.000	0.12797	0.000	0.02685	0.000	0.03253
coin flip	1.000	0.06773	1.000	0.02544	1.000	0.01944
best label-free heuristic	1.000	0.00472	1.000	0.02558	1.000	0.01864
oracle bound at shipped yield	0.668	0.00020	0.521	0.00298	0.535	0.00424
$\mathrm{FA_u}$ at κ=1	0.0000 ($\mathrm{W_{95}}$ 0.0006)		0.0049 ($\mathrm{W_{95}}$ 0.0178)		0.0021 ($\mathrm{W_{95}}$ 0.0117)	
regret ratio, best fixed / KGA	**5.00**×		**2.90**×		**0.57**× (KGA loses)	

not achievable by abstaining. On ImageNet-C the ratio is 2.90×. **On ImageNet-R it goes the other way: the certificate is** 1.76× ***worse*** **than always-adapt** (0.0112 vs. 0.0064), paying 0.48 accuracy points per cell to buy $\mathrm{FA_u}$ 0.0021 against 0.2021 on a track where adaptation essentially never hurts. That is the clearest counterexample in the paper to a blanket "uniformly no-harm" reading, and it comes from an artifact we ourselves promote.

Is the shipped operating point efficient? Mostly not, but only just. CIFAR-10-C pooled is dominated by $\kappa = 0.9$ (regret 0.00134 at yield 0.684), ImageNet-C pooled likewise (0.00763 at 0.551), and ImageNet-C EATA is dominated by nine more permissive values of κ; ImageNet-R is on the frontier of its own family. The sharpest instance is ImageNet-C SAR, where $\kappa = 0.5$ gets 46% more decisions at 39% less regret (yield 0.874 vs. 0.600, regret 0.0066 vs. 0.0108) for a price of one extra false adaptation (1/135 → 2/135, both far under α). The gap is 10–40% of regret, not an order of magnitude. $\alpha = 0.10$ **is defensible as a pre-registered safety choice; it is not regret-optimal, and we do not claim it is.** Against a *perfect* ordering of cells by the regret a correct commitment removes, the certificate sits 7.50× / 2.79× / 2.64× above the achievable floor at its own yield: there is real headroom in *which* cells it commits on, not only in how many.

Against cheap label-free heuristics. Four single-signal, no-training references drawn from the same evidence panel (`marginal_KL` as a batch-statistics drift proxy, `entropy_drop`, confidence gain, `pbal_drop`) are scored two ways: with a hindsight-optimal threshold, and with a threshold fitted on labelled seeds $\{0, 1, 2\}$ and scored on $\{3, 4\}$. On CIFAR-10-C the certificate beats a hindsight-tuned drift heuristic held to the same false-adapt budget by 3.1× on regret (0.00150 vs. 0.00472) while making zero false adaptations against that heuristic's 324/6480. On ImageNet-C it wins only against heuristics held to its own budget: the fair-split confidence-gain heuristic reaches *lower* regret (0.00689 vs. 0.00905 on held-out seeds) by adapting 93% of the time at $\mathrm{FA_u} = 0.185$. On ImageNet-R all four heuristics degenerate to always-adapt and beat the certificate on the same held-out seeds

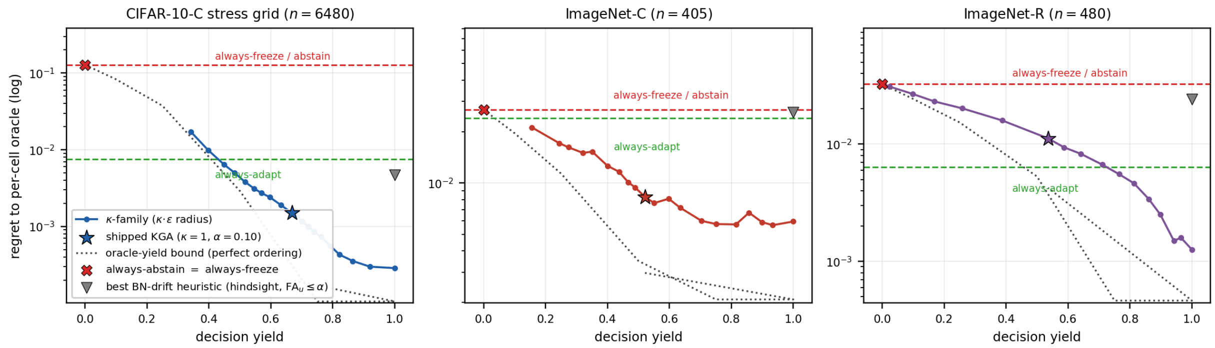

Figure 4: **What the certificate buys.** Regret against decision yield as the conformal radius is scaled by κ; the star is the shipped α=0.10 operating point, which reproduces the released decisions exactly. Always-abstain is the leftmost point and coincides with always-freeze: abstention is not free. The certificate sits well below both fixed policies on CIFAR-10-C and ImageNet-C and *above* always-adapt on ImageNet-R, which is reported as a negative result. The dotted curve is the best regret achievable at each yield by a perfect ordering of cells; the grey triangle is the batch-statistics drift heuristic (`marginal_KL`) with its threshold chosen using the test labels subject to $\mathrm{FA_u} \le \alpha$, which is the best of the four heuristics on CIFAR-10-C but not on the other two tracks (Table 25).

(0.00677 vs. 0.01617) while adapting on every cell at $\mathrm{FA_u} = 0.217$, against the certificate's 0.000 on those seeds and 0.0021 pooled. **The certificate's advantage over cheap heuristics is entirely an advantage in false-adapt rate, and it converts into a regret advantage only where adaptation is genuinely two-sided.**

Value per committed decision. Among the cells where the certificate commits, we measure the regret removed relative to the *better* fixed policy, against a permutation null that holds the number of ADAPT and FREEZE commitments fixed and reassigns them to a uniformly random subset of cells (5,000 draws). On CIFAR-10-C a committed decision is worth +0.0094 accuracy against a null of −0.0185 ($p < 0.0002$), and commitments land on cells whose true effect is **24.5**× larger than the cells it declines ($p < 0.0002$): it fires on high-stakes cells, not at random. **The single ImageNet-C Tent commitment—the critique's own example—is worth +0.081 accuracy, 2.4× the average cell effect, against a null of −0.004 ($p = 0.016$).** One decision out of 135, on the right cell. **The counter-case is ImageNet-R**, and we state it plainly: per backbone, the value per committed decision is exactly 0.0000 on 9 of 10 backbones, the $|\Delta|$ concentration ratio is 0.71–1.57 with permutation p between 0.054 and 1.000 (not one backbone reaches $p < 0.05$), and the pooled share of value attributable to committed cells is −0.77. The mechanism is that on those backbones adaptation always helps, so every commitment is an ADAPT that merely reproduces always-adapt, and all of the certificate's effect lives on the cells it abstains on, where it freezes and loses.

Why the abstaining tracks abstain. A cell abstains iff $|\widehat{\Delta}| \le \varepsilon$, which can mean the estimator is too noisy or that the effect is genuinely smaller than the radius. The discriminator is the *yield ceiling* $\mathbb{P}(|\Delta| > \varepsilon)$: the maximum yield any estimator could reach at that radius. Table 26 and Fig. 5 report it. On four of five per-cell slices the certificate commits on 93–102% of what is committable; **the abstention is the radius being wider than the effect, not conservatism.** The one genuine estimator failure is ImageNet-C Tent, at 4% of its ceiling. We state the limit of this statistic rather than lean on it: because $\widehat{\Delta}$ and Δ have nearly the same marginal spread, a share near 1 is

Table 26: **Abstention decomposition.** The ceiling $\mathbb{P}(|\Delta| > \varepsilon)$ is the maximum yield *any* estimator could reach at that radius; "share" is achieved yield over ceiling. Natural-track rows use the published ε and decision counts against per-cell Δ from the same artifact; $\mathbb{P}(\Delta > \varepsilon)$ is the fraction of cells on which an ADAPT is even available.

track / candidate	ε	$\varepsilon/\overline{\lvert\Delta\rvert}$	yield	ceiling	share
CIFAR-10-C pooled	0.0179	0.13	0.668	0.675	**0.99**
ImageNet-C EATA	0.0053	0.15	0.956	0.933	1.02
ImageNet-C SAR	0.0294	0.35	0.600	0.644	0.93
ImageNet-C Tent	0.0534	**1.59**	**0.007**	0.178	**0.04**
ImageNet-R pooled	0.0365	0.94	0.535	0.527	1.02
			$\mathbb{P}(\Delta>\varepsilon)$		reg. adapt
RxRx1 Tent (0/12/0)	0.0496	0.29	**0.000**		0.1689
RxRx1 SAR (0/12/0)	0.0432	0.16	**0.000**		0.2718
iWildCam Tent (0/39/105)	0.1369	1.20	0.007		0.1075
iWildCam EATA (0/47/97)	0.1257	1.21	**0.000**		0.0975
O-Home tent-aggr. (0/0/35)	0.0454	2.32	0.086		0.0100
O-Home `labelshift` (0/35/0)	0.0549	0.11	0.000		**0.4958**

close to the null—replacing $\widehat{\Delta}$ by Δ plus a *reshuffled* residual, which destroys the per-cell pairing entirely, still gives a share of 1.05 (CIFAR-10-C) and 1.02 (ImageNet-C)—above the observed values (`adversarial_share_of_ceiling_probe.json`). The share therefore discriminates only at the extreme, which is exactly where it is used (ImageNet-C Tent, 0.04); the evidence that abstention is a radius problem rather than timidity is the $\varepsilon/\overline{|\Delta|}$ column, and the evidence that commitments land on the right cells is the permutation test above, not this ratio. On the natural tracks the critique names, the answer is sharper. On **RxRx1**, $\mathbb{P}(\Delta > \varepsilon) = 0.000$ for all three candidates: not one cell of 36 has a true adaptation benefit exceeding the radius, so "0 ADAPT" is correct rather than timid—and meanwhile the certificate commits 12/12 FREEZE on both promoted candidates against an always-adapt regret of 0.169–0.272. On **iWildCam**, $\mathbb{P}(\Delta > \varepsilon) \leq 0.007$ across all six candidates, and the certificate commits FREEZE on 22–33% of cells against an always-adapt regret of 0.098–0.108. On **Office-Home `tent_online_aggressive`**—the 0/0/35 case—$\varepsilon = 0.0454$ is 2.3× the mean absolute effect and, decisively, always-adapt regret (0.0100) and always-freeze regret (0.0096) differ by 0.0004: total abstention costs four hundredths of an accuracy point because there was no decision worth making. The same certificate commits on 8.6–37.1% of cells on the other Office-Home candidates, so 0/0/35 is a property of that adapter, not of the method. The rows where the certificate is unambiguously earning are the ones where it commits FREEZE against an always-adapt regret of 0.496 (Office-Home `labelshift`, 35/35 FREEZE) and 0.272 (RxRx1 SAR): unguarded adaptation would have destroyed the model there.

Power analysis: what would have to improve. Modelling $\widehat{\Delta} = \Delta + c\,r$ with Δ and the residual r resampled from their empirical distributions and the radius recomputed as the exact-rank order statistic the real certificate uses, c is an evidence-quality multiplier (c=1 is today's estimator). Residuals are *not* Gaussian—a Gaussian radius would be 3.6× the true one on ImageNet-C SAR—so the empirical form matters. **Yield above** 50% **at** α=0.10 **requires the margin to separate** sign Δ **at AUC** $\approx$ 0.94–0.99; CIFAR-10-C (0.982) and ImageNet-C EATA (0.957) are past that phase transition, ImageNet-C Tent (0.741) is nowhere near it. Two diagnostics locate the binding constraint. First, the radius floor: Δ is a paired accuracy difference on n_D samples, so ε cannot fall below $\Phi^{-1}(0.95)\,\mathrm{sd}(\Delta)$. On CIFAR-10-C the observed $\varepsilon = 0.0179$ sits at 0.84 of the upper end of a $[0.0110, 0.0213]$ bracket at n_D=2000—**already at the finite-sample floor**, so no better estimator can shrink it and yield is anyway at 99% of its ceiling. On ImageNet-C Tent the radius is 3.3×

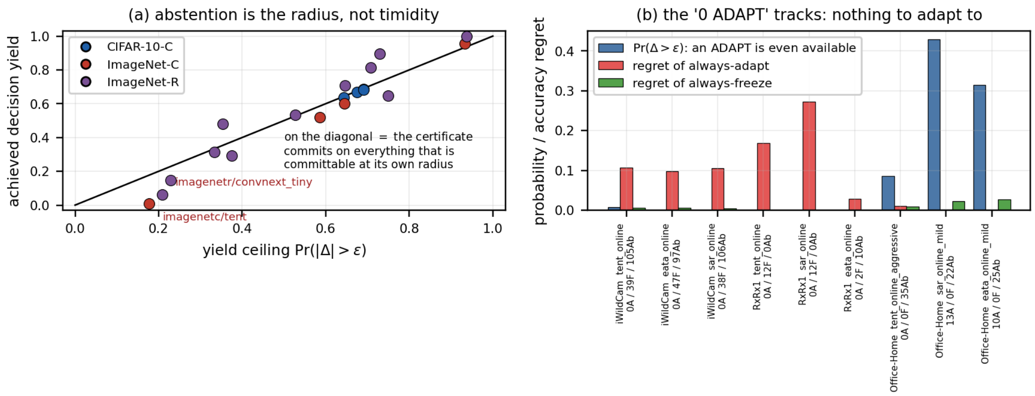

Figure 5: **Abstention is the radius, not timidity.** *(a)* Achieved decision yield against the ceiling $\mathbb{P}(|\Delta| > \varepsilon)$; points on the diagonal commit on everything committable at their own radius. *(b)* On the natural tracks the critique names, the fraction of cells on which an ADAPT is even available at the published radius, against the regret of the two fixed policies. On RxRx1 and iWildCam that fraction is ≤ 0.007, and the certificate's value is carried entirely by its FREEZE commitments.

the floor, and that excess is estimator error no amount of extra data removes. Second, shrinkage: the plug-in regresses on Δ with slope 0.995 on CIFAR-10-C but 0.322 on ImageNet-C Tent, where shrinkage accounts for 55% of the residual variance and hurts twice (it pushes $|\widehat{\Delta}|$ toward zero *and* inflates the radius). Removing the shrinkage alone—making the estimator merely unbiased, noise unchanged—takes ImageNet-C Tent from 0.7% yield to 45% at $\mathrm{FA_u} = 0.005$. **The binding constraint is estimator calibration, not sample size**, and that is the concrete target for future work. These are retrospective design calculations, not estimators, and are labelled as such.

The scoped claim. The certificate's value is not "no harm"—abstention alone would deliver that, since an ABSTAIN realizes the frozen model. Its value is decision yield at a controlled false-adapt rate, and that value is concentrated where adaptation is genuinely two-sided. On the CIFAR-10-C stress grid it commits on 66.8% of 6,480 cells with zero false adaptations, cutting regret 5.0× below the better fixed policy and 3.1× below a hindsight-tuned label-free drift heuristic at the same false-adapt budget, with commitments landing on cells whose true effect is 24.5× larger than those it declines. On natural shifts it buys far less: on iWildCam and RxRx1 no cell has a true benefit exceeding the conformal radius, so the zero ADAPT decisions are correct rather than timid and the value is entirely in the FREEZE commitments; on ImageNet-R, where adaptation almost never hurts, the certificate is 1.76× worse than always-adapt and its committed decisions are statistically indistinguishable from a random subset. **K-Bound buys decision yield under two-sided risk, not uniform improvement; on one-sided natural shifts its guarantee is close to free but also close to empty.** Threats to these conclusions are recorded in §11.

10 Ablations and Sensitivity

10.1 Value of the Uncertainty Radius

Table 17 isolates the radius: KGA without the radius has lower point regret but a higher empirical false-adapt rate, whereas the full certificate sacrifices coverage to eliminate observed harmful commitments on the locked grid. This comparison separates benefit-prediction quality from certification.

Table 27: Ablation (i): safety-level α sweep (exact-rank radius), CIFAR-10-C Tent (n=432). $\mathrm{FA_u}$=0 at every certified α; only the radius-free row breaks it.

α	regret↓	$\mathrm{FA_u}$	adapt	coverage
0.01	0.0036	0.000	0.45	0.48
0.05	0.0021	0.000	0.49	0.63
0.10	0.0019	0.000	0.50	0.67
0.20	0.0010	0.000	0.53	0.74
no radius	0.0004	0.051	0.68	1.00

Table 28: Ablation (ii): benefit estimator (exact-rank radius), CIFAR-10-C Tent (α=0.10). The radius keeps $\mathrm{FA_u}\approx 0$ regardless of estimator.

estimator	regret↓	$\mathrm{FA_u}$	adapt	coverage
gradient boosting	0.0019	0.000	0.50	0.67
random forest	0.0017	0.002	0.51	0.65
ridge (linear)	0.0043	0.000	0.45	0.47
small MLP	0.0143	0.000	0.36	0.36

10.2 Alpha, Evidence, Estimator, and Transfer Sensitivity

All ablations reuse the logged CIFAR-10-C stress evidence (per-condition Z and true benefit B; 432 conditions/candidate, seed 0) and the deployed rule with an *exact-rank* empirical residual radius $\varepsilon = r_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$ (locked exact-rank ablation artifact with input hashes, App. L). At the deployed operating point (Tent, α=0.10) the harness returns regret 0.0019, $\mathrm{FA_u}$=0, adapt 0.50, coverage 0.67 — the same $\mathrm{FA_u}$ as the locked gate row of Table 17 and the same decision profile to within one point of coverage, but not the identical number: the ablation harness cross-fits the benefit model in 8 folds while Table 17 refits it leave-one-cell-out (431 fits per cell), and the cheaper fit costs 0.0003 of regret. The two are consistent, not interchangeable.

(i) Safety level α. Sweeping α (Table 27) trades decision coverage for regret while holding $\mathrm{FA_u}$=0 at every certified level; only the radius-free variant breaks the guarantee ($\mathrm{FA_u}$ = 0.051). **(ii) Estimator.** Swapping the benefit regressor (Table 28) leaves the false-adapt guarantee intact: gradient boosting, random forest, and ridge all keep $\mathrm{FA_u}\leq 0.002$; a small MLP over-abstains on this small calibration set. **(iii) Evidence family.** Dropping any single evidence family (frozen, adapted, change, drift, or update) keeps regret in $[0.0012, 0.0019]$ at $\mathrm{FA_u}\leq 0.002$: no single family is load-bearing for regret. The one family that is not free to drop is *change*: without it the radius admits a false adapt ($\mathrm{FA_u}$=0.002, 1 cell of 432), the only non-zero entry in this sweep. **(iv) Cross-adapter transfer.** Fitting the benefit model on one adapter and applying it to another (Table 29) *breaks* the false-adapt bound in most directions (SAR→Tent $\mathrm{FA_u}$ = 0.25), which is exactly why the deployed method refits per adapter. **(v) Drift budget β.** The population-frontier β sweep is §5, and it is kept distinct from the empirical radius ε: ε is a conformal quantity and is not an estimate of β.

Table 29: Ablation (iv): cross-adapter transfer (exact-rank radius, α=0.10). Transferring one benefit model across adapters violates $\mathrm{FA_u} \leq \alpha$ in most directions—per-adapter calibration is required.

fit → apply	regret↓	$\mathrm{FA_u}$	adapt	coverage
Tent → EATA	0.0022	0.000	0.54	0.63
EATA → Tent	0.0010	0.007	0.55	0.69
Tent → SAR	0.0173	0.125	0.50	0.72
EATA → SAR	0.0236	0.192	0.57	0.62
SAR → Tent	0.0085	0.255	0.84	0.84
SAR → EATA	0.0027	0.116	0.75	0.75

11 Limitations and Failure Modes

Scope of the guarantee. K-Bound does not guarantee that a newly encountered deployment is exchangeable with the calibration data, that label-free evidence is risk-aligned, or that a fitted benefit estimator transfers. Its guarantee is conditional: *if* the declared benefit interval attains marginal target coverage, *then* strict adaptation on $\widehat{\Delta} - \varepsilon > 0$ controls the unconditional false-adapt event at level α, and strict freezing on $\widehat{\Delta} + \varepsilon < 0$ the false-freeze event. Deployment diagnostics can identify support shift, instability, leakage and insufficient effective sample size, but passing them does not prove coverage. When required checks fail or cannot be evaluated, the prescribed output is ABSTAIN or FREEZE — not a certified adaptation decision.

Scope of empirical decision evidence. On the stress grid KGA exercises selective adapt/freeze/abstain behavior for Tent/EATA (Table 15); CIFAR-10-C SAR is withheld. On ImageNet-C the pattern flips by candidate—Tent/EATA are robust while SAR collapses and KGA retains low regret under a mechanism-faithful re-run (Table 20). Per-corruption CIFAR-10-C and helpful natural shifts are insurance regimes, not policy wins.

Real-shift scope. The uniform panel deliberately distinguishes three outcomes, and the no-harm claim covers only the first. Camelyon17 genuine OOD and RxRx1 are no-harm results: KGA ties the better fixed policy at zero observed false-adapt — with the caveats that Camelyon17's promoted n=18 triple, although recomputable from the release as of 2026-07-26, is scored on a subset with *no harmful cell* (all 18 have $\Delta > 0$) and on which KGA adapts 18/18, so it is behaviourally always-adapt there; and that RxRx1 makes zero ADAPT decisions. Neither exercises the guarantee. Office-Home and iWildCam are locked from their out-of-fold bootstrap seal (App. L; Office-Home promotes OOF no-harm only; LOO beats-both is not claimed); iWildCam makes one ADAPT decision, and the record file behind its promoted triple is absent from the release, so that row is a sealed summary rather than a reproduced result. PACS and ImageNet-R are completed, locked null diagnostics in which KGA *loses* to always-adapt — by 2.45× on PACS and on 7 of 10 ImageNet-R backbones — and CIFAR-10.1 is a locked diagnostic fail. We state the losses in the limitations section as well as the results section because the abstract of an earlier version of this paper asserted no-harm across "every natural distribution shift we test", which these three tracks contradict. The three-source mixture is a constructed routing aggregate; it is not a transfer result and does not change the per-dataset conclusion.

Theory and calibration. The conformal radius assumes exchangeable calibration pairs; sign bracketing without structure is impossible (Theorems 1 and 2; Appendix C). KGA's value scales with how often catastrophic, detectable harm occurs, not with universal accuracy gains. On helpful-dominated shifts the resulting near-*tie* with always-adapt reflects the certificate abstaining rather than committing harmful updates: KGA approximately matches the better policy, at most paying a small abstention cost when it declines a helpful but uncertifiable update, so the value proposition is conditional, like insurance against detectable harm. *Setting β.* The budget is *declared*, not estimated from unlabeled data — and provably so: no label-free procedure supplies a budget below the fibre radius, which for the deployment class is β itself (Theorem 3, Corollary 3). Every audit in Appendix P purchases the first term of Proposition 2 with labels at an anchor and sets the third to zero by declaration; none of them computes a budget from unlabeled deployment data, and Theorem 4 says that labels at an anchor with no declared coupling to the deployment domain move the minimax budget by exactly zero. The only route to a budget available to a deployer without deployment labels is the episode route of §6, which needs $K \geq 1/\alpha - 1$ logged deployments with retrospectively revealed outcomes and pays for them with an exchangeability assumption that §6.8 measures failing. Absent all of that, β is fixed from domain knowledge, historical dev-to-deployment gaps, or an explicit stress envelope, and is a declaration. The choice β=0 is the strongest zero-drift assumption and may be reported only as a plug-in baseline; it is not an assumption-free or conservative default. When no credible bound is justifiable, the population frontier cannot certify robustness to calibration drift, and the safe operational response is to abstain or freeze. Larger β widens the abstain band monotonically, trading decision coverage for robustness.

β is not estimable from development data, and the frontier's operational reading is withdrawn. This is the paper's own negative result; §5 reports it in full and Theorem 4 says why it had to happen. We repeat here only the consequences. (i) We do not claim $|M| > \beta$ as a deployable rule and never ran it as one. (ii) We keep Theorem 1 and the budget-impossibility results, whose proofs are untouched by any of this. (iii) The sensitivity is itself adverse: over $\beta \in [0, 0.30]$ regret spans 26× to 192×, and at a single fixed $\widehat{\beta}$ decision yield varies by 21 to 52 points across adapters and seeds, so β behaves as a free knob rather than a specification. (iv) A further limitation, new in this revision: episode exchangeability (E2) is *not verifiable, only falsifiable* (Corollary 5), and its label-free observable component is only a one-sided screen — the separability AUC correlates with realised coverage at $r = -0.35$ over $n = 470$ splits, so a low reading is informative and a high reading is only a warning. Limitations of the β-sweep test itself: two benchmarks with only 6 and 3 corruption families, so family-clustered intervals are reported but are not primary; the benefit rather than disagreement scale, since $\mu_T(D)$ is not recorded in these artifacts; $\beta^{\min}_{\text{sound}}$ is an oracle diagnostic no deployer could compute; and regret resolves ABSTAIN to FREEZE, which penalizes abstention heavily—the soundness conclusions do not depend on that convention, the regret comparisons do.

The escape is marginal, not conditional. The episode budget of §6 controls a long-run rate across deployments; it provides no protection *at* a deployment (Proposition 3(a), Remark 9). Conditional on the current episode being one of the adversarial ones, the audited frontier commits and is wrong, and the α it was run at says nothing about that. This is Remark 13 one level up, and a deployer who is told "β was estimated" should be told this as well.

The probe dominates at one deployment. Proposition 5(b) is the uncomfortable ranking. At a *single* deployment, buying $k = \Theta(\varepsilon^{-2})$ labeled probes strictly dominates estimating a budget: the probe certifies $\text{sign}(M + \gamma)$ directly, the budget route certifies only $|M| > |\gamma|$. The budget route is worth taking because it amortises over future deployments, and amortisation is exactly what E2 licenses. If E2 fails — and §6.8 measures it failing on the axis a novel deployment lives on — the budget route loses both its guarantee and its economic rationale.

Scope of the decision-value analysis. The frontier of §9.7 is a replay, not a re-run: no model is refit, and it inherits every limitation of the artifacts it replays. Four specific threats. n_D (2000/4000) is inferred from accuracy denominators rather than read from the files, so the radius-floor conclusions are stated as brackets. The heuristic references are given either a hindsight-optimal threshold or a labelled-seed-split threshold, both generous relative to a deployable heuristic—where the certificate wins it wins against a handicapped-in-its-favour opponent, and where it loses (ImageNet-R) it loses to a heuristic that is also achievable. The power model resamples Δ and the residual independently, dropping their empirical coupling; the de-shrunk variant, not the naive curve, is the corrected reading. And the natural tracks publish no per-cell $\widehat{\Delta}$, so their rows use the published ε and decision counts against per-cell Δ: $\mathbb{P}(\Delta > \varepsilon)$ and the regret columns are exact, but the $|\widehat{\Delta}|$-versus-ε split is available only on the three per-cell tracks. Finally, α=0.10 is a pre-registered safety choice and is *not* regret-optimal: κ=0.9 dominates it on two of three pooled tracks, and on ImageNet-C SAR κ=0.5 obtains 46% more decisions at 39% less regret for one extra false adaptation.

Detectability and transfer. ImageNet-R is a weak, split-specific evidence regime: no candidate survives both comparisons across four seeds. CIFAR-10.1 fails the cross-seed certificate bar ($\mathrm{FA_u}$ = 0.167 and descriptive $\mathrm{FA_c}$ = 0.444). These are useful boundary cases, not failed headline experiments: they show that a low-margin signal should not be promoted merely because one split or one candidate looks favorable.

Theory limits versus empirical limits. Some abstention is compatible with the population frontier, whereas baseline faithfulness, seed count, estimator quality, and replay status are ordinary empirical limitations requiring further validation. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding*: one cannot beat an oracle that is already always-adapt. Declining ImageNet-R is consistent with a weak-evidence, low-margin regime; the null result does not by itself establish structural non-identifiability under Theorem 2 (it could also reflect weak evidence, poor benefit estimation, or calibration-transfer failure). Abstention in the structurally ambiguous closed band prevents a uniformly sound strict commitment over the declared class. Empirical abstention, however, may instead stem from an overly wide radius or a weak estimator and therefore does not establish that a tested shift lies in that structural band. Richer label-free evidence may improve benefit estimation and reduce avoidable abstention, while better-justified or more data-efficient calibration can narrow finite-sample uncertainty; neither removes the population limitation without strengthening the declared assumptions or evidence.

Baseline scope and recomputation. On the ImageNet-C SAR configuration, the committal confidence gate adapts on harmful cells while KGA abstains on the low-margin band. The POEM/AETTA comparison is complete only for protocol-matched ports; official implementations remain future baseline work. The Office-Home/iWildCam summaries have a separate saved recomputation lock, but no external-lab replication. PACS and ImageNet-R are completed registered diagnostics and are retained as null/weak-evidence results rather than tuned into wins. The frozen external procedure is documented in `INDEPENDENT_REPLICATION_PROTOCOL.md`. A physical-camera study is pre-registered in Appendix H and *no physical result is claimed*: a template is not evidence, and we report it as a protocol only. Abstention lowers *decision* coverage but never withholds a prediction: it retains the frozen fallback f_0.

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; wins on helpful-dominated shifts where always-adapt is already oracle-optimal; that the drift budget β can be declared from development data (§5 measures the attempt and it fails); that β can be supplied by *any* label-free procedure (Theorem 3 makes this a theorem, not a concession); that episode exchangeability holds at a novel deployment (§6.8 measures it false, coverage 0.626 and 0.454); that weighted conformal repairs that (§6.8 measures it valid and vacuous, coverage 1.000 at yield 0.000); that $|M| > \beta$ is the rule KGA runs or should run; or that α=0.10 is regret-optimal.

11.1 Why Natural Shifts Mainly Produce No-Harm

A single natural benchmark is often one-sided: adaptation helps almost everywhere or harms almost everywhere. In the first case, always-adapt is already near-oracle; in the second, always-freeze is near-oracle. A three-way router gains a strict margin over both fixed policies only when helpful

Table 30: Deployment failures, diagnostics, and safe responses.

Failure	Diagnostic	Safe response
Calibration transfer fails	support/residual audit	recalibrate, widen radius, or abstain
True drift exceeds β	external domain monitoring	revise class/budget; freeze meanwhile
Evidence out of support	distance or density check in Z-space	abstain
Adapter/checkpoint changed	configuration hash mismatch	require recalibration
Batch too small	minimum-size rule	wait for more data or freeze
Continual state accumulates error	checkpoint/stream monitor	rollback to last certified state
Missing/invalid evidence	schema and numerical validation	freeze

and harmful conditions coexist and the evidence distinguishes them. Natural no-harm results and mixed-regime wins are therefore different operating points of the same decision geometry.

11.2 Four Sources of Abstention

Not every abstention has the same interpretation. *Structural non-identifiability* means opposite-benefit worlds share the same admissible evidence law. *Finite-sample uncertainty* means the sign is identifiable in principle but the interval still crosses zero. *Estimator inadequacy* means the chosen model fails to extract an available signal. *Calibration failure* means a development relationship does not transfer to deployment. We distinguish these causes throughout the evaluation rather than treating every null result as evidence of structural unknowability. §9.7 now separates them quantitatively, using the yield ceiling $\mathbb{P}(|\Delta| > \varepsilon)$ as the discriminator. Four of five per-cell slices commit on 93–102% of what is committable at their own radius, so their abstention is *finite-sample*: the radius is wider than the effect, and on CIFAR-10-C it is already at the n_D floor. ImageNet-C Tent is the one clear case of *estimator inadequacy*—4% of its ceiling, with 55% of the residual variance attributable to shrinkage, and removing that shrinkage alone would take yield from 0.7% to 45%. And §5 supplies the missing measurement of *calibration failure*: the drift a source-like calibration fails to anticipate is 1.4× to 50× the budget it declares. What we still cannot do is demonstrate *structural* non-identifiability on any real track; empirical abstention never establishes it.

11.3 Operational Failure Modes

Table 30 maps each deployment failure to a diagnostic and a safe response.

11.4 When to Deploy K-Bound

K-Bound is most valuable when harmful updates are plausible, the cost of a bad commit is high, a frozen fallback is available, and historical or structural information supports calibration. It is less useful when adaptation is uniformly benign, when no credible drift class or calibration set exists, or when the candidate update cannot be evaluated and rolled back safely.

12 Reproducibility

12.1 Artifacts and Result Lineage

All locked/reconciled rows in Table 23 link to saved raw cells or seed aggregates in the released repository, with two documented exceptions: the Office-Home and iWildCam promoted regrets, whose named record files are absent while other scorings of the same track survive and disagree

(App. L). Those two rows are marked *not reproducible from the release* rather than *locked*. The Camelyon17 n=18 triple was the third such row until 2026-07-26; its sealed reconciliation directory is still absent, but the row was recomputed from `camelyon17_richZ_F_v1/_partial.json`, which carries the full 5 seeds $\times$ 6 candidates $\times$ 3 domains $\times$ 6 grid the reconciliation partitioned. Calibrating on dev seeds $\{0, 1\}$ and scoring test seeds $\{2, 3, 4\}$ reproduces all four routes recorded in the sealed protocol — pooled n=54 to seven significant figures (3.616898×10^{-5}/1.320168×10^{-3}/0.0749240, $\mathrm{FA_c}$ exactly 1/39), n=36 freeze regret 0.11228, and the promoted OOD-only n=18 freeze regret 0.138129. That row is now *recomputable*, and it is also weaker than it looks: see below. Provisional rows are retained only to document completed evidence and the required replay; they are not reused as headline claims. The core certificate module is packaged as an installable module in the same repository. **Formal verification and claim-to-support map.** The Lean formalization inventory, its explicit non-coverage list, and the claim-to-support map are in Appendix K and Appendix O.3. Neither is evidence for an empirical claim; both are provenance.

13 Conclusion

A label-free adapt/freeze/abstain decision is possible only relative to a declared bound on calibration drift, and that bound cannot come from the deployment. This is a theorem with an exact value, not a limitation: for every declared class the least label-free-certifiable budget is the fibre radius Γ_z, attained by a constant audit that reads none of its data, and for the paper's own deployment class $\Gamma_z(\mathcal{C}_\beta) = \beta$ — the best possible label-free audit of the budget returns its own input, and the number it returns is simultaneously the radius of the band on which a strict commitment is unsound. One lemma carries this, the matched-evidence impossibility, and the abstention band, as readings of the same one-parameter family at three radii; labels at another domain move the minimax budget by exactly zero, which is why declaring β from development data fails on 6,885 real cells (§5) rather than merely being hard to estimate. The impossibility's configuration is not only constructed: in logged deployments there are pairs whose label-free evidence cannot be told apart while their benefit has resolvably opposite sign, and roughly 30% of evaluable CIFAR-10-C cells sit in fibres whose sign genuinely goes both ways — though such pairs are 39$\times$ *rarer* than chance, so the evidence is informative and merely insufficient (§4.7).

A budget can be obtained from a *population* of deployments, at a price we state exactly: between $1/(e\alpha) - 1$ and $1/\alpha - 1$ logged episodes with retrospectively revealed outcomes, three to nine at $\alpha = 0.10$, first feasible at exactly $K = 9$ on real data. The escape is one logical move — average over the episode law instead of supremum over the fibre — paid for with historical labels, and neither ingredient works alone. The guarantee it buys is marginal across deployments, not conditional on the one in front of you, and the exchangeability that buys it is what a shifted deployment violates: coverage collapses from a nominal 0.900 to 0.626 and 0.454 when the corruption family or severity changes, and weighted conformal restores it to 1.000 only by driving decision yield to 0.000 — the same vacuity reappearing inside the method that was supposed to escape it.

What remains is the deployable object: a finite-sample certificate that trades decision yield for a false-adapt guarantee, with the frontier measured. The guarantee is exercised with genuine statistical power on one track, the CIFAR-10-C stress grid, where the certificate commits on 66.8% of 6,480 cells at $\mathrm{FA_u} = 0/6480$, cuts regret 5.00$\times$ below the better fixed policy, places its commitments on cells whose true effect is 24.5$\times$ larger than the ones it declines, and still returns zero false adaptations in all ten leave-one-corruption-out runs. Elsewhere it ties the better fixed policy on the four one-sided natural tracks — three of those zeros weakly, as §11 records — and where it buys nothing we say so: PACS, ImageNet-R (1.76$\times$ worse than always-adapt) and CIFAR-10.1 stay in

Table 31: Base 11-dim evidence panel (stress grids). **Only 9 of the 11 coordinates are independent**: the definitions column shows that `entropy_drop` and `pbal_drop` are exact linear functions of four others, so every covariance of this panel is exactly singular (§4.7). No result in this paper depends on the two redundant coordinates, but a re-implementer should not count them as evidence.

Feature	Definition	Family
`pre_entropy`	$\mathrm{mean}_n H(p_{0,n})$	frozen output
`pre_conf`	$\mathrm{mean}_n \max_c p_{0,n}$	frozen output
`pre_pbal`	$H(\bar{m}_0)/\log C$	frozen marginal
`post_entropy`	$\mathrm{mean}_n H(p_{a,n})$	adapted output
`post_conf`	$\mathrm{mean}_n \max_c p_{a,n}$	adapted output
`post_pbal`	$H(\bar{m}_a)/\log C$	adapted marginal
`pbal_drop`	`pre_pbal`−`post_pbal`	change
`entropy_drop`	`pre_entropy`−`post_entropy`	change
`frac_highconf`	$\mathrm{frac}_n(\max_c p_{a,n}>0.9)$	collapse
`marginal_KL`	$\sum_c \bar{m}_{a,c} \log(\bar{m}_{a,c}/\bar{m}_{0,c})$	drift
`update_norm`	$\lVert\Delta\theta_{\mathrm{aff}}\rVert_2$	update

the panel as a conservative null, a loss, and a failed transfer bar.

The broader implication is that an adaptation system should expose two components: an update mechanism and a validity layer that decides whether the update is supportable under the available evidence and assumptions. Strengthening K-Bound therefore requires not only better adapters, but also richer risk-aligned evidence, more defensible calibration across shifts, and either logged deployment populations or deployment labels — not a better label-free audit, which §4 rules out. The measurements above make one requirement concrete: the binding constraint on decision yield is estimator *calibration*, not sample size—the phase transition sits at a margin AUC of roughly 0.95, and de-biasing an already-noisy plug-in is worth more than any feasible increase in evaluation batch size.

A Calibration and held-out evaluation

Algorithm 2 gives the evaluation protocol behind the uniform nine-track panel (Table 23), which is the primary numeric evidence. Its two branches are *not* interchangeable: stress grids are scored cell-by-cell with leave-one-cell-out cross-fitting (no single deployment estimator exists), whereas natural-shift tracks fit one estimator and radius on the development split and score the untouched held-out domains exactly once.

B Evidence feature schema and adapter hyperparameters

Two label-free evidence panels are used, matched to the two calibration regimes of §7. Notation: p_0, p_a are the frozen and adapted softmax rows on the adaptation batch; $\bar{m}_0 = \mathrm{mean}_n\, p_{0,n}$ and $\bar{m}_a$ are the predicted-class marginals; $H(p) = -\sum_c p_c \log p_c$; C is the number of classes. The stress-grid tracks (CIFAR-10-C, ImageNet-C, PACS) use the 11-dimensional base panel of Table 31 (`kbound_pkg/kbound/evidence.py`). The natural-shift tracks (Camelyon17, iWildCam, Office-Home, RxRx1) use the 16-dimensional panel of Table 32: a 10-statistic frozen-logit summary plus a 6-feature rich panel (`scripts/run_wilds_camelyon17.py`). All features are target-label-free; the ATC feature calibrates its threshold on *source* labels only.

Algorithm 2 Calibration and evaluation for K-Bound (grid vs. natural-shift branches)

Require: Conditions $\mathcal{C}$; frozen f_0; adapters $\mathcal{A}_1, \ldots, \mathcal{A}_K$; track type (stress grid or natural shift); level α

Ensure: Regret-to-oracle, false-adapt rate, coverage, and action composition

for all $c \in \mathcal{C}$, adapters $\mathcal{A}_k$ **do** ▷ shared evidence logging
    $f_a^{(k,c)} \leftarrow \mathcal{A}_k(f_0, X_c)$; $Z_{k,c} \leftarrow \phi(X_c, f_0, f_a^{(k,c)})$; $\Delta_{k,c} \leftarrow R_c(f_0) - R_c(f_a^{(k,c)})$
end for
if stress grid (CIFAR-10-C, ImageNet-C, PACS): $\mathcal{C}$ is one grid of cells **then**
    for all cells $i \in \mathcal{C}$ **do**
        Fit $\widehat{\Delta}_{-i}$ on the other $N-1$ cells; $\rho_i \leftarrow |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|$ ▷ leave-one-cell-out
    end for
    for all cells $i \in \mathcal{C}$ **do**
        $\varepsilon_i \leftarrow \rho_{(k)}$ over $\{\rho_j\}_{j \neq i}$, $k = \lceil N(1-\alpha) \rceil$; $\varepsilon_i \leftarrow +\infty$ if $k > N-1$ ▷ **leave-one-out-of-pool** (§8.3); unclamped rank rule, so an infeasible k returns $+\infty$ and forces ABSTAIN. Pool is a jackknife, not a clean split: **NOT exact split conformal**, coverage here is empirical
    end for
    for all cells $i \in \mathcal{C}$ **do**
        Run Algorithm 1 with $(f_0, \mathcal{A}_k, X_i, \widehat{\Delta}_{-i}, \varepsilon_i)$ ▷ each cell scored by the estimator that never saw it (cross-fitted)
    end for
else natural shift (Camelyon17, iWildCam, Office-Home, RxRx1): split $\mathcal{C} = \mathcal{C}_{\text{cal}} \,\dot{\cup}\, \mathcal{C}_{\text{test}}$ across domains
    Form out-of-fold residuals on $\mathcal{C}_{\text{cal}}$; $\varepsilon \leftarrow \rho_{(k)}$, $k = \lceil (N+1)(1-\alpha) \rceil$ ($+\infty$ if $k > N$) ▷ **cross-validated residuals applied to a full-data estimator; NOT exact split conformal.** The CV-vs-CV+ mismatch means no finite-sample guarantee attaches; jackknife+ at level $1-2\alpha$ would, and is not implemented
    Fit deployment estimator $\widehat{\Delta}(Z)$ on all of $\mathcal{C}_{\text{cal}}$; freeze $(\widehat{\Delta}, \varepsilon)$
    for all $c \in \mathcal{C}_{\text{test}}$ **do**
        Run Algorithm 1 with $(f_0, \mathcal{A}_k, X_c, \widehat{\Delta}, \varepsilon)$ ▷ untouched held-out domains, scored once
    end for
end if
Record decision, regret-to-oracle, coverage, false-adapt indicator per condition
Aggregate metrics vs. always-adapt, always-freeze, per-condition oracle

C Complete core proofs

This appendix is not required for the main empirical claim; it preserves the full theorem stack and validator mapping behind the three main-paper theorems: matched-evidence impossibility (Theorem 1), the benefit-sign frontier (Theorem 2), and the finite-sample certificate (Theorem 11). The main text contains the complete algebraic proofs; the material below records only the supporting matched-evidence form and zero-drift interpretation used by those proofs.

Core-theory inventory. The main text gives complete proofs of disagreement reduction, interior impossibility, boundary abstention, the strict-commitment frontier, and the coverage-based certificate. This appendix records the matched-evidence full form and the zero-drift interpretation. Minimax, one-bit, capacity, and rate extensions are kept in the long manuscript and are not part of the

Table 32: Rich 16-dim panel (natural shift): 10 frozen-logit statistics + 6 rich features. H is softmax entropy, $s = \max_c$ softmax the confidence, $E(x) = -\log \sum_c e^{z_c}$ the free energy; subscripts s, t denote source/target pools.

Feature	Definition	Family
Frozen-logit summary (10)		
ent mean/std	mean/std H	frozen
ent p25/p75	25/75 pct. of H	frozen
conf mean/std	mean/std s	frozen
frac hi-conf	frac(s>0.9)	frozen
frac hi-ent	frac($H > \log 2$)	frozen
marg. max	$\max_c \text{mean}_n\, p$	frozen marginal
marg. std	$\text{std}_c \text{mean}_n\, p$	frozen marginal
Rich features (6)		
`disagree`	frac($\arg\max f_0 \neq \arg\max f_a$)	disagreement
`ent_gap`	mean $H(f_0)$ − mean $H(f_a)$	change
`en_shift`	mean E_t − mean E_s	drift
`bn_kl`	$\text{mean}_{\text{ch}}\, \text{KL}(\mathcal{N}_{\text{run}} \,\Vert\, \mathcal{N}_{\text{batch}})$	drift
`atc`	$\text{frac}_t(s \geq t)$, t src-calib.	accuracy proxy
`conf_drop`	mean s_s − mean s_t	drift

Table 33: Per-track adapter operating points, read back from the archived run manifests (`argv` plus the driver's resolved constants) rather than from the intended configuration. Adapted parameters are BN/LayerNorm-affine only *except* on ImageNet-C, where `layer4` is also adaptable; TTA optimizer is Adam except SAR, which uses SGD(momentum 0.9) under SAM. EATA uses an entropy filter e_{thr}=0.4 ln K (≈ 0.92 for CIFAR-10, 2.76 for ImageNet); SAR adds SAM + entropy reset. Backbones for CIFAR-10-C, ImageNet-C, PACS, and Camelyon17 are verified from the run code; the remaining WILDS/DomainBed tracks use the standard baseline.

Track	Backbone	Adapter(s)	Batch / steps / lr (as run)
CIFAR-10-C stress	ResNet-18	Tent/EATA/SAR	both levels crossed per cell: mild = (10 steps, 10^{-3}) and aggressive = (50 steps, 2.5×10^{-3}); batch regimes large-iid / small / tiny
ImageNet-C	ResNet-50	Tent/EATA/SAR (SAM+reset)	**aggressive only, small batch only**: batch 16, 50 steps, lr $\mathbf{4\times10^{-3}}$, `layer4` adaptable. This is 16× SAR's published 2.5×10^{-4} and the block Niu et al. freeze is not frozen (§9.4)
PACS (LODO)	ResNet-18	Tent/EATA/SAR	mild and aggressive crossed: $10/10^{-3}$ and $50/2.5\times10^{-3}$
Camelyon17	DenseNet-121	EATA online	10 / 10^{-3}
iWildCam	ResNet-50	Tent episodic	10 / 10^{-3}
Office-Home	ResNet (DomainBed)	SAR online aggr.	50 / 2.5×10^{-3}
RxRx1	ResNet-50	episodic	5 steps, batch 64

An earlier version of this table recorded "mild & aggressive (per cell)" for ImageNet-C. That was wrong: the archived `argv` for every promoted ImageNet-C seed contains `-batch-regimes small -aggressiveness aggressive -adapt-lr 0.004`, i.e. a single operating point, and `SAR_FREEZE_LAYER4` is left at its default `False`. The corrected row is above.

short-paper claim stack.

Total variation. For probability measures P, Q on a common measurable space, $\text{TV}(P, Q) := \sup_A |P(A) - Q(A)| \in [0, 1]$; equivalently $\text{TV} = \frac{1}{2}\|P - Q\|_1$ when both are dominated by a common measure.

Theorem 12 (Matched evidence impossibility (full form))**.** *Theorem 1 establishes part (i) of the following (only part (i); (ii) and (iii) are proved here). Fix* $\alpha \in (0, \frac{1}{2})$*.*

(i) *Witness worlds with* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* $\operatorname{sign} \Delta_1 = -\operatorname{sign} \Delta_2 \neq 0$*.*

(ii) *For any test* g *deciding between the two worlds from* Z*, writing* $\mathrm{err}(g) := \mathbb{P}_1[g{=}2] + \mathbb{P}_2[g{=}1]$ *for the* sum *of the two error probabilities (not their average, and unrelated to the observable margin* M*),* $\inf_g \mathrm{err}(g) = 1 - \mathrm{TV}\big(\mathrm{Law}(Z \mid P_T^1), \mathrm{Law}(Z \mid P_T^2)\big)$*. On the witness pair of (i) the two laws coincide, so* $\mathrm{TV} = 0$ *and* $\inf_g \mathrm{err}(g) = 1$*.*

(iii) *Forced abstention at rate* $\geq 1 - 2\alpha$ *under dual error control.*

Proof. Part (i) is Theorem 1, itself Lemma 1 at one radius; part (iii) is Corollary 1, proved in the main text on exactly this witness pair. Only part (ii) is new. Let μ dominate both laws with densities p_1, p_2. For any (possibly randomized) test with rejection function $\psi \in [0, 1]$, $\mathrm{err} = \int \psi p_1 + \int (1 - \psi) p_2 = 1 - \int \psi (p_2 - p_1)$, which is minimized pointwise by the Neyman–Pearson choice $\psi = \mathbf{1}\{p_2 > p_1\}$, giving $\mathrm{err} = 1 - \int (p_2 - p_1)^+ = 1 - \mathrm{TV}$. Randomization on the tie set does not change the value. □

Corollary 6 (Closed-band abstention under dual error control)**.** *In the* $|M| \leq \beta$ *wedge of Theorem 2, Theorem 12(iii) implies any label-free rule with false-adapt and false-freeze* $\leq \alpha$ *must abstain with probability at least* $1 - 2\alpha$ *on matched evidence. Opposite nonzero signs are required only for* $|M| < \beta$*; at* $|M| = \beta > 0$ *zero-versus-strict ambiguity suffices, and at* $M = \beta = 0$ *the benefit is identically zero so neither strict action is sound.*

Proof. This is Proposition 1, restated at the level of the witness pair: apply Theorem 12(iii) to the pair supplied by Theorem 1 when $|M| < \beta$, and to the zero-benefit boundary law when $|M| = \beta > 0$ (there the "error" event in the zero world is $\{\Delta \leq 0\} \cap \{\Delta \geq 0\}$, which both strict actions incur). No construction is added. □

Remark 16 ($\beta = 0$ face). On the $\beta = 0$ face the class $\mathcal{C}_0$ forces $\gamma = 0$, so Lemma 2 gives $\operatorname{sign} \Delta = \operatorname{sign} M$ exactly, and the frontier of Theorem 2 specializes as follows. If $M \neq 0$, the strict action of sign M (ADAPT for $M > 0$, FREEZE for $M < 0$) is uniformly sound over $\mathcal{C}_0$. If $M = 0$, then $\Delta = 0$ identically, so under the strict-decision semantics neither strict action is sound and the maximal sound action is ABSTAIN. Existing label-free estimators may be interpreted as $\beta = 0$ plug-ins only when their estimand realizes the stated disagreement-score decomposition; assuming $\gamma = 0$ alone does not establish their soundness.

D Episode-budget proofs, full tables, and implementation disclosures

This appendix holds the proofs deferred from §6, the complete coverage grid behind Table 8, and two implementation bugs found in our own first version of this analysis. Everything below is regenerated by `experiments/kbound/frontier_sweep_v1/beta_estimability/{episode_beta.py,make_tables.py,compare_to_sweep.py,theory_selfcheck.py}`.

D.1 Proofs

Proof of Theorem 7. (a) $\beta^\star(\alpha)$ is by construction the $(1-\alpha)$-quantile of $|\gamma|$ under Q, hence a functional of the (W,γ)-law; and $\{|\gamma|\le b\} = \{P \in \mathcal{C}_b\}$ by the definition of $\mathcal{C}_b$, so the quantile characterisation is immediate. For the decision consequence, fix the history and condition on $\{|\gamma_{K+1}| \le b\}$. On that event Theorem 2(ii) applies, so a strict commitment made at margin $|M_{K+1}| > b$ has the correct sign; a false strict commitment therefore requires $\{|\gamma_{K+1}| > b\}$, an event of probability at most α. (This is Theorem 14 with the audit event supplied by Q instead of by a concentration bound.)

(b) Fix the W-marginal Q_W. By Lemma 1, for any episode with observables (μ, f_0, f_a, s) — hence any value of W — and any $\bar{a} \in [0,1]$ there is a label kernel on D realising $\bar{a}$ while leaving (Z, M) unchanged. The achievable drift set at matched evidence is $I_M = [-(M+\frac{1}{2}), \frac{1}{2} - M]$, an interval of length 1 containing 0 whose endpoints have absolute values $\frac{1}{2}+M$ and $\frac{1}{2}-M$, the larger being $\frac{1}{2}+|M| \ge \frac{1}{2}$. Hence for every $b \in [0,\frac{1}{2}]$ at least one of $+b, -b$ lies in I_M; concretely, set $\theta(W) = +1$ if $b \le \frac{1}{2} - M$ and $\theta(W) = -1$ otherwise, which is admissible because $b > \frac{1}{2} - M$ forces $M + \frac{1}{2} > 1 - b \ge b$ for $b \le \frac{1}{2}$. Define Q_b by drawing $W \sim Q_W$ and realising $\gamma = \theta(W)\, b$ by the corresponding kernel. Then Q_b has W-marginal Q_W, and $|\gamma| = b$ Q_b-almost surely, so $Q_b(|\gamma| \le b') = 0 < 1-\alpha$ for every $b' < b$ while $Q_b(|\gamma| \le b) = 1$; hence $\beta^\star_{Q_b}(\alpha) = b$ for every $\alpha \in (0,1)$. Since b was arbitrary in $[0,\frac{1}{2}]$, the W-marginal carries no information about $\beta^\star$. ☐

Proof of Theorem 8. Define the counterfactual score $S_{K+1} = |\widehat{\gamma}_{K+1}|$ that a size-n label sample from the current episode would produce. Under E1–E2 the vector $(S_1, \ldots, S_{K+1})$ is exchangeable: $V_{1:K+1}$ is exchangeable, the label samples are conditionally i.i.d. of common size n given the episodes, and S_e is the same measurable functional of (V_e, labels_e) for every e. Split conformal on an exchangeable sequence gives $\mathbb{P}(S_{K+1} \le S_{(j)}) \ge j/(K+1) \ge 1 - \alpha + \delta$. Independently, Hoeffding on the n i.i.d. variables $u_i \in [-1,1]$ gives $\mathbb{P}(|\widehat{\gamma}_{K+1} - \gamma_{K+1}| > t(n,\delta)) \le \delta$. On the intersection, $|\gamma_{K+1}| \le S_{K+1} + t \le S_{(j)} + t$. A union bound gives failure probability at most $(\alpha - \delta) + \delta = \alpha$. Feasibility is $j \le K$, i.e. $\alpha \ge \frac{1}{K+1} + \delta$. ☐

Proof of Theorem 9. For $p \in (0,1)$ let Q_p be the i.i.d. episode law with $|\gamma_e| = b$ with probability p and $\gamma_e = 0$ otherwise (realisable at fixed observables by Lemma 1, as in Theorem 7(b)). Suppose $h(0,\ldots,0) < b$. The event $A = \{\gamma_1 = \cdots = \gamma_K = 0\} \cap \{|\gamma_{K+1}| = b\}$ has Q_p-probability $p(1-p)^K$ and on A the budget misses. Hence $\alpha \ge \sup_p p(1-p)^K$. Setting $p = 1/(K+1)$ gives $\alpha \ge \frac{1}{K+1}(1 - \frac{1}{K+1})^K$. Finally $K \ln(1 - \frac{1}{K+1}) = -K \ln(1 + \frac{1}{K}) \ge -1$ since $\ln(1+x) \le x$, so $(1 - \frac{1}{K+1})^K \ge e^{-1}$. ☐

Proof of Proposition 4. The Bernstein statement is standard for $u \in [-1,1]$ (range 2); the sufficient n follows by bounding each term by $\kappa\beta/2$. For the identity, calibration gives $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = \mathbb{E}[\mathbb{E}[(C-s)^2 \mid X]] = \mathbb{E}[\mathrm{Var}(C \mid X)] = \mathbb{E}[s(1-s)]$, so $\sigma^2 = \mathbb{E}[s(1-s)]$. Also $\mathrm{Var}(C) = \mathbb{E}[C] - \mathbb{E}[C]^2 = \mathbb{E}[s] - \mathbb{E}[s]^2 = \mathbb{E}[s(1-s)] + (\mathbb{E}[s^2] - \mathbb{E}[s]^2) = \mathbb{E}[s(1-s)] + \mathrm{Var}(s)$. ☐

Proof of Proposition 5. (a) Consider the two worlds $\bar{a} = \frac{1}{2} \pm \varepsilon$ realised at matched observables by Lemma 1; by Lemma 2 they have $\Delta > 0$ and $\Delta < 0$ respectively. Each probe is Bernoulli$(\bar{a})$ for the event $\{f_a(X) = Y\}$. In the $+$ world, $\mathbb{P}_+[\textsc{freeze}] \le \alpha$ and $\mathbb{P}_+[\textsc{abstain}] \le \eta$, so $\mathbb{P}_+[\textsc{adapt}] \ge 1-\alpha-\eta$; in the $-$ world $\mathbb{P}_-[\textsc{adapt}] \le \alpha$. Hence $\mathrm{TV}(\mathrm{Ber}(\frac{1}{2}+\varepsilon)^{\otimes k}, \mathrm{Ber}(\frac{1}{2}-\varepsilon)^{\otimes k}) \ge 1 - 2\alpha - \eta$. Pinsker and tensorisation give $\mathrm{TV} \le \sqrt{k\,\mathrm{KL}/2}$ with $\mathrm{KL} = 2\varepsilon \ln \frac{1/2+\varepsilon}{1/2-\varepsilon} = 2\varepsilon \cdot 2\,\mathrm{artanh}(2\varepsilon) \le 2\varepsilon \cdot \frac{4\varepsilon}{1-4\varepsilon^2} \le \frac{32\varepsilon^2}{3}$ for $\varepsilon \le \frac{1}{4}$. Thus $1 - 2\alpha - \eta \le 4\varepsilon\sqrt{k/3}$, which rearranges to the claim. (b) is arithmetic on the two label counts, using $n \asymp \sigma^2 \log(1/\delta)/\beta^2$ and $k \asymp \log(1/\alpha)/\varepsilon^2$. ☐

Proof of Theorem 10. (a) Write $\mathcal{H}$ for the history. By Theorem 8, $\mathbb{E}_{\mathcal{H}}[Q(|\gamma| > \widehat{\beta}(\mathcal{H}))] \le \alpha$. Since $Q'(A) \le \rho\, Q(A)$ for every measurable A, $\mathbb{P}_{Q'}(\text{miss}) = \mathbb{E}_{\mathcal{H}}[Q'(|\gamma| > \widehat{\beta})] \le \rho\, \mathbb{E}_{\mathcal{H}}[Q(|\gamma| > \widehat{\beta})] \le \rho\alpha$. The feasibility condition at level α/ρ is $\alpha/\rho \ge 1/(K+1)$. (b) Under the stated factorisation the scores $S_1, \dots, S_{K+1}$ are weighted exchangeable in the sense of [34] with weight function w; their Theorem 2 gives exact $(1-\alpha)$ coverage for the weighted quantile $\inf\{s : \sum_k p_k \mathbf{1}\{S_k \le s\} \ge 1-\alpha\}$, $p_k = w(W_k)/(\sum_i w(W_i) + w(W_{K+1}))$. Estimability of w from unlabeled data is the standard two-sample density-ratio reduction: $w \propto \frac{\pi(W)}{1-\pi(W)}$ for $\pi(W) = \mathbb{P}(\text{episode is current} \mid W)$, and W contains no labels. $\square$

Proof of Theorem 5. Take $s \equiv \frac{1}{2}$ on D, so $M = 0$, and for $t \in [0, \frac{1}{4}]$ let P_t be the law of Lemma 1 with $\eta \equiv \frac{1}{2} + t$; then $u_t \equiv t$ is the constant t, $P_t \in \mathcal{C}_{\mathcal{U}}$ whenever $t \le \tau(\mathcal{U})$, and $\gamma(P_t) = t$, while P_0 is the null law with $\gamma = 0$. The audit's data are n i.i.d. Bernoulli$(\frac{1}{2} + t)$ variables together with label-free quantities whose law does not depend on t (Lemma 1); write Q_t for the induced law of the audit's input and $A := \{\widehat{\beta} \ge t\}$.

Suppose, for contradiction, that $\kappa < t$ for some $t \le \min\{\tau(\mathcal{U}), \frac{1}{4}\}$ with $\mathrm{TV}(Q_0, Q_t) < \frac{1}{2}$. Validity at P_t gives $Q_t(A) \ge 1 - \delta \ge \frac{5}{6}$; usefulness at P_0 gives $Q_0(A) \le Q_0(\widehat{\beta} > \kappa) \le \frac{1}{3}$. Hence $\mathrm{TV}(Q_0, Q_t) \ge Q_t(A) - Q_0(A) \ge \frac{5}{6} - \frac{1}{3} = \frac{1}{2}$, a contradiction. Therefore $\kappa \ge t$ for every such t.

It remains to bound TV. The t-dependent part of Q_t is a product of n Bernoulli laws, so by the tensorisation of KL, then $\mathrm{KL} \le \chi^2$, then Pinsker,

$$\mathrm{TV}(Q_0, Q_t) \le \sqrt{\tfrac{1}{2}\mathrm{KL}(Q_t \| Q_0)} = \sqrt{\tfrac{n}{2}\,\mathrm{KL}(\mathrm{Ber}(\tfrac{1}{2}+t) \,\|\, \mathrm{Ber}(\tfrac{1}{2}))} \le \sqrt{\frac{n}{2} \cdot \frac{t^2}{\frac{1}{2}(1-\frac{1}{2})}} = t\sqrt{2n},$$

using $\chi^2(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) = (p-q)^2/(q(1-q))$ with $q = \frac{1}{2}$. Thus $\mathrm{TV} < \frac{1}{2}$ whenever $t < (2\sqrt{2n})^{-1}$. Taking the supremum of admissible t over $t < \min\{\tau(\mathcal{U}), \frac{1}{4}, (2\sqrt{2n})^{-1}\}$ gives the claim. (The conditions of Pinsker's and the $\mathrm{KL} \le \chi^2$ inequalities hold: both laws are mutually absolutely continuous for $t < \frac{1}{2}$.) $\square$

D.2 The full coverage grid

Table 34 is the complete version of Table 8: both benchmarks, both estimators, with the false-adapt columns and the weighted-conformal columns. $\mathrm{FA_u}$ and $\mathrm{FA_c}$ are the marginal and conditional false-adapt rates of the audited frontier run at $\widehat{\beta}_{\mathrm{epi}}$; $\mathrm{infl_{med}}$ and $\mathrm{infl_{max}}$ are the median and maximum ratio of the budget a sound decision would have required to the budget the estimator returned. **The null coverage is** 0.900 **throughout.**

The K-sweep. Table 35 reproduces the feasibility floor of Corollary 4 on real data. At $\alpha = 0.10$ the budget is infeasible at $K = 3$ and $K = 5$ and first feasible at exactly $K = 9 = \lceil 1/\alpha \rceil - 1$, where it is already at nominal coverage. Increasing K beyond 9 buys precision (the standard deviation falls from 0.088 to 0.021), not validity. *The binding constraint on this route is the number of logged deployments, not the sample size within one.*

D.3 Two implementation bugs in our own first version

Both were found by us, both are fixed in the released code, and both are recorded here because anyone reusing these artifacts will meet the second one.

1. *A nested cross-fit manufactured coverage of* 1.000. The first implementation fitted the benefit model with an inner cross-fit that made history residuals systematically larger than deployment

Table 34: **Episode-budget coverage, complete grid** ($\alpha = 0.10$, null 0.900). Source: `beta_estimability/episode_beta_results.json`, rendered by `make_tables.py`. `LOCO` does *not* fail on ImageNet-C because its three held-out "families" are Gaussian, shot and impulse noise — one kind of corruption, so the hold-out does not break exchangeability.

Benchmark / est.	axis	n_{dep}	coverage	95% CI	$\widehat{\beta}$	infl$_{\text{med}}$	yield	FA$_{\text{u}}$	cov$_w$	yield$_w$
CIFAR-10-C / M_{atc4}										
	`RAND`	24300	0.912	[0.908, 0.915]	0.0561	0.96	0.446	0.0072	0.911	0.447
	`LOSO`	32400	0.890	[0.886, 0.893]	0.0537	0.91	0.444	0.0134	0.879	0.438
	`LOPROT`	32400	0.897	[0.894, 0.901]	0.0543	1.05	0.436	0.0085	0.903	0.435
	`LOBATCH`	32400	0.897	[0.894, 0.901]	0.0554	0.97	0.437	0.0087	0.913	0.432
	`LOMO`	32400	0.864	[0.860, 0.868]	0.0532	1.12	0.460	0.0164	0.904	0.315
	`LOCO`	32400	0.866	[0.863, 0.870]	0.0540	0.94	0.438	0.0095	0.959	0.172
	`LOSTR`	32400	0.768	[0.763, 0.773]	0.0468	1.93	0.500	0.0207	0.952	0.285
	`LOSEV`	32400	**0.595**	[0.589, 0.600]	0.0516	8.63	0.424	0.0143	**1.000**	**0.000**
CIFAR-10-C / M_{gbm}										
	`RAND`	24300	0.905	[0.901, 0.908]	0.0196	0.99	0.647	0.0002	0.905	0.647
	`LOSO`	32400	0.875	[0.872, 0.879]	0.0192	0.97	0.647	0.0113	0.872	0.645
	`LOPROT`	32400	0.877	[0.874, 0.881]	0.0192	1.08	0.657	0.0002	0.880	0.655
	`LOBATCH`	32400	0.862	[0.858, 0.866]	0.0190	0.99	0.655	0.0078	0.861	0.658
	`LOMO`	32400	0.804	[0.800, 0.808]	0.0188	1.67	0.630	0.0216	0.880	0.414
	`LOSTR`	32400	0.721	[0.717, 0.726]	0.0176	2.60	0.594	0.0292	0.951	0.381
	`LOCO`	32400	**0.626**	[0.621, 0.631]	0.0192	2.37	0.645	0.0041	0.849	0.241
	`LOSEV`	32400	**0.454**	[0.449, 0.460]	0.0181	18.85	0.615	0.0022	**1.000**	**0.000**
IMAGENET-C / M_{atc4}										
	`RAND`	1515	0.871	[0.853, 0.888]	0.1077	1.24	0.050	0.0086	0.891	0.035
	`LOSO`	2025	0.880	[0.866, 0.894]	0.1105	0.76	0.065	0.0020	0.959	0.010
	`LOPROT`	2025	0.841	[0.824, 0.857]	0.1326	0.67	0.110	0.0025	0.851	0.089
	`LOMO`	2025	0.729	[0.709, 0.749]	0.1156	4.30	0.227	0.0025	0.898	0.039
	`LOCO`	2025	0.903	[0.889, 0.915]	0.1098	0.96	0.038	0.0030	0.959	0.009
	`LOSEV`	2025	0.875	[0.859, 0.889]	0.1191	1.14	0.075	0.0153	0.955	0.029
IMAGENET-C / M_{gbm}										
	`RAND`	1515	0.886	[0.869, 0.901]	0.0385	1.11	0.304	0.0046	0.890	0.299
	`LOSO`	2025	0.892	[0.878, 0.905]	0.0411	0.85	0.341	0.0020	0.960	0.107
	`LOPROT`	2025	0.771	[0.752, 0.789]	0.0355	1.19	0.393	0.0035	0.801	0.360
	`LOMO`	2025	0.712	[0.691, 0.731]	0.0338	4.80	0.504	0.0119	0.921	0.436
	`LOCO`	2025	0.888	[0.873, 0.901]	0.0395	1.08	0.307	0.0054	0.967	0.040
	`LOSEV`	2025	**0.606**	[0.585, 0.628]	0.0350	2.54	0.214	0.0207	0.943	0.035

Table 35: K-sweep, CIFAR-10-C / M_{gbm}, 200 draws per K, $\alpha = 0.10$. The "coverage" entries at $K = 3, 5$ are the vacuous 1.000 that an infeasible (infinite) budget returns and are not measurements.

K	3	5	**9**	10	20	50	200	1000
feasible fraction	0.00	0.00	**1.00**	1.00	1.00	1.00	1.00	1.00
mean coverage	—	—	**0.907**	0.909	0.912	0.905	0.897	0.903
sd	—	—	0.088	0.084	0.059	0.043	0.031	0.021
mean $\widehat{\beta}$	—	—	0.0292	0.0287	0.0246	0.0211	0.0196	0.0197

residuals, so the conformal quantile was inflated and the exchangeable control returned perfect coverage. The three-pool design of §6.8 — fit pool, history, deployment, disjoint, with M fit once and applied identically — fixes it, and the control then reads 0.905, as it must.

2. *An unshuffled* `cross_val_predict` *produced a separability AUC of* 0.24. The episode-weight fit called `cross_val_predict(..., cv=5)` without shuffling. `StratifiedKFold` then took contiguous blocks, and the deployment rows arrive in the artifact's native (method, seed, condition) order, so each fold was a coherent block of conditions; the separability AUC

Table 36: Numerical self-checks of the closed forms in §4 and §6.

	what is checked	result
C1	conformal coverage = $\lceil(1-\alpha)(K+1)\rceil/(K+1)$, infeasible iff $K < 1/\alpha - 1$ (Theorem 8)	PASS; 27 (α, K) cells, max deviation < 0.012
C2	$\sup_p p(1-p)^K$ attained at $p = 1/(K+1)$, value $\geq 1/(e(K+1))$ (Theorem 9)	PASS; 10 values of K
C3	control variate $\mathrm{Var}(C-s) = \mathrm{Var}(C) - \mathrm{Var}(s) = \mathbb{E}[s(1-s)]$ (Proposition 4)	PASS; $\mathrm{Var}(u) = 0.20001$ vs. $\mathbb{E}[s(1-s)] = 0.20001$, $\mathrm{Var}(C) = 0.25000$, $\mathrm{Var}(s) = 0.04999$; label saving factor 0.800
C4	$\mathrm{KL}(\mathrm{Ber}(\frac{1}{2}+\varepsilon)\|\mathrm{Ber}(\frac{1}{2}-\varepsilon)) \leq 32\varepsilon^2/3$ for $\varepsilon \leq \frac{1}{4}$, and the oracle likelihood-ratio test respects the k lower bound (Proposition 5(a))	PASS
C5	miss rate $\leq \rho\alpha$ (Theorem 10(a))	PASS *and tight*: measured 0.0978/0.1452/0.1968/0.2935 against bounds 0.10/0.15/0.20/0.30

came out at 0.24 on splits whose true separability is 0.5. With `shuffle=True` the control reads 0.497 ± 0.033. **Anyone reusing `cross_val_predict` on these artifacts should shuffle.** The superseded run is retained in the tree as `*_UNSHUFFLED_SUPERSEDED.*` rather than deleted.

A third, smaller correction: seeding used `hash(name)`, which Python salts per process. It was replaced by `zlib.crc32`, and the script is now bit-reproducible.

E Algebra self-checks

Every closed-form constant in §4 and §6 is checked numerically so that none of them is typed from memory (Table 36). Source: `beta_estimability/{theory_selfcheck.py,theory_selfcheck_results.json}`. All five pass.

F Multiclass and regression derivations

For multiclass 0/1 loss, errors cancel outside $D = \{f_0 \neq f_a\}$, giving $\Delta = \mu_T(D)(p_a - p_0)$ with $p_j = \mathbb{P}_T(f_j = Y \mid D)$. In contrast to binary classification, p_0 need not equal $1 - p_a$. If $p_a - p_0 = M_{\mathrm{mc}} + \gamma_{\mathrm{mc}}$ and $|\gamma_{\mathrm{mc}}| \leq \beta_{\mathrm{mc}}$, then $|M_{\mathrm{mc}}| > \beta_{\mathrm{mc}}$ is sufficient for a strict commitment; necessity additionally needs the declared class to realize the admissible evidence-identical drifts.

For squared-loss regression, let $m_T(x) = \mathbb{E}_T[Y \mid X = x]$. Direct expansion gives

$$\Delta = \mathbb{E}_T[(f_a - f_0)(2m_T - f_0 - f_a)]\,.$$

The integrand vanishes where $f_a = f_0$, so the comparison is again disagreement-local. For an observable proxy $u(x)$, define

$$M_{\mathrm{reg}} = \mathbb{E}_T[(f_a - f_0)(2u - f_0 - f_a)],$$
$$\gamma_{\mathrm{reg}} = 2\mathbb{E}_T[(f_a - f_0)(m_T - u)].$$

Then $\Delta = M_{\mathrm{reg}} + \gamma_{\mathrm{reg}}$. A declared bound $|\gamma_{\mathrm{reg}}| \leq \beta_{\mathrm{reg}}$ yields the same sufficient strict-commitment rule $|M_{\mathrm{reg}}| > \beta_{\mathrm{reg}}$; a converse again requires a richness assumption on the target class.

F.1 Multiclass and Regression Scope

The binary 0–1 loss case gives the cleanest frontier because, on $D = \{x : f_0(x) \neq f_a(x)\}$, exactly one predictor is correct. For multiclass 0–1 loss,

$$\Delta = P_T(D)(p_a - p_0), \qquad p_j = P_T(f_j(X) = Y \mid D),$$

so the decision becomes an ordinal comparison of the two disagreement-region accuracies. For squared-loss regression, the risk difference reduces to a disagreement-local moment. The main theorem uses a split-observable decomposition $\operatorname{sign}(\Delta) = \operatorname{sign}(M + \gamma)$; the multiclass bridge is stated below and the regression derivation appears in Appendix F.

Proposition 6 (Multiclass benefit decomposition)**.** *For multiclass* $0/1$ *loss,* $\Delta = \mu_T(D)(p_a - p_0)$ *with* $p_j = \mathbb{P}_T(f_j(X){=}Y \mid D)$*. If the disagreement-region accuracy gap admits a split-observable decomposition* $p_a - p_0 = M_{\mathrm{mc}} + \gamma_{\mathrm{mc}}$ *with* $|\gamma_{\mathrm{mc}}| \leq \beta_{\mathrm{mc}}$*, then* $|M_{\mathrm{mc}}| > \beta_{\mathrm{mc}}$ *implies* $\operatorname{sign}\Delta = \operatorname{sign} M_{\mathrm{mc}}$*. The converse strict-commitment result additionally requires a richness assumption: the declared class must contain augmented-evidence-identical laws whose admissible* γ_{mc} *values straddle* $-M_{\mathrm{mc}}$*, together with the zero-benefit boundary law. At* $M_{\mathrm{mc}} = \beta_{\mathrm{mc}} = 0$*, the sign is identified as zero.*

Proof. Off D the two predictors agree, so their $0/1$ losses cancel pointwise and only D contributes: $\Delta = \mu_T(D)\big(\mathbb{P}_T(f_a{=}Y \mid D) - \mathbb{P}_T(f_0{=}Y \mid D)\big) = \mu_T(D)(p_a - p_0)$. Since $\mu_T(D) > 0$, $\operatorname{sign}\Delta = \operatorname{sign}(p_a - p_0) = \operatorname{sign}(M_{\mathrm{mc}} + \gamma_{\mathrm{mc}})$; if $M_{\mathrm{mc}} > \beta_{\mathrm{mc}} \geq |\gamma_{\mathrm{mc}}|$ the sum is positive, and symmetrically for the negative case. The converse repeats the label-kernel construction of Theorem 1 with the multiclass conditional, which requires the declared class to contain the resulting evidence-identical laws — hence the richness hypothesis. Full derivation in Appendix F. □

In the multiclass experiments KGA estimates Δ directly through the benefit proxy $\widehat{\Delta} = h_\theta(Z)$ (not M_{mc} itself); the binary frontier is the identifiability backbone, and the multiclass results are evaluated through this benefit-estimation route.

G Historical protocol reconciliation

The main text contains the complete empirical accounting in the uniform nine-track panel. Historical route studies and superseded per-dataset tables are deliberately excluded from this short-paper build. Their raw logs remain in the repository for audit, but they are not independent empirical claims. The declared evaluation unit is a condition or evaluation cell (cross-fitted on the stress grids, held-out on the natural-shift tracks), never an arbitrary pool of frames or domains; all policies replay the same unit stream, and labels are revealed only after the KGA decision for offline scoring.

The retired synthetic wiring check. An earlier version opened the results with a synthetic run in which the generator supplies M, $\gamma \sim \mathrm{U}(-\beta, \beta)$ and hence $\operatorname{sign}\Delta$. It cannot come out negative: the evidence is four noisy copies of M, so the conformal radius converges to $q_{0.9}(|\mathrm{U}(-\beta, \beta)|) = 0.9\beta$ by algebra and the commit rule reduces to $|M| > 0.9\beta$. It is retained in the repository as a unit test and is not evidence. §5 supersedes it.

H Physical-study pre-registration

The two-phone package-inspection protocol is locked before fresh capture, in two tables: R1 fixes the split and deployment design, R2 fixes the primary endpoint layout. **No physical result is**

Table 37: **Controlled multimodal Protocol D33 (130 conditions).** Accuracy is averaged over the fixed corruption grid. The superiority probabilities are paired-bootstrap quantities saved by the locked analysis. This is a controlled mechanism confirmation, not a natural-shift result.

Policy or diagnostic	Result
Always fuse (adapt)	0.5832
Always single-A (freeze)	0.8536
KGA routed	**0.8568**
Oracle	0.8573
Actions (adapt / freeze / abstain)	9/119/2
Observed false-adapt	0/130
$P(\text{KGA} > \text{fuse})$ / $P(\text{KGA} > \text{single-A})$	1.000/1.000

claimed and every result cell of both tables is blank, and remains blank until the exact physical inventory, source-model gate, development seal, held-out chronology, replication replay, and anti-leakage audit pass the machine-readable publication gate. Browser previews, mock clips, and reconstructed media are software diagnostics, not evidence.

The locked table layouts themselves are released, unmodified, as `docs/research/kbound/edge/kbound_camera_main_tables.tex`; they are reproduced there rather than typeset here because every one of their result cells is empty.

I Controlled multimodal mechanism check

Protocol D33 is a pre-registered controlled two-view MNIST experiment: each digit is split into left- and right-half modalities, and the right modality is corrupted by Gaussian noise over 13 severities and 10 batches, yielding 130 conditions. This construction deliberately creates the named regime in which fusion changes from helpful to harmful and the change is visible in label-free reliability evidence. Table 37 reports the locked result.

The result confirms that the controller can exercise all three actions and improve on both fixed policies when the helpful-to-harmful transition is present and detectable. Its scientific scope is narrow: the corruption is injected, the gain over the safe single-modality policy is 0.0032, and the experiment does not establish a natural multimodal win or universal accuracy improvement. The protocol was pre-registered in `research_lock/CONTROLLED_MULTIMODAL_PROTOCOL_D33_v1.yaml` (sealed 2026-06-15, before any result was computed); the authoritative artifact is `experiments/kbound/results/controlled_multimodal_d33/results.json`, and the claim ledger records this result as `KB-CLAIM-027` (supported).

J Runtime details

The adapter update requires forward and backward computation before the controller can decide. The saved controller microbenchmark reports evidence extraction, benefit-model evaluation, rollback copy memory, and benefit-model size; these are the added-cost quantities reported in Table 38. A complete end-to-end profile separating frozen inference, candidate adaptation, candidate inference, rollback time, and total window latency remains pending and is not inferred from the controller-only microbenchmark.

Table 38: KGA controller *added* cost (ResNet-18/CIFAR; controller-only microbenchmark). Candidate adaptation/inference are unchanged by KGA and are *not* measured here; the end-to-end component profile is pending (App. J).

Component	Cost
Evidence extraction (per batch)	0.074 ms
Benefit-model evaluation (per decision)	0.127 ms
Controller added latency	**0.20 ms**
Rollback model copy (memory)	44.8 MB
Benefit model size	195 KB

J.1 Computational Cost

KGA cannot avoid the cost of proposing an update because candidate adaptation includes forward and backward computation before commitment. The controller's *added* cost is measured by a controller-only microbenchmark (Table 38): evidence extraction 0.074 ms and benefit-model evaluation 0.127 ms per decision (0.20 ms total), plus one model copy (44.8 MB, ResNet-18/CIFAR) for rollback; the benefit model itself is 195 KB. Candidate adaptation and inference are unchanged by KGA—f_a is evaluated regardless—and are expected to dominate end-to-end latency, but this microbenchmark does not measure them: a complete end-to-end component profile (frozen inference, candidate adaptation and inference, rollback time, total window latency) remains pending (App. J).

K Formalization inventory

The repository contains a kernel-checked Lean 4 formalization of indexed algebraic, finite-decision, measure-containment, and conditional uniform-rank results, listed in `formal/KBound/TheoremMap.lean`, with no `sorry`, `admit`, or project-defined axioms. It checks frontier sufficiency, all three deterministic branches, algebraic witnesses for the closed and open bands, both zero-versus-strict boundary witnesses, and the marginal false-adapt/false-freeze implication once coverage is assumed.

Successful compilation does not by itself certify a whole paper theorem, because each such theorem also contains unformalized clauses. The current development does *not* formalize the target-law witness construction, equality of its induced evidence laws, membership of those laws in $\mathcal{C}_\beta$, realization of the algebraic witnesses by measurable target laws, the zero-versus-strict boundary construction, distributional frontier necessity or pointwise maximality, risk alignment, multiclass identifiability, calibration transfer, or the lift from a conditional uniform-index model to arbitrary exchangeable deployment processes. These clauses are proved only in the manuscript, and compilation of the mapped helper declarations is not described as kernel verification of the complete paper theorem. The implementation theorem uses the same exact rank rule as the current clean-split scorer; historical benchmark artifacts produced with an interpolated empirical quantile are not presented as kernel-certified split-conformal experiments. Machine-checked learning-theory results are an emerging area—Lean formalizations of generalization-error bounds exist [43]—and to our knowledge machine-checked conformal-coverage results remain uncommon; no priority claim is made.

L Claim-to-artifact map

The machine-readable manifest `paper/generated/kbound_result_manifest.json` is the authoritative index for every promoted number, seed count, protocol, quantile convention, and known replay caveat. It was regenerated for this revision, and three of its `source` fields were wrong or absent before that: `cifar10c_tent` and `cifar10c_eata` named `stress_grid_multiseed_v1/LOCKED_ANALYSIS_RESULTS.json`, which holds 0.0016259/0.0079757/0.1239368 — *not* the values this paper prints; the promoted values come from `mixed_headtohead_v1/` and are now recomputed cell by cell from its five per-condition dumps. `cifar10_1_K` and `rxrx1_J` carried no `source` field at all and now name existing artifacts. `camelyon17_ood` named a directory that does not exist, and carried `NOT_REPRODUCIBLE_FROM_RELEASE` for that reason until 2026-07-26; the directory is still absent, but the row is recomputable without it and now carries `RECOMPUTED_FROM_RELEASE` and points at `experiments/kbound/results/camelyon17_richZ_F_v1/_partial.json`, which holds the full $5 \times 6 \times 3 \times 6$ grid the sealed reconciliation partitioned; calibrating on dev seeds $\{0, 1\}$ and scoring test seeds $\{2, 3, 4\}$ reproduces all four of its recorded routes, the pooled n=54 one to seven significant figures. `iwildcam_H_v2` moved the other way: its promoted `regret` triple mixed two scoring paths — the $\text{regret}_{\text{KGA}}$ slot carried the always-freeze value from the OOF bootstrap file, while the decision counts beside it came from the Protocol H v2 scorer — and it now reports the Protocol H v2 scorer throughout (0.0037, not 0.0041) with `NOT_REPRODUCIBLE_FROM_RELEASE`, because its held-out record file is absent. Every row that was recomputed says so, and every row that could not be says why.

The derived audit `paper/generated/empirical_audit/decision_metrics.json` adds adapt/freeze/abstain counts and rates, the exact source artifact, and a reproduction command. Three changes were made to it in this revision. First, its CIFAR-10-C rows recorded 1,109 and 1,243 ADAPT decisions where this paper prints 1,113 and 1,244. Both numbers were real, and of different things: 1,109 is the `stress_grid_multiseed_v1` five-seed aggregate under the *archived interpolated* quantile, assembled as seed 0's stored summary (221 adapts) plus seeds 1–4 recomputed (888). That seed-0 summary can now be checked directly: the seed-0 per-condition dump was an unmaterialised placeholder when this paragraph was first written and is readable as of 2026-07-26, and its own committed decisions are exactly 221 ADAPT / 74 FREEZE / 137 ABSTAIN, with its stored ε equal to `np.quantile`$(\rho, 0.9)$ and not to the exact-rank order statistic — direct confirmation that the 1,109 aggregate is the interpolated one. 1,113 is the `mixed_headtohead_v1` five-seed aggregate under the declared exact-rank rule, recomputed from all five dumps. The paper promotes the latter, so the audit now does too, and its reproduction command reproduces it. (The old glob was also self-inconsistent: it matched four files, 1,728 cells, while the row claimed $n = 2160$.) Second, it now carries one-sided Clopper–Pearson 95% upper bounds on the *conditional* false-adapt rate, which is the statistic that constrains the guarantee once the marginal rate is 0 (§9.6, Table 24). Second, the 95% Wilson intervals previously attached to `interval_coverage_observed` have been **deleted**: under in-sample rank calibration that count is $N - (N - k)$, a deterministic function of N and α rather than a binomial draw (it is $1940/2160 = 0.8981$ at N=432×5 and $24/27$ at N=27 for *any* data), so a binomial interval on it has no frequentist interpretation. Wilson intervals are retained only where the count is a genuine draw, e.g. the PACS false-adapt count. Third, the ImageNet-C rows were single-seed scorings under the superseded interpolated rule and are now the five-seed, 135-cell aggregates under the declared leave-one-out-of-pool exact-rank rule; note that under that pool the coverage count is *not* a deterministic function of n, which is why the ImageNet-C SAR row's FA_{u} is a measured 1/135 rather than a forced 0. Observed interval-hit coverage remains marked *empirical only*; it neither verifies exchangeability nor establishes the theorem's coverage premise or risk alignment. Historical natural-shift summaries that did not retain interval-hit records are marked `not_retained`, not assigned a synthetic coverage value. The compact claim matrix `paper/generated/empirical_audit/claim_matrix.md` records claim, dataset, seeds, artifact, paper consumer, and closure status. The reader-facing summary is the guarantee-accounting table of §O.3.

The two analyses added in this revision. Both live in `experiments/kbound/frontier_sweep_v1/`, are seeded at 20260726, require no GPU and no network, and rewrite their result JSONs byte-identically on re-run. *(i)* The β-sweep frontier test of §5: `beta_sweep.py` (with `fs_common.py`) reads the 15 CIFAR-10-C and 15 ImageNet-C per-condition dumps and writes `beta_sweep_results.json`; `leakage_check.py` is the adversarial label-firewall check whose six configurations must all pass before the results are read; `summarize.py` emits the tables. Every number in Tables 6 and 7 is a field of `beta_sweep_results.json`. *(ii)* The decision-value analysis of §9.7: `decision_value.py` writes `decision_value_results.json` and carries a `--self-test` mode that demonstrates its run-time label guard firing on B, on a copy of B, on a view of B, and on a_{oracle}, and that states the guard's own limitation (it catches copies and views, not arbitrary functions of a label, so the evaluation-only annotations still require structural review). Its integrity record shows the replayed κ=1 rule reproducing the released `kga_decision` on 7,365/7,365 cells. Figures 2, 4 and 5 are regenerated from those two JSONs alone by `docs/research/kbound/figures/source/make_frontier_sweep_figs.py`; that script contains no hard-coded results, and re-running it rewrites all three PNGs byte-identically.

(iii) Adversarial re-test of (i) and (ii). An independent check, written against the premise that the two analyses above are circular until proven otherwise, lives in the same directory and produced the caveats recorded in §5 and §9.7. `adversarial_ablations.py` re-runs the whole β pipeline end to end under four input perturbations—labels shuffled, Z replaced by Gaussian noise, Z row-shuffled, and M replaced by noise at the real M's scale—and additionally reports (a) the correlation of every Z coordinate with B, a_0 and a_{adapted}, (b) the overlap between the cells that declare $\widehat{\beta}$ and the cells that are scored, (c) the correlation between cells the splits treat as independent, and (d) the by-construction value of the coverage column. `adversarial_stability_probes.py` reports the GBM random-state sensitivity of the GBM rows and the null value of the share-of-ceiling statistic. Results are in `adversarial_ablations_results.json`, `adversarial_gbm_seed_stability.json` and `adversarial_share_of_ceiling_probe.json`. The headline separations survive: under a full label shuffle the out-of-fold AUC on the primary CIFAR-10-C configuration falls from 0.875 to 0.505 and the rule stops committing at all; with the labels left intact but M replaced by noise at the same scale, the commit-error rises from 3.1% to 27.8%, and with Z replaced by noise it rises to 25.6%. The reported numbers are therefore measurements, not arithmetic. The residual weaknesses the same check found are stated in the two sections rather than here.

M ImageNet-C multi-seed detail

Table 39 reports the SAR row of Table 20 per seed, under the declared unclamped exact-rank radius calibrated leave-one-out-of-pool (§8.3, §8.3).

Correction 1: the rule. Earlier versions of this table were computed with the interpolated empirical quantile `np.quantile`$(\rho, 0.9)$ while the main-text aggregate used the exact rank statistic, and the two rules are 2.4× apart on this track (pooled regret 0.0122 versus 0.0289; 61 versus 13 ADAPT decisions out of 135). The sentence "point estimates improve both fixed-policy regrets on every seed" was true only under the interpolated rule and is **false** under the promoted one: on seeds 0, 1, and 3 KGA never adapts, so its regret is bit-identical to always-freeze and it beats neither. The corrected statement is that point estimates improve both fixed-policy regrets on 2 of 5 seeds (seeds 2 and 4), and the pooled point estimate rests on 13 ADAPT decisions of which 10 come from seed 2.

Table 39: **ImageNet-C SAR per seed under the declared unclamped exact-rank rule, calibrated leave-one-out-of-pool** (n=27 conditions per seed, α=0.10). Regret is against the per-cell oracle $\max(a_0, a_a)$. "≡" marks a seed on which KGA never adapts, so its regret is bit-identical to always-freeze. AD/FR/AB are the action counts. The final row is the superseded in-pool scoring, printed so the reader can price the leakage; it is not the promoted row.

seed	KGA	adapt	freeze	$\mathrm{FA_u}$	AD	FR	AB	beats both
0	0.0319	0.0625	0.0319 ≡	0.000	0	3	24	—
1	0.0312	0.0595	0.0312 ≡	0.000	0	4	23	—
2	0.0226	0.0425	0.0284	0.037	10	2	15	yes
3	0.0290	0.0441	0.0290 ≡	0.000	0	3	24	—
4	0.0297	0.0561	0.0389	0.000	3	3	21	yes
pooled (promoted)	**0.0289**	0.0529	0.0319	**0.0074**	13	15	107	point est. only; Tab. 21
pooled, in-pool (superseded)	*0.0264*	0.0529	0.0319	*0.000*	12	14	109	—

Correction 2: the pool. The values previously printed here (0.0264 pooled, $\mathrm{FA_u} = 0$, 12/14/109) were computed with each cell's own residual inside the pool that produced its radius. Removing the scored index changes 2 of 135 decisions and gives the promoted row: 0.0289, $\mathrm{FA_u} = 1/135$, 13/15/107. The superseded in-pool values are kept in the last row of Table 39 for comparison, not for citation.

What the per-seed zeros mean, and the one that is not zero. At $n = 27$ the in-sample exact-rank ceiling is $(n - k)/n = 1/27 = 0.037$, so under an in-pool radius "$\mathrm{FA_u} \leq \alpha$" would be arithmetically forced (§9.6). Under the promoted leave-one-out pool it is not forced, and it is not 0: seed 2 takes 10 ADAPT decisions and 1 of them is on a cell with $\Delta \leq 0$, giving $\mathrm{FA_u} = 1/27 = 0.037$ on that seed and $1/135 = 0.0074$ pooled — inside $\alpha = 0.10$, but measured rather than forced. Conditional on the adapts actually taken, the one-sided Clopper–Pearson 95% upper bounds on $\mathrm{FA_c}$ are 0.394 for seed 2 (1/10) and 0.632 for seed 4 (0/3); seeds 0, 1, 3 make no adapt decision and admit no bound at all. The pooled bound is 0.316 on 1/13.

N Pre-registered arm inventory and the multiplicity family

§8.5 defines the confirmatory family prospectively as the pre-registered beats-both bars. This appendix publishes the inventory so a reader can see the denominator.

Campaign census. Across the committed result and protocol artifacts there are 1,387 recorded `beats_both` determinations (326 true, 1,049 false; a 23.5% base rate), spread over 70 distinct scored result units and 69 pre-registered protocol files in `research_lock/`, produced by campaigns named `win_hunt_v2`–`v5`, `win_finder_v*`, `win_loop_v1`, and `hard_dataset_win_loop_v1`. The overwhelming majority are development screens on dev splits. They are recorded here because a reader is entitled to know that the search happened, not because they belong in an inferential family.

Confirmatory family. Table 40 lists every (track, candidate) pair that reached a declared held-out scoring under a sealed protocol, with its verdict — including the pairs that failed. Holm at 0.05 is applied over this inventory, not over the subset that won.

Table 40: Confirmatory arm inventory: pairs scored once on declared test cells under a sealed protocol. "BB" = beats both fixed policies at the declared resampling unit.

Track	Candidate	Verdict on declared test cells
CIFAR-10-C stress	Tent	BB, CI excludes zero vs. both *at every resampling unit down to 6 corruption clusters*
CIFAR-10-C stress	EATA	BB point estimate; CI excludes zero vs. freeze at every unit, vs. adapt *only* at the cell / twin-pair unit ($[-0.0044, +0.0004]$ at 6 corruption clusters)
CIFAR-10-C stress	SAR	**quarantined**; loses to always-adapt on seeds 1–4
ImageNet-C	SAR	**no CI-supported BB**; point-estimate no-harm vs. always-freeze (0.0289 vs. 0.0319); both condition-level intervals include zero under the promoted leave-one-out pool
ImageNet-C	Tent	no BB; ties always-freeze exactly, 0/135 adapts
ImageNet-C	EATA	no BB; *trails* always-adapt 0.0007 vs. 0.0001
Camelyon17	EATA	no-harm, and vacuous: recomputable from release (2026-07-26), but all 18 cells are helpful and KGA adapts 18/18
Camelyon17 (multi-seed)	Tent/EATA/SAR	no BB; SAR over-freezes
iWildCam H v2	Tent	no-harm, ties freeze; 1 adapt
Office-Home M v2	SAR	no-harm; LOO beats-both deliberately *not* promoted
RxRx1 J	online	no-harm, ties freeze; 0 adapts
PACS (LODO)	fixed	**null**; loses to always-adapt 2.45×
ImageNet-R D	10 backbones	**null**; 0/10 CI beats-both; worse than adapt on 7/10
CIFAR-10.1 K	Tent/EATA/SAR	**fails** the transfer bar, $\mathrm{FA_u} = 0.167$
Three-source mixture	constructed	routing aggregate, not a transfer arm
Multimodal D33	constructed	BB on an injected-corruption grid; 9 adapts

O Panel detail, constructed routing, and consolidated claim accounting

This appendix holds three blocks moved out of the results section in the 2026-07 restructure (this is Appendix O): the per-track variance behind the panel means, the constructed heterogeneous routing aggregate (which is explicitly not a transfer result), and the consolidated claim and guarantee accounting, which is the same object as the claim-to-support map at a different granularity and is therefore presented once.

O.1 Variance behind the panel means

Three of the panel's aggregate rows average over a spread wide enough that the mean misdescribes the track. We report the spread rather than the mean alone.

PACS. The panel mean 0.0431 is an average over 12 domain–seed cells whose KGA regret ranges from 0.00529 to 0.15344 (median 0.03616, s.d. 0.03895). The adapt rate is *zero* on 11 of those 12 cells: PACS's entire adaptation evidence is 12 ADAPT decisions taken on the single cell art_painting/seed 1, and 2 of the 12 are false. That cell's own $\mathrm{FA_u}$ is $0.1111 > \alpha$, and art_painting/seed 2 abstains on all 18 cells (decision coverage 0.0). Pooled over all 216 cells, $\mathrm{FA_u} = 2/216 = 0.0093$ (Wilson 95% $[0.0025, 0.0331]$), but $\mathrm{FA_c} = 2/12 = 0.167$ with a Clopper–Pearson 95% upper bound of 0.438. PACS is a null in which KGA loses to always-adapt by 2.45×; it is reported, not withdrawn.

ImageNet-R. The mean-across-backbone row hides a reversal on most backbones (Table 41). KGA is worse than always-adapt on 7 of 10 backbones, by up to 434× (swin_b) and 43.9×

Table 41: **ImageNet-R per backbone** (4 seeds $\times 12$ conditions each). Regrets KGA / always-adapt / always-freeze; "harm" is the harmful base rate. Decomposition is from the archived per-backbone scoring; the panel row above is the exact-rank aggregate. Under either quantile rule KGA's mean exceeds always-adapt's 0.0064.

backbone	KGA	adapt	freeze	harm
convnext_base	0.0000	0.0000	0.0693	0.0%
convnext_tiny	0.0207	0.0015	0.0221	14.6%
efficientnet_b0	0.0000	0.0506	0.0000	100.0%
efficientnet_b3	0.0186	0.0000	0.0501	0.0%
resnet101	0.0152	0.0005	0.0238	8.3%
resnet152	0.0088	0.0000	0.0421	0.0%
resnext101_32x8d	0.0059	0.0000	0.0520	0.0%
swin_b	0.0226	0.0000	0.0375	2.1%
swin_t	0.0043	0.0106	0.0047	60.4%
vit_b_16	0.0160	0.0004	0.0237	6.2%

(vit_b_16), and 4 backbones have a degenerate 0% harmful base rate — always-adapt is exactly oracle there, so any abstention costs regret and the comparison is uninformative. On the two backbones with a substantial harmful base rate (efficientnet_b0 at 100%, swin_t at 60.4%) KGA is better than always-adapt, which is the regime map's prediction; the panel mean averages these opposite regimes together and should not be read as a single verdict.

CIFAR-10-C stress grid. Here the spread is small and it is worth saying so: per-seed KGA regret is 0.001567/0.001631/0.001639/0.001440/0.001853 for Tent (ε coefficient of variation 0.030) and 5/5 seeds beat both fixed policies. EATA is likewise 5/5 with ε c.v. 0.068. The withheld SAR arm's ε c.v. is 0.390, an order of magnitude larger, which is the quantitative form of the instability that put it in quarantine (§9.1).

O.2 Constructed Heterogeneous Routing

The three-source OOF mixture combines separately calibrated Office-Home, iWildCam, and Camelyon17 conditions. It demonstrates routing value under a researcher-constructed heterogeneous stream, not single-dataset natural-shift dominance or transfer to unseen shift types. On its $n = 143$ conditions the regrets are 0.0059 (KGA) against 0.0632 (always-adapt) and 0.0342 (always-freeze), under the declared exact-rank radius.

Throughout the panel, "CI utility" against a named policy means a paired bootstrap at the declared resampling unit whose interval excludes zero. ImageNet-C SAR has no such policy on either side: both of its condition-level gaps include zero under the promoted leave-one-out pool, so its entry is a point estimate against always-adapt *and* against always-freeze.

O.3 Consolidated Claim and Guarantee Accounting

Statement	Status	Requirement / observed
Abstention needed in matched-evidence opposite-benefit worlds	theorem	stated model class
The minimax label-free budget is the fibre radius Γ_z, and $\Gamma_z(\mathcal{C}_\beta) = \beta$	theorem	Thm. 3, Cor. 3 (§4); the best label-free audit returns its own input
Labeled data at a non-deployment anchor lowers the minimax budget	**false**	Thm. 4: it moves it by exactly zero absent a declared coupling
β estimable from K exchangeable logged episodes with revealed outcomes	theorem	Thm. 8 (§6); $K \geq 1/\alpha - 1$, guarantee is episode-marginal
E2 (episode exchangeability) holds at a novel deployment	**false, measured**	coverage 0.626 (corruption family) and 0.454 (severity) against a null of 0.900; 5/10 severity splits infeasible (§6.8)
Weighted conformal repairs the E2 failure	**valid and vacuous**	`LOSEV` coverage 0.454 → 1.000 at decision yield 0.615 → 0.000 (§6.8)
$\mathrm{FA}_u \leq \alpha$ (unconditional false-adapt)	theorem	measurable marginal coverage; exchangeability/shift correction only justifies coverage
$\mathrm{FA}_u \leq \alpha$ *as measured on the grids*	**forced, not measured**	under in-sample rank calibration the ceiling is $(n-k)/n < \alpha$ for every n used here (§9.6)
$\mathrm{FA}_c \leq \alpha$ (conditional false-adapt)	**not claimed**	CP_{95} upper bound is 0.0027 on CIFAR-10-C but 0.13–0.95 elsewhere
An observed $\mathrm{FA_u}$ exceedance exists in this paper	**yes, disclosed**	Camelyon17 SAR seed 0, $1/9 = 0.111$ under the archived interpolated rule; CP_{95} interval $[0.003, 0.429]$, i.e. inside binomial noise; PACS art_painting seed 1, $2/18 = 0.111$
Nine-track panel	**accounting**	sealed lock inventory (`NINE_TRACK_LOCK_SEAL_v1`); incomplete tracks stay diagnostic
Stress-grid action validity and routing utility (Tent/EATA)	locked	locked Protocol A v1 stress grid; 6 of 15 corruptions, severities $\{1, 5\}$
CIFAR-10-C *Tent* beats both, cluster-robustly	locked	adapt-gap CI excludes zero at all four units down to 6 corruption clusters $[-0.0095, -0.0026]$; survives leave-one-corruption-out refit (Table 16)
CIFAR-10-C *EATA* beats both, cluster-robustly	**not claimed**	adapt-gap CI includes zero at 12 clusters $[-0.0048, +0.0004]$ and 6 clusters $[-0.0044, +0.0004]$; two adverse corruption families. Point beats-both and the freeze-side CI stand
ImageNet-C SAR beats always-freeze	**point estimate only**	0.0289 vs. 0.0319; under the promoted leave-one-out pool the condition-level CI $[-0.0085, +0.0038]$ and even the i.i.d.-135 CI $[-0.0079, +0.0036]$ include zero; 2/5 seeds beat both
ImageNet-C SAR beats always-adapt	**point estimate only**	condition-level CI $[-0.076, +0.018]$ includes zero
Natural safety (RxRx1)	locked	held-out test artifacts; 0 adapts, so the bound is untested
Natural safety (Camelyon17 n=18)	**recomputable and vacuous**	recomputed 2026-07-26 from `camelyon17_richZ_F_v1/_partial.json`; all 18 cells are helpful and KGA adapts 18/18, so the zero false-adapt rate is untested
Office-Home and iWildCam OOF summaries	locked	OOF bootstrap seal (no-harm only; OH LOO BB not promoted)
Three-source mixed routing aggregate	locked	constructed OOF mixture, not transfer
Universal improvement	**not claimed**	impossible in general
No-harm on *every* natural shift	**withdrawn**	holds on the four one-sided locked tracks; PACS and ImageNet-R lose to always-adapt
ε estimates drift budget β	**false**	β is declared; ε is empirical
$\lvert M\rvert > \beta$ is a deployable rule at a declarable β	**withdrawn (measured)**	$\widehat{\beta}$ from source-like dev cells is 1.4–50× too small on CIFAR-10-C (24–73% of cells outside $\mathcal{C}_\beta$, commit error 0.4–16.6% against a promised 0%) and returns 0 commitments on all 405 ImageNet-C cells in 5 of 10 configurations (§5)
Population frontier matches the empirical rule at some $\beta > 0$	**false under held-out** M	fails on both yield and regret in 15 of 18 configurations; the 3 exceptions are the full-Z GBM channel on the 405-cell ImageNet-C grid at a β the declaration procedure does not return
β-thresholding raises yield over the interval at fixed evidence	**yes, scoped**	CIFAR-10-C, $M{:=}\widehat{\Delta}$, $\widehat{\beta}$=0.0097: yield 0.749 vs. 0.668, regret 0.00081 vs. 0.00150; separated at k=432 condition clusters, *not* at k=6 family clusters; costs 18 sign errors vs. 1

P Auditable drift budgets: the positive side

The impossibility half of the budget question is Theorem 3 and Corollary 2 in the main text. This appendix collects the *positive* side: the structural purchases under which a budget can be computed, and the rate each one achieves. Their role after §4 is precise: **each is an instance of Proposition 2. Each purchases the first term, $|\gamma_{\mathrm{cal}}|$, with labels at an anchor; each sets the third term ρ to zero by a declared invariance; each estimates the middle term in a class-specific way; and none of them omits the first term.** That last fact is the non-vacuity certificate for §4: if some audit here computed a budget with no labels anywhere, Theorem 3 would be false.

Write $u = \eta_a - s \in [-1, 1]$ on D and $\gamma(P) = \mathbb{E}_{\mu_P}[u \mid D]$. **Proof status.** Theorems 13, 14, and 15 are *stated here without proof* and are proved in the long version of this manuscript (appendix "Auditable drift budgets"); we mark them as such rather than let a reader assume a refereed proof is available, and no claim in the main text depends on them. Their synthetic validation is the repository artifact `audit_validation/results_audited_drift.json`.

Theorem 13 (Computed budgets under purchasable structure)**.** *Each of the following side informations yields a computable $\widehat{\beta}$ with $\mathbb{P}(|\gamma(P_T)| \le \widehat{\beta}) \ge 1 - \delta$:* (i) *$n$ deployment labels (Hoeffding; width $\sqrt{2\ln(2/\delta)/n}$);* (ii) *an L-Lipschitz drift on a bounded 1-d representation (Kantorovich–Rubinstein + DKW, using only unlabeled batches and labeled* calibration *data);* (iii) *unknown-direction index drift $u = g(\theta_*^\top \psi)$ on $\|\psi\| \le B$ (rotation-uniform adaptive max-sliced audit at rate $O(\sqrt{(d + \ln(1/\delta))/n})$ via halfspace-VC uniformity; sharper constants from known max-sliced concentration). More generally, any symmetric witness class of Rademacher/VC complexity $O(\sqrt{v/n})$ yields a* useful *root-n integral-probability-metric audit whenever the unknown drift lies in that class (long-manuscript Theorem Aud-H); full Lipschitz W_1 witnesses are excluded from that rate. The structural assumptions are declared and falsifiable, not verifiable—some such purchase is unavoidable by Corollary 2, and each of (i)–(iii) is the middle term of Proposition 2 with the first bought at an anchor and the third declared zero.*

Theorem 14 (Composition; fully empirical rule)**.** *For any audit with $\mathbb{P}(|\gamma| \le \widehat{\beta}) \ge 1 - \delta$: the audited rule (adapt iff $M > \widehat{\beta}$, freeze iff $M < -\widehat{\beta}$, else abstain) falsely strict-commits with probability $\le \delta$; replacing M by the batch estimate $\widehat{M} \pm t_M(m, \delta')$ gives a rule computable entirely from data with false-commit probability $\le \delta + \delta'$. On the synthetic sweep: $0/12$ false commits (empirical rule) and $0/13$ (population rule) versus $3/20$ for the β=0 plug-in.*

Theorem 15 (Domain-level verifiability with an exact floor)**.** *With K calibration domains exchangeable with deployment, common per-domain labeled size, and a drift predictor independent of all K+1 domains, a domain-level conformal bound certifies $|\gamma_{K+1}| \le |\gamma_{\mathrm{pred}}(Z_{K+1})| + s_{(j)} + e$ at level $1 - \alpha$, feasible iff $\alpha \ge \frac{1}{K+1} + \delta$. Alignment is thus verifiable exactly at the rate one can afford exchangeable domains (K=5: nothing below $\alpha \approx 0.17$+δ); per-deployment verification remains impossible.*

Open (long manuscript). (1) *Maximal auditable class:* useful $O(\sqrt{d/n})$ audits exist for rotation-closed classes whose dual witnesses have root-n complexity (Aud-F/Aud-H), while full Lipschitz/W_1 is rate-excluded; whether anything strictly larger than adaptive MSW$_1$-ridge admits a useful root-n audit—and whether one-sided audits can beat the spiked-transport estimation line—remains open (estimation lower bounds alone do not finish the question: Wasserstein testing can be easier than estimation). (2) *Certified poly-time supremum:* whether a poly(d, n) algorithm can output a sound and null-complete upper bound on linear max-sliced W_1 is open; only net certificates (not gradient-ascent surrogates) are admissible in Aud-F until settled.

References

[1] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell. Tent: Fully test-time adaptation by entropy minimization. *ICLR*, 2021.

[2] J. Liang, D. Hu, J. Feng. Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation. *ICML*, 2020.

[3] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, M. Hardt. Test-time training with self-supervision for generalization under distribution shifts. *ICML*, 2020.

[4] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, M. Tan. Efficient test-time model adaptation without forgetting. *ICML*, 2022.

[5] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan. Towards stable test-time adaptation in dynamic wild world. *ICLR*, 2023.

[6] Q. Wang, O. Fink, L. Van Gool, D. Dai. Continual test-time domain adaptation. *CVPR*, 2022.

[7] M. Zhang, S. Levine, C. Finn. MEMO: Test time robustness via adaptation and augmentation. *NeurIPS*, 2022.

[8] M. Schirmer, M. Jazbec, C. A. Naesseth, E. Nalisnick. Monitoring risks in test-time adaptation. *NeurIPS*, 2025. arXiv:2507.08721.

[9] A. Podkopaev, A. Ramdas. Tracking the risk of a deployed model and detecting harmful distribution shifts. *ICLR*, 2022. arXiv:2110.06177.

[10] Y. Bar, S. Shaer, Y. Romano. Protected test-time adaptation via online entropy matching: A betting approach. *NeurIPS*, 2024. arXiv:2408.07511.

[11] S. Garg, S. Balakrishnan, Z. Lipton, B. Neyshabur, H. Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. *ICLR*, 2022. arXiv:2201.04234.

[12] A. T. Kalai, V. Kanade. Towards optimally abstaining from prediction with OOD test examples. *NeurIPS*, 2021. arXiv:2105.14119.

[13] J. Miller, R. Taori, A. Raghunathan, S. Sagawa, P. W. Koh, V. Shankar, P. Liang, Y. Carmon, L. Schmidt. Accuracy on the line: On the strong correlation between out-of-distribution and in-distribution generalization. *ICML*, 2021. arXiv:2107.04649.

[14] C. Baek, Y. Jiang, A. Raghunathan, J. Z. Kolter. Agreement-on- the-line: Predicting the performance of neural networks under distribution shift. *NeurIPS*, 2022. arXiv:2206.13089.

[15] T. Lee, S. Chottananurak, T. Gong, S.-J. Lee. AETTA: Label-free accuracy estimation for test-time adaptation. *CVPR*, 2024. arXiv:2404.01351.

[16] J. Steinhardt, P. Liang. Unsupervised risk estimation using only conditional independence structure. *NeurIPS*, 2016. arXiv:1606.05313.

[17] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. Wortman Vaughan. A theory of learning from different domains. *Machine Learning* 79(1–2):151–175, 2010.

[18] E. Rosenfeld, S. Garg. (Almost) provable error bounds under distribution shift via disagreement discrepancy. *NeurIPS*, 2023. arXiv:2306.00312.

[19] Y. Geifman, R. El-Yaniv. Selective classification for deep neural networks. *NeurIPS*, 2017.

[20] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei. ImageNet: A large-scale hierarchical image database. *CVPR*, 2009.

[21] D. Hendrycks, T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*, 2019.

[22] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, et al. WILDS: A benchmark of in-the-wild distribution shifts. *ICML*, 2021.

[23] P. Bándi, O. Geessink, Q. Manson, M. van Dijk, M. Balkenhol, M. Hermsen, et al. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Trans. Med. Imaging* 38(2):550–560, 2019.

[24] J. Taylor, S. L. Earnshaw, B. M. Mabey, M. Victors, M. N. Annis, D. Y. Torigian, et al. The RXRX1 dataset and challenge: Unsupervised classification of cellular morphological changes. *arXiv:1907.04758*, 2019.

[25] H. Venkateswara, J. Eusebio, S. Chakraborty, S. Panchanathan. Deep hashing network for unsupervised domain adaptation. *CVPR*, 2017. (Office-Home dataset release).

[26] I. Gulrajani, D. Lopez-Paz. In search of lost domain generalization. *ICLR*, 2021.

[27] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, M. Hein. RobustBench: A standardized adversarial robustness benchmark. *NeurIPS Datasets and Benchmarks*, 2021.

[28] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. Do CIFAR-10 classifiers generalize to CIFAR-10? *ICML*, 2019.

[29] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al. PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*, 2019.

[30] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *JMLR* 12:2825–2830, 2011.

[31] R. Xie, A. Odonnat, V. Feofanov, W. Deng, J. Zhang, B. An. MaNo: exploiting matrix norm for unsupervised accuracy estimation under distribution shifts. *NeurIPS*, 2024. arXiv:2405.18979.

[32] W. Deng, Y. Suh, S. Gould, L. Zheng. Confidence and dispersity speak: characterizing prediction matrix for unsupervised accuracy estimation. *ICML*, 2023. arXiv:2302.01094.

[33] J. Liang, R. He, T. Tan. A comprehensive survey on test-time adaptation under distribution shifts. *IJCV* 133(1):31–64, 2025. arXiv:2303.15361.

[34] R. J. Tibshirani, R. F. Barber, E. Candès, A. Ramdas. Conformal prediction under covariate shift. *NeurIPS*, 2019. arXiv:1904.06019.

[35] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics* 51(2):816–845, 2023. arXiv:2202.13415.

[36] I. Gibbs, E. Candès. Adaptive conformal inference under distribution shift. *NeurIPS*, 2021. arXiv:2106.00170.

[37] S. Park, E. Dobriban, I. Lee, O. Bastani. PAC prediction sets under covariate shift. *ICLR*, 2022. arXiv:2106.09848.

[38] S. Bates, A. Angelopoulos, L. Lei, J. Malik, M. I. Jordan. Distribution-free, risk-controlling prediction sets. *JACM* 68(6), 2021. arXiv:2101.02703.

[39] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei. Learn then test: calibrating predictive algorithms to achieve risk control. *Annals of Applied Statistics*, 2025. arXiv:2110.01052.

[40] T. Hoang, D. M. Vo, M. N. Do. Persistent test-time adaptation in recurring testing scenarios. *NeurIPS*, 2024. arXiv:2311.18193.

[41] J. Lee, D. Jung, S. Lee, J. Park, J. Shin, U. Hwang, S. Yoon. Entropy is not enough for test-time adaptation: from the perspective of disentangled factors. *ICLR*, 2024. arXiv:2403.07366.

[42] T. Lim, J. Hwang, K. Lee. When and where to reset matters for long-term test-time adaptation. *ICLR*, 2026. arXiv:2603.03796.

[43] S. Sonoda, Y. Mizuno, K. Tsukamoto, N. Onda. Lean formalization of generalization error bound by Rademacher complexity and Dudley's entropy integral. 2025. arXiv:2503.19605.

[44] S. Sreeram, Y. D. Kwon, C. Mascolo. Tempora: Characterising the time-contingent utility of online test-time adaptation. *ICML*, 2026. arXiv:2602.06136.

[45] A. Corrada-Emmanuel. The logic of NTQR evaluations of noisy AI agents: Complete postulates and logically consistent error correlations. 2023. arXiv:2312.05392.

[46] C. Gupta, A. Ramdas. Top-label calibration and multiclass-to-binary reductions. *ICLR*, 2022. arXiv:2107.08353.

[47] S. Ben-David, R. Urner. On the hardness of domain adaptation and the utility of unlabeled target samples. *ALT*, 2012.

[48] Z. C. Lipton, Y.-X. Wang, A. Smola. Detecting and correcting for label shift with black box predictors. *ICML*, 2018.

[49] S. Garg, Y. Wu, S. Balakrishnan, Z. C. Lipton. A unified view of label shift estimation. *NeurIPS*, 2020.

[50] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Trans. Information Theory*, 16(1):41–46, 1970.

[51] C. Cortes, G. DeSalvo, M. Mohri. Learning with rejection. *ALT*, 2016.

[52] D. Madras, T. Pitassi, R. Zemel. Predict responsibly: improving fairness and accuracy by learning to defer. *NeurIPS*, 2018.